MINDSET 2.0
Model Documentation

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This document provides a comprehensive review of MINDSET 2.0, including theoretical underpinnings, mechanisms and practical considerations. Two related documents released alongside this main documentation describe parameter estimation of the behavioural parameters driving much of the model and the process of obtaining and compiling the data for these estimations. These documents can be found as:

1. Brancher, M., Byambasuren, B., Kiss-Dobronyi, B., Lin, X., Hartvig, A.D., Dorband, I.I., Pollitt, H. and Lehr, U. (2025): Estimation of Behavioral Parameters for the MINDSET Model

2. Brancher, M., Kiss-Dobronyi, B., Byambasuren, B., Lin, X., Hartvig, A.D., Dorband, I.I., Pollitt, H. and Lehr, U. (2025): Data Documentation for MINDSET Parametrization

The MINDSET 2.0 model code is open-source and under continued development, this document reflects the state of the model as of February 2026. The latest version of the model's code is available at https://github.com/cpmodel/MINDSET_dynamic/.

Contents

Acronyms	Full Name
CBAM	Carbon Border Adjustment Mechanism
CCS	Carbon capture and storage
CES	Constant elasticity of substitution
COICOP	Classification of Individual Consumption According to Purpose
COMTRADE	United Nations Commodity Trade Statistics Database (link)
CGE	Computable General Equilibrium
CPAT	Climate Policy Assessment Tool (World Bank & IMF, link)
CPI	Consumer Price Index
DSGE	Dynamic Stochastic General Equilibrium
E3ME	Environment-Energy-Economy Macro-Econometric Model (link)
ECM	Error Correction Model
ENVISAGE	Environmental Impact and Sustainability Applied General Equilibrium Model (link)
EORA	Eora Global Supply Chain Database (link)
EXIOBASE	EXIOBASE Multi-Regional Environmentally Extended Input-Output Database (link)
FAO	United Nations Food and Agriculture Organization
FTT	Future Technology Transformations models
GDP	Gross Domestic Product
GEM-E3	General Equilibrium Model for Economy-Energy-Environment (link)
GFCF	Gross fixed capital formation
GHG	Greenhouse gas
GINFORS	Global Interindustry Forecasting System model (link)
GLD	World Bank's Global Labor Database (link)
GLORIA	Global Resource Input Output Assessment database (link)
GTAP	Global Trade Analysis Project (link)
GVA	Gross Value Added
HBS	Household Budget Survey
IEA	International Energy Agency
ILO	International Labour Organization
IMF	International Monetary Fund
ISIC	International Standard Industrial Classification
KLEMS	Capital (K), Labour (L), Energy (E), Materials (M), and Services (S) Inputs Database
kW	Kilowatt
LCOE	Levelised cost of electricity
LFS	Labour force surveys
MANAGE	Mitigation, Adaptation and New Technologies Applied General Equilibrium Model
MFMod	World Bank's Macroeconomic and Fiscal Model
MINDSET	Model of Innovation in Dynamic Structural Economic and Employment Transformations
MRIO	Multi-Regional Input-Output table
MW	Megawatt
MWh	Megawatt-hour
ORBIS	ORBIS Company Database (link)
REITI	Research Institute of Economy, Trade and Industry (Japan)
SNA	System of National Accounts
TJ	Terajoule
UNDESA	United Nations Department of Economic and Social Affairs
UNIDO	United Nations Industrial Development Organization
WDI	World Bank's World Development Indicators

List of Acronyms

Acronyms	Full Name
WIIW	Vienna Institute for International Economic Studies
WIOD	World Input-Output Database (link)

List of Acronyms

Introduction

MINDSET 2.0 is a global, demand-driven, recursively dynamic macroeconometric simulation model that is used to evaluate how climate change, climate policies, and other development and trade policies affect macroeconomic, sector-level, and labour market outcomes. MINDSET stands for Model of Innovation in Dynamic Low-Carbon Structural Economic and Employment Transformations. This manual describes version 2.0 of MINDSET. The previous version of MINDSET (1.0), under development since 2021, is described in Dorband et al. (forthcoming).[1]

The goal of MINDSET is to provide actionable policy advice and to enable informed decisions that are tailored to country-specific opportunities and risks, while adopting a whole-of-economy and people-centred strategy. Model simulations can assess a range of fiscal, price-based, and regulatory policies, as well as climate change impacts (Table 1.1). The model has a high level of disaggregation, covering 163 countries and regions, with 120 sectors in each region. Hence, it supports the analysis of detailed sector- and country-specific, single- or multi-country policy and climate scenarios. MINDSET's results include standard macroeconomic indicators, sector-level employment, greenhouse gas (GHG) emission levels, and prices. By linking to countries' labour force and household survey data, impacts on workers and consumers can be analysed across income strata, skill levels, occupation types, and sub-national provinces, providing relevant information for active labour market and re-skilling interventions and industry-specific or regional government support.

The remainder of this introduction discusses the key features (Section 1.1) and main model mechanisms (Section 1.2), and compares MINDSET to other models (Section 1.3).

Policy and climate impact scenarios	**Metrics of interest for policy advice**
• Acute climate impacts: e.g., floods, storms, or wildfires, quantified by direct production losses • Chronic climate impacts: e.g., heat stress or loss of productivity • Carbon prices: commodity or carbon tax, subsidy, and tariff changes, differentiated by commodity & sector • Carbon Border Adjustment Mechanism (CBAM): single- or multi-country trade scenarios, differentiated by sectors • Income support or tax reforms • Sector-specific public spending, cross-funded or additional • Sector-specific public investment, cross-funded or additional • Electricity and transport sector policies (with sectoral models)	• Sectoral job losses/gains by: – Occupations & skill level – Income groups – Gender & age – Spatial: province/region • GDP, output (sectoral: GVA, output) • Tax revenue collection (by sector) • Emissions effects (GHG) • Sectoral consumer price changes & consumption incidence across income groups • Competitiveness and industry-specific import and export changes

Table 1.1: Policy and climate impact scenarios and results metrics

[1]The key differentiating aspects are that the 1.0 version solves only for a single period and that there are limited endogenous price adjustments as impacts on factor prices are not modelled. While their specific setup can differ, the 1.0 version includes most of the same key behavioural mechanisms as the 2.0 version, namely price and income effects of household demand, energy efficiency and substitution in production and demand, changes in trade relations, output-employment reactions, and induced investment.

1.1 Key features of MINDSET 2.0

MINDSET combines the strengths of input-output analysis with those of a demand-led macro-econometric model. The multi-regional input-output network links sectors and regions, while the macro-econometric equations determine behavioural dynamics. Several of MINDSET's features make it well-suited for assessing climate impacts and policies, and their distributional effects. However, its general economic structure also allows for application in other policy areas. Table 1.2 provides examples of how these features enable and support the modelling of certain policies and impacts.

Tracing cross-country linkages and spillovers. Its multi-regional design enables analysis of cross-country spillovers, third-country effects, and trade policies, capturing how shocks and policy changes propagate through global value chains and affect competitiveness and sectoral dynamics across interconnected economies. The detailed representation of supply chains allows for the analysis of knock-on, or cascading, impacts from shocks and supply constraints.[2] Cross-sectoral and cross-country spillovers may be important when considering global climate impacts or domestic climate policies that affect countries' competitiveness.

Capturing short-term adjustment frictions through empirical foundation. Its empirically grounded structure captures country- and sector-specific adjustment frictions and path dependency, allowing the model to simulate the immediate economic and distributional effects of policies or shocks before economies have fully reallocated resources. Behavioural linkages are quantified through econometrically estimated parameters based on historically observed relationships, underpinned by a theoretical understanding of how economies work, instead of assuming perfect information or rationally optimizing behaviour. (Section 6.3 and the separate Estimation Documentation provide further details).

Allowing for spare capacity through demand-driven setup. While production is subject to endogenous capacity constraints (cf. Section 8.1), the level of production is determined primarily by the level of effective demand. The model is thus suitable for producing short-term impact estimates, in line with the immediate needs of policymakers. Whether supply constraints are reached and to what extent they might be binding in the short term depends on the policy scenario and the sector-country-specific market conditions.[3] See Section 8.1 for a summary and Chapter 4 for all details.

Allowing for involuntary unemployment. If the level of effective demand lies below the level of potential output, some production factors may not be used. This can lead to involuntary unemployment in the labour market, both in the short and long runs, providing relevant information for active labour market and re-skilling interventions and industry-specific or regional government support funds.

Stranded assets. Stranded assets occur if technological advances destroy the economic viability of existing assets. Stranded assets can occur in all sectors and are a natural part of economic progress, as defined in Schumpeter (1934). The pace of change required to meet global greenhouse gas emissions targets means that the potential for stranded assets is substantial (Semieniuk et al., 2022). MINDSET's capital stock is defined by sector, with no movement of fixed capital between sectors (Section 4.1.2.2).

[2] An extensive strand of literature has used input-output methods, e.g., for simulating extended effects of pandemics (Pichler et al., 2022; J. Santos, Meng, and Mozumder, 2024), natural disasters (Hallegatte, 2008; Hallegatte, 2014; Okuyama and J. R. Santos, 2014; MacKenzie, J. R. Santos, and K. Barker, 2012; Rafi et al., 2024) or other exogenous shocks (J. R. Santos and Haimes, 2004). The approach can also support the analysis of inoperability (J. Santos, Meng, and Mozumder, 2024; J. R. Santos and Haimes, 2004) and criticality of production inputs (Pichler et al., 2022)

[3] This setup is, not least, motivated by empirical evidence emerging in lower-income countries: capacity underutilization, or economic slack, can be structural - or long-term; it differs by sector and is more prevalent in lower-income environments; lastly, it can support large production increases while exerting minimal price pressures (Walker et al., 2024; Egger et al., 2022).

Policy / impact modelled	Some relevant features
Climate change related damages	• **Capital immobility** means that assets destroyed in one industry cannot be replaced by assets from another industry • **Dynamic nature with path dependency** means that damages in one year may have persistent effects in the economy in subsequent years as well • **Endogenous productivity** means that, if climate damages destroy assets, this impact can reduce productivity also in subsequent years • **Multi-regional linkages** mean that climate damages do not stop at borders, they channel through value-chain linkages
Climate and development policies	• **Econometric parameters estimated on historical behaviour** mean that reactions to policy changes and other disruptions can be heterogeneous across sectors and countries • **Endogenous money and involuntary employment** means that **if** there are un- or underutilized capacities to implement policies, they can be used • Resilient and sustainable transitions induce changes along domestic and global supply-chains; the **MRIO approach** captures many of these changes
Trade-related policies, e.g., CBAM	• Trade relations are **path dependent and change gradually over time**, which means that trade policies will have an effect over time, but does not change consumption or trade-relations overnight • The **MRIO structure** enables capturing multi-regional spill-over effects to understand competitiveness effects • Specific targeting is possible due to **high granularity**, e.g. targeting products with high carbon content
Other industrial policies, e.g., subsidies or direct investment	• Increased investments can induce productivity increases due to **endogenous technology and productivity mechanics** • **Industry-specific competitiveness effects** can arise from varying price trajectories across countries that interact through the MRIO structure • Sector-level and overall employment reacts to policy support; **involuntary unemployment** means that increasing overall employment through sectoral interventions is possible

Table 1.2: Relevant MINDSET features associated with different policy impacts

Explicitly representing technology. Many development goals depend on technology development. MINDSET includes a focus on innovation by making technology and productivity part of the model's internal dynamics. It links to the Future Technology Transformation (FTT) models to provide a wide range of price-based and regulatory policy options. Real-world features like learning rates, path dependency, and increasing returns to scale are incorporated (Arthur, 1989). See Section 5.2 for the description of how the electricity sector can be included.

Tracing physical energy and emissions linkages. MINDSET maintains consistency between monetary and physical accounts of domestic and global energy flows, by combining energy purchases in monetary terms from economic national accounts with purchases in physical terms from countries' energy balances. This is essential for accurately capturing the feedbacks between macroeconomic activity, energy demand, and environmental outcomes, such as greenhouse gas emission levels. The approach used is similar to that adopted in systems dynamics modelling (see Section 3.2).

Capturing environmental double dividends. If production levels fall below potential, policy interventions can help increase levels of production. It is thus possible to get a double dividend effect, where a policy both simultaneously increases output and employment while reducing environmental pressures.

Beyond these technical characteristics, MINDSET also has some important practical features:

- **Substantial detail in labour market and consumer outcomes**: Distributional effects include disaggregation by occupation, income level, location, and gender, when linked to countries' household survey data (Chapter 7).[4]
- **Providing transparency and flexibility:** By solving iteratively rather than simultaneously, MINDSET preserves the stepwise transmission of shocks through the economy. This structure allows users to *trace results back to their drivers*. Built on *open-source data* which are structured in accessible spreadsheets and implemented in Python, MINDSET promotes *country-specific adjustment of assumptions* and does not rely on proprietary tools.
- **Ensuring broad applicability and cost-effectiveness:** The model's *relatively simple setup* has two practical advantages: first, it is broadly applicable to low-income and data-scarce contexts, *bringing industry-specific and distributional analysis to a wider range of countries*, and, second, it allows for *rapid, low-cost applications*, as country-specific baseline setup and data adjustments are not time consuming[5] and the broad range of supported scenarios as well as user-friendly parameter input sheets facilitate speedy sensitivity analyses.

1.2 How MINDSET 2.0 works

At its core, MINDSET integrates a multi-regional input–output (MRIO) framework with a set of empirically estimated macro-econometric equations that capture impacts on key components, including income, wages, investment, government activity, trade, and energy use. This hybrid structure enables the model to simulate direct, indirect, and induced effects of shocks and policies across sectors, regions and global supply chains. The model ensures macroeconomic consistency through accounting identities that reconcile economic output, income, and expenditure, as well as taxes and external balances. Figure A.2 provides a high-level schematic of its core components.

The model begins with the specification of policy instruments and exogenous drivers, including changes in taxes, government spending, external demand and prices, interest rates, technological trends, and demographic shifts. These inputs shape the formation of final demand, which includes household consumption, government expenditure, investment, and exports. Investment determines capital accumulation and capacity for future production. Imports are determined based on domestic demand and relative prices. Figure 1.2 shows the econometric interactions within years (circles & hexagons). Stocks and lagged variables carry values between years (diamonds & hexagons), creating path dependency.

Final demand then propagates through the MRIO framework, which translates it into sectoral gross output and

[4]The distributional routines in MINDSET are simplified, not intended for long-run distributional and poverty incidence analyses. Instead, the sectoral and labour market detail in MINDSET is intended to facilitate linkages with more sophisticated microsimulation tools for detailed poverty assessments.

[5]See Section 6.2 for a detailed description of how country-specific data are incorporated.

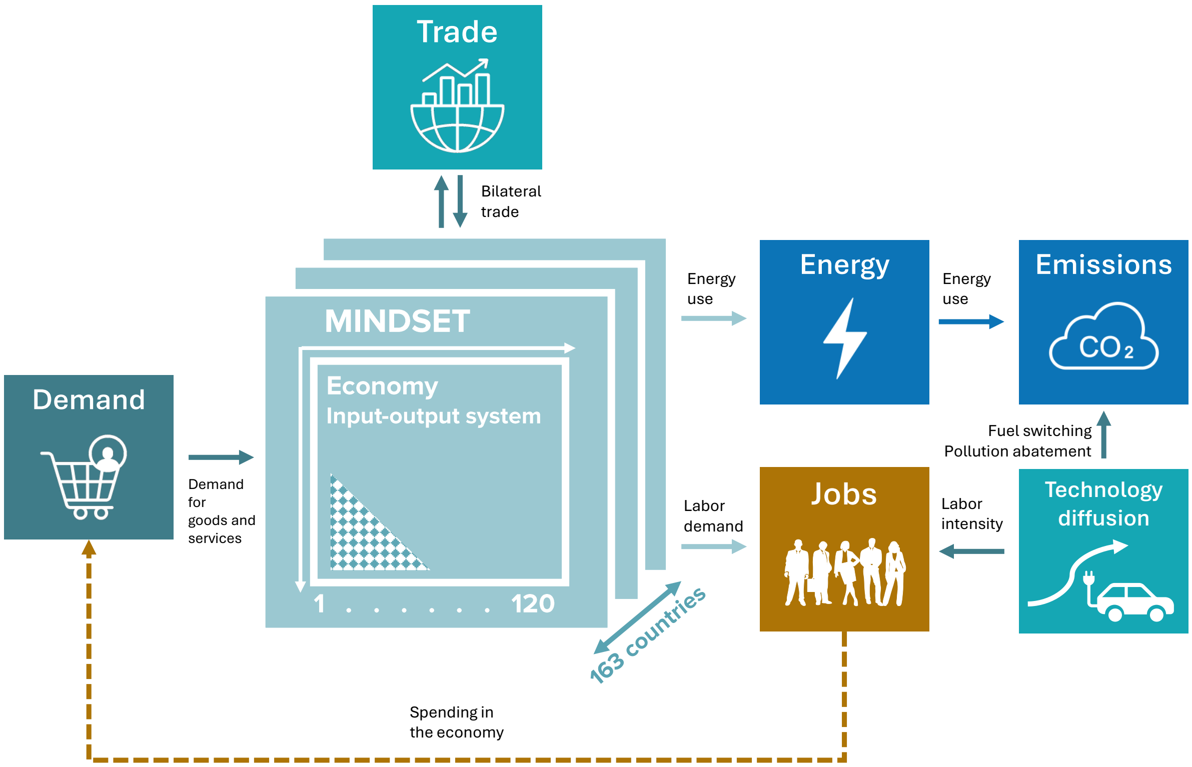

Figure 1.1: High-level overview of linkages in MINDSET

intermediate input requirements. This propagation captures both direct and indirect effects across industries and regions. Given the resulting levels of produced output, the model determines employment, wages, unit labour costs, capacity utilization, and normal (expected) output. Prices are formed based on costs - including wages, intermediate inputs, and taxes - as well as margins and import prices. Profits are calculated as the residual between prices and costs.

Incomes accrue to households, firms, and the government, and the model ensures that all accounts are balanced. Energy demand and emissions are computed based on sectoral output and technology assumptions. At the end of each period, stocks such as capital and labour force are updated, along with lagged variables and policy paths, setting the stage for the next period[6]. Section 6.4 provides a detailed description of the solution cycles in the two-timescale hybrid model.

[6] Figure A.1 in the Appendix, provides a more detailed version of the model flowchart, including more mechanisms and the expected effect signs.

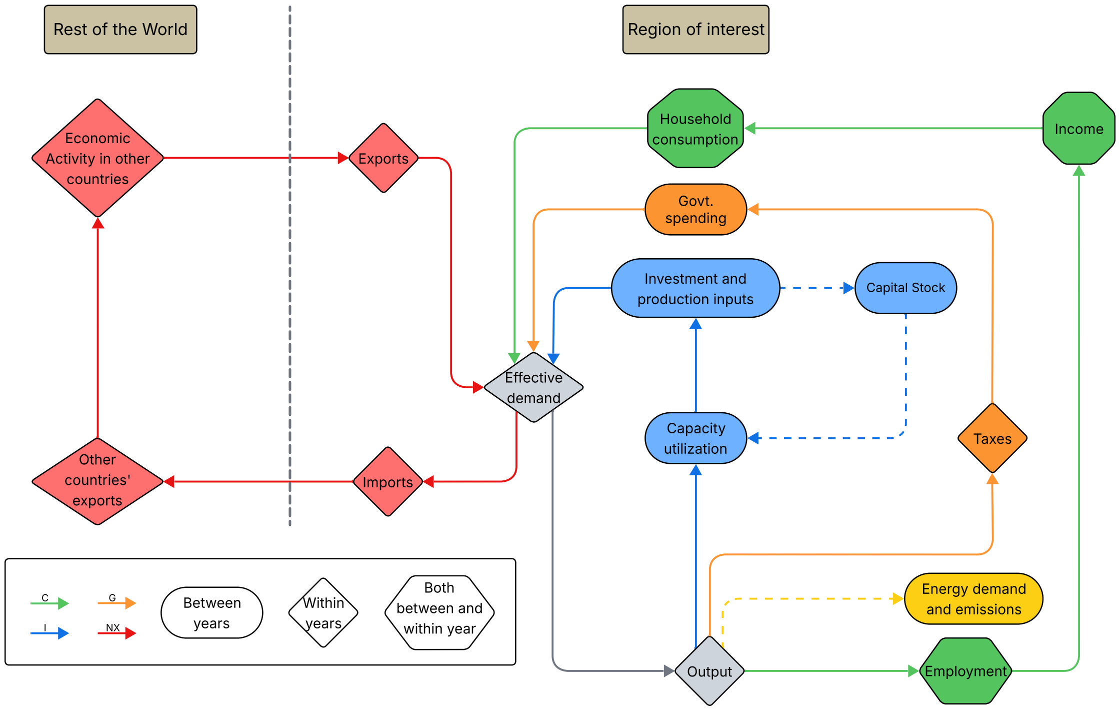

Figure 1.2: Model cycle flowchart

1.3 How MINDSET compares to other macroeconomic models

MINDSET complements other macroeconomic modelling approaches by combining features that are especially relevant for sectoral transformations and for applied, industry-specific or multi-country policy analysis. Three specific characteristics are important to consider when comparing MINDSET to alternative approaches:

- A high level of sectoral disaggregation that can reflect the different ways that energy is used across sectors;
- Global geographical coverage, including both developed and developing countries;
- An empirical approach that defines human behaviour based on the observed reality and allows analysis of short-term impacts and disruptions.

Computable General Equilibrium (CGE) models may include substantive sectoral disaggregation (e.g., GEM-E3 (Capros et al., 2013) or MANAGE (Beyene et al., 2024)) or regional detail (e.g., ENVISAGE (Mensbrugghe, 2024)), thereby meeting the first two characteristics. However, their assumptions about rational, optimizing behaviour, perfect information and market clearing are not empirically backed. CGE models can only inform long-term impacts once an equilibrium has been reached, so the third characteristic is not met. These assumptions also preclude the assessment of many climate policies because demand-driven dynamics and transitional impacts cannot be assessed (Pollitt et al., 2024).

In contrast, Dynamic Stochastic General Equilibrium (DSGE) and other aggregated macro-structural models, such as

MFMOD (Burns et al., 2019), may provide short-term macroeconomic assessments and meet the third characteristic. But their limited sectoral disaggregation precludes detailed analysis of industry-specific climate and energy policies (Varga, Roeger, and 't Veld, 2022) and they rarely cover more than one geographical region.

MINDSET is closest in structure to macro-econometric Economy-Energy-Environment (E3) models such as GINFORS (Lutz, B. Meyer, and Wolter, 2010) and E3ME (Cambridge Econometrics, 2022). These approaches combine an empirical approach with substantial sectoral detail, fully meeting the first and third characteristics. Such an approach has been shown to be essential for acceptance and implementation in Lehr (2024). They are also global in scope, but require extensive data sets, which limits their suitability for countries with limited data. Hybrid sectoral models like IMACLIM (Sassi et al., 2010), which relax some of the CGE-type equilibrium assumptions in the short run, provide similar advantages and disadvantages in the context of the three characteristics above.

By including these three characteristics, the aim of MINDSET is to enable policy analysis that jointly captures short-term sectoral dynamics and the propagation of shocks and policies along global value chains. The model allows policymakers to understand better domestic distributional impacts alongside cross-country spillovers and third-country effects in an interconnected global economy. Coverage of impacts in countries with restricted data is an important advantage.

All models have their respective strengths and weaknesses (some of MINDSET's limitations are described in Section 8.3). The choice of model used should be determined by the policy question at hand (World Bank, 2022b). There is also value in comparing results between models, to sharpen or condition policy advisory and guidance to government counterparts. This practice is illustrated by recent publications that consider comparison of results from primarily demand-led with primarily supply-driven models (Hartvig et al., 2025; European Commission et al., 2022; Dafnomilis et al., 2022). Using different types of models together can provide a more credible range of estimated impacts (Thomassin, 2018).

1.4 Outline of this manual

The remainder of this manual describes the basic structure of MINDSET, its equations, parametrization, and setup. Chapter 2 discusses the underlying economic theory. Chapter 3 provides the background of the core accounting framework and input-output relationships. The various sections of Chapter 4 detail all behavioural equations and underlying parametrization. Chapter 5 describes how physical energy and emission flows are treated in the model. Chapter 6 presents the data and software used, how the model is parametrized and how it is solved. Chapter 7 describes how the model is linked to household surveys to disaggregate employment and income results by gender, occupation, and skill level. Chapter 8 provides a discussion of the modelling assumptions and their limitations, including a summary of supply-side mechanisms.

The underlying economic theory

As with all models, MINDSET represents a simplification of reality. In formulation, MINDSET's priority is to be based on empirics as much as possible. The underlying theory ensures consistency in approach across different parts of the model.

In the tradition of input-output and demand-driven modelling, the starting point is an acknowledgment of fundamental uncertainty (Knight, 1921; Keynes, 1921). This means not only that agents cannot predict future events, but also that there are potential outcomes that they have not considered. The uncertainty arises from the complex interactions between individuals and firms in the economy. Even if economic agents are fully aware of their own choices, they cannot predict the behaviour of other agents and therefore never have full knowledge with which to make decisions (Keynes, 1936; Keynes, 1937; Mercure et al., 2016).

Under these conditions, it is not possible for agents to optimize their decision-making, even if they behave in a rational manner. Regardless of the presence of market frictions, it is therefore not possible to obtain economic equilibrium in the sense that all available factors of production are fully, let alone optimally, utilized and all markets clear. Notably, because of uncertain future outcomes, agents typically maintain a level of liquid resources (i.e., precautionary savings) that can be drawn upon in the event of an unexpected shock (Keynes, 1936; Keynes, 1937; Dimand, 1988).

In the model, the economy is thus formed as a mixture of economic matches and mismatches. The matches are represented by economic transactions that are recorded in the national accounts. The mismatches are a combination of products and factors of production that do not meet other agents' demands, demand deficiencies due to higher propensities to save, or the non-availability of products.

Market frictions may further increase the ratio of mismatches to matches. These frictions are recognized in the model empirically, with the model parameters implicitly extrapolating historical frictions into the projection period.

This perspective makes MINDSET broadly consistent with the post-Keynesian school of economics (Lavoie, 2014; King, 2015). Post-Keynesian economics has been well established since the 1970s (Kregel, 1973), with Eichner and Kregel (1975) focusing on four key components: 1) The dynamic and path-dependent nature of the economy; 2) The importance of distributional, as well as aggregate, outcomes; 3) Money and the financial system; 4) A microeconomic base to complement macro-level outcomes.

Modern post-Keynesian economics has expanded on these roots but they remain important to quantitative modelling. Interest in dynamics and path dependency has grown in response to the development of evolutionary and complexity economics (Nelson and Winter, 1982; Arthur, 1999; Kirman, 2018), but Keynes provides many references to what would now be regarded as complex systems. Post-Keynesian modelling often abstracts from complexity for pragmatic reasons but, as discussed below, important areas of complexity like technology development may be augmented to the basic model structure.

Distributional issues remain important to post-Keynesian analysis. The recognition that wealthier households save a larger share of their income than poorer households is sufficient to make inequality a determinant of macro-level outcomes in a Keynesian system (e.g., Carvalho and Rezai (2016)). Again, post-Keynesian modelling sometimes downplays the issue because of a lack of household disaggregation in national accounts data, meaning that the link is one-way from macro to distributional (Stockhammer, 2015). The current version of MINDSET also adopts this treatment but leaves open the possibility for a two-way interaction in future.

The propensity to save is especially important because of the post-Keynesian treatment of money and the financial system. The size of the money supply is endogenous, with expansion every time a bank extends a loan and contraction every time a debt is repaid. Loans are advanced when entrepreneurs and banks see profitable opportunities, and the volume of investment determines the volume of saving, rather than vice versa. This theory of credit creation has been shown as empirically accurate (Werner, 2014; Werner, 2016) and there is overwhelming anthropological evidence that money results from debt creation (Graeber, 2012). Various central banks have now acknowledged this as the correct representation of money; in inflation-targeting regimes, the central bank determines the policy rate rather than the money supply (e.g., McLeay, Radia, and Thomas (2014)).

Finally, the microeconomic base has lost traction amongst some post-Keynesian economists (e.g., Chick (2016)). Some economists cite the difficulty in reconciling the micro and macro levels in the context of emergent properties. Others refer to the pragmatic reason of working with macro-sectoral level data. MINDSET follows the current approach of working primarily with macro-sectoral data, but leaving open the possibility of incorporating micro-level insights.

These characteristics may be critical to understanding the impacts of climate policies, and can be used to explain the shortcomings of assessments based on neoclassical models (Pollitt et al., 2024). The definition of money, coupled with not assuming markets are at full capacity, can explain the direction of GDP impacts in model results from climate policy (Pollitt and Mercure, 2018). Post-Keynesian models may suggest more positive GDP impacts from climate policies, but uncertainty can also lead to a misallocation of investment that cannot be shifted to other purposes (Cohen and Harcourt, 2003) and leads to *'stranded assets'* (Semieniuk et al., 2022; Chester et al., 2024).

Finally, it is important to note that MINDSET goes beyond current post-Keynesian theory, which offers limited insight on innovation and economic growth (Keen, 2013). MINDSET acknowledges that knowledge is the key driver of productivity growth and a source of increasing returns to scale that follows an evolutionary process (Mercure et al., 2025; Mercure et al., 2016). By linking to the Future Technology Transformation (FTT) model of technology diffusion (Mercure, 2012b), the model can incorporate learning rates of key technologies at global level (Farmer and Lafond, 2016), including spillovers between countries.

Accounting frameworks

The core of MINDSET is a multi-regional input-output table, combined with the accounting identities defined in the System of National Accounts (SNA) framework and augmented with physical flows of energy, or fuels, and greenhouse gas emissions. Global MRIO systems of monetary flows capture all inter-country inter-industry flows of the goods and services used as intermediate inputs to production and for final demand from households, government, and businesses. In other words, MRIOs represent global value chains through these network linkages. Projecting national accounting identities on the input-output system follows the tradition of Richard Stone and models such as E3ME, GINFORS or INFORUM models (D. S. Meade, 2017; Cambridge Econometrics, 2022; Lutz, B. Meyer, and Wolter, 2010). Recently developed MRIO systems[7] are constructed to represent countries' officially published statistics to the maximum extent possible: main economic aggregates, such as GDP, final demand, and exports and imports, are benchmarked to countries' National Accounts, particularly the expenditure, output and income accounts. Sectoral production recipes, output, and bilateral trade by industry are based on countries' input-output tables and trade statistics (Yamano et al., 2023). This section introduces the basic MRIO setup, the relevant accounting identities, and describes the corresponding physical energy flows.

3.1 Multi-regional input-output monetary flows

Originally developed by Wassily Leontief (Miller and Blair, 2009; D. S. Meade, 2017), input-output tables reflect all monetary flows between economic sectors as well as flows towards final demand and trade relations. A simple two-sector economy can be represented in a simplified input-output table:

	Sector 1	Sector 2	Final demand	Exports	Total output
Sector 1	$z_{1,1}$	$z_{1,2}$	c_1	x_1	q_1
Sector 2	$z_{2,1}$	$z_{2,2}$	c_2	x_2	q_2
Value added	v_1	v_2			
Imports	m_1	m_2	m_c		
Total output	q_1	q_2	$C+I+G$	X	

Table 3.1: Simplified IO table of a two-sector economy
(based on Miller and Blair, 2009, p. 14)

Table 3.1 shows how parts of the IO matrix represent a complete set of income and product accounts for an economy (Miller and Blair, 2009). Domestic final demand from household consumption, investment, and government ($C+I+G$) is served by producing sectors 1 and 2 (c_1, c_2) and by imported goods (m_c); total gross output ($\sum x$) is the sum of intermediate demand (z), final demand (c) and exports (x), such as $Q = \sum z + \sum c + \sum x$. Gross domestic product, GDP, can be calculated as the sum of value added (v) components.

International input-output models or multi-regional input-output models extend this concept to achieve a representation of relationships between economic sectors in several countries at the same time by including import and export flows between countries and sectors (Miller and Blair, 2009; MacKenzie, J. R. Santos, and K. Barker, 2012). Trade

[7] Such as EORA (Lenzen et al., 2013; Lenzen et al., 2012), EXIOBASE (Stadler et al., 2018), GTAP (Aguiar et al., 2022), WIOD (Timmer et al., 2015) or MARIO (Guilhoto et al., 2023).

flows are, therefore, broken down by country and industry of origin. This provides a powerful tool for the analysis of global production networks or global shock propagation, such as from climate damages (Timmer et al., 2015; Wenz et al., 2014). The structure of the MRIO table, contrasting the national IO table shown in Table 3.1 is shown in Table 3.2.

		Country A		Country B			
		Sector 1	Sector 2	Sector 1	Sector 2	Final demand	Total output
Country A	Sector 1	$z_{1,1}^{A,A}$	$z_{1,2}^{A,A}$	$z_{1,1}^{A,B}$	$z_{1,2}^{A,B}$	c_1^A	q_1^A
	Sector 2	$z_{2,1}^{A,A}$	$z_{2,2}^{A,A}$	$z_{2,1}^{A,B}$	$z_{2,2}^{A,B}$	c_2^A	q_2^A
Country B	Sector 1	$z_{1,1}^{B,A}$	$z_{1,2}^{B,A}$	$z_{1,1}^{B,B}$	$z_{1,2}^{B,B}$	c_3^B	q_3^B
	Sector 2	$z_{2,1}^{B,A}$	$z_{2,2}^{B,A}$	$z_{2,1}^{A,A}$	$z_{2,2}^{B,B}$	c_4^B	q_4^B
	Value added	v_1	v_2	v_3	v_4		
	Total output	q_1^A	q_2^A	q_3^B	q_4^B		

Table 3.2: Simplified MRIO table of two economies both with two-sectors

The main elements of MRIO tables comprise a matrix of domestic and cross-border sector-to-sector monetary flows of direct intermediate inputs ($\mathbf{Z}$), a vector capturing final demand ($\mathbf{y}$), a vector of total output by country and industry ($\mathbf{q}$), and a vector representing value added generated in production by country and industry ($\mathbf{VA}$). Final demand is differentiated between households (C), government (G), and fixed capital formation (FCF), or 'investment', but for the purposes of the MRIO calculations they are summed together. Value-added components are typically compensation of workers (wages), taxes less subsidies on production, and gross operating surplus (profits). Taxes minus subsidies on both intermediate and final products are reported separately to ensure that intermediate and final demand are recorded at basic prices. The row and column sums each represent total sectoral output (Miller and Blair, 2009; Yamano et al., 2023).

The matrix of direct intermediate inputs ($\mathbf{Z} = [z_{m,n}^{i,j}] \in \mathbb{R}^{m \times n \times i \times j}$) specifies country-industry-country-industry flows which thus incorporate imports and exports in production. The final demand vectors are three-dimensional, differentiating the country of origin and of consumption for each sector (or product). Given that imports ($\mathbf{m}$) and exports ($\mathbf{e}$) are broken down into trade flows, these disappear from the table. Note too that the distinction between domestic and foreign flows does not exist in the MRIO table.

A set of identities and transformations defines the MRIO model structure, as discussed in more detail in Miller and Blair (2009). In a single country setting, the total output vector $\mathbf{q}$ of industries m, the row-sums of the intermediate inputs matrix ($\mathbf{1}_m^\intercal \mathbf{Z}$), and the aggregate final demand vector $\mathbf{y}$ make up the identity Equation 3.1. The technical, or direct-input, coefficient matrix ($\mathbf{A} = [a_{m,n}^{i,j}] \in \mathbb{R}^{m \times n \times i \times j}$) depicts the so-called *"production recipes"*, see Equation 3.3, i.e., how much of a sector's output is used *directly* per unit of output of another sector.

Finally, the Leontief matrix, or **Leontief inverse** $(\mathbb{I} - \mathbf{A})^{-1}$, describes the *total input requirements* throughout the whole upstream value chain, to produce one unit of final demand (Leontief, 1986; Miller and Blair, 2009). The Leontief quantity model links final demand (Y) to output by industry via the identity matrix and the input coefficient matrix, calculated by subtracting the $\mathbf{A}$ matrix from the $\mathbb{I}$ identity matrix and then calculating the inverse of the resulting matrix (see Equation 3.4). The Leontief quantity model delivers a *demand-pull effect*, or the output (or demand) multiplier, i.e. the production required from each sector and country to satisfy a given final demand:

$$\mathbf{q} = \sum_{n,i,j} z_{m,n,i,j} + \mathbf{y} \tag{3.1}$$

$$q_{m,i} = \sum_{n,j} q_{m,n,i,j} \tag{3.2}$$

$$a_{m,n,i,j} = \frac{z_{m,n,i,j}}{q_{m,i}} \tag{3.3}$$

$$\mathbf{q} = (\mathbb{I} - \mathbf{A})^{-1}\mathbf{y} \tag{3.4}$$

MINDSET operates with such an MRIO table as a starting point, including the structural identities of the input-output table. Consider the national accounting framework of the IO model, projected onto the MRIO table (Lequiller and Blades, 2014; Lewney, Pollitt, and Mercure, 2019):

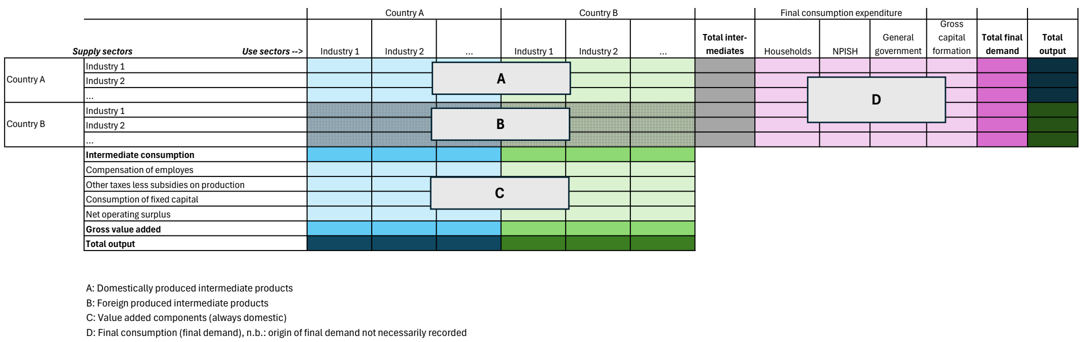

Figure 3.1: Elements of the national accounts system projected onto the MRIO structure

Given this structure, total output is calculated from final demand. This assumes that the accounting identities hold and that intermediate coefficients (technical coefficients calculated for sub-matrices A and B in Figure 3.1) are constant.[8] Thus, assuming that the GVA ratios are constant[9], gross value added (GVA) and its components are also directly computed from this setup. See more on the calculation of value added in Section 4.7.

3.2 Physical energy and GHG emission flows

Similar to the matrices of monetary transactions, MINDSET tracks physical flows of energy and energy-borne greenhouse gas (GHG) emissions for analysing the economic impacts of emissions-related legislation. For example, when simulating a domestic upstream energy price change relative to energy carriers' greenhouse gas (CO_2eq) content, such as through a carbon tax, the aggregate revenues collected in the model need to correspond to the aggregate GHG emissions released in domestic production. In the context of economic modelling, this means that the price change applied to the monetary (or economic) flows of energy must be such that the product of the price change and the monetary flow will equal the product of the policy-implicated carbon price level (USD/CO_2eq), the volume of

[8] Although, as discussed below, this assumption is relaxed in the model.

[9] Which again is an assumption that is relaxed in the model, as described in Chapter 4.

energy flow (toe) and its emissions factor (CO_2eq/toe). If this is not the case, in this example, the carbon price that is (implicitly) modelled does not equal the carbon price of the policy scenario (cf. Hubacek et al. (2017) on this issue).

Hence, to map physical to monetary flows in the input-output system, a corresponding matrix of physical energy and energy-borne emissions flows is constructed, similar to Guevara and Domingos (2017) and connected to the monetary accounting system of MINDSET. The matrix is based on energy balances statistics published by the International Energy Agency (IEA), United Nations Department of Economic and Social Affairs (UNDESA), and Emissions Database for Global Atmospheric Research (EDGAR) data (Crippa et al., 2023; UNDESA, 2025; IEA, 2024), as well as fuel-, sector- and country-specific emission factors from the Greenhouse Gas and Air Pollution Interactions and Synergies (GAINS) model (IIASA, 2020). The process of compiling these matrices, as well as the data sources are described in detail as part of the Github repository[10] and in the Appendix Constructing physical energy and GHG emission flows (Section A.4).

[10] https://github.com/cpmodel/energy-IEA; access upon request.

Economic behavioural equations

As a demand-led model, MINDSET estimates final demand components based on an endogenous structure. Labour demand leads to income generation, which, alongside price changes, drives final consumption of households. Capacity utilization, together with the cost of credit, drives fixed capital formation (investment demand). Meanwhile, production factors change: profit and wage shares in value-added components vary, and the assumption of fixed input-output coefficients is relaxed, allowing for substitution across types of energy sources and across trade partners. These behaviours are driven by econometrically estimated equations and parameters, described in the following sections, whereas the notation used in the document is described in Table 4.1.

Notation	Description	Notation	Description
i	Purchasing region	GDP	Gross domestic product (economy level)
j	Producing region	GVA	Gross value added (sectoral)
m	Purchasing sector	ι	Interest rate
n	Producing sector	u	Capacity utilization rate
t	Period	z	MRIO flow
Δ	Absolute change	N	Employment
δ	Relative change	U	Rate of unemployment
C	Household consumption / final demand	LF	Labour force
G	Government consumption / final demand	w	Annual average wage
g	Component of government consumption	K	Stock of fixed assets / capital
I	Investment / fixed capital formation	E	Energy use
FCF	Fixed capital formation (final demand approach)	e	Fuel use of specific fuel
D	Final demand	P	Economy-wide price
Q	Output	p	Producer price
β	Estimated behavioural parameter	γ	Depreciation rate

Table 4.1: Commonly used notation in this document

4.1 Final demand

Final demand, as captured by MINDSET, consists of three main components, in line with the main SNA components (Lequiller and Blades, 2014): (1) household final demand, (2) fixed capital formation, and (3) general government expenditures[11]. All final demand vectors are three-dimensional, differentiating the country of origin and of consumption for each sector. This section introduces the calculation of the final demand components and how they are affected by various impact channels in the model.

[11]Note that Non-Profit Institutions Serving Households (NPISH) is missing from the final demand components, given the small share of NPISH demand compared to the other components and the non-market drivers inducing NPISH consumption. NPISH consumption, however, is captured in the consumption residuals discussed in Section 4.1.

Final demand for producing sector n in producing country j is the sum of household consumption, fixed capital formation (investment), and government spending, all in real values (^{R})[12].

$$D_{j,n,t} = \sum_i C^R_{i,j,n,t} + \sum_i FCF^R_{i,j,n,t} + \sum_i G^R_{i,j,n,t} \tag{4.1}$$

4.1.1 Household demand

Household consumption (C) reacts primarily to changes in income and prices, reflecting observed household behaviours such as consumption inertia, habits, and limited substitutability between goods, as well as market imperfections. The level and composition of household final demand, or consumption, is driven by two main channels: (1) the household income effect, and (2) the household price effect. Beyond the general demand reaction to price changes, trade and energy substitution are addressed separately (see sections on trade substitution and fuel substitution).

Household consumption income effect

First, the *household income effect* is given by:

$$\Delta C^R_{i,m,t} = \delta\Upsilon^R_{i,t} \times \beta^1_{I,m} \times C^R_{i,m,t-1} \tag{4.2}$$

$$\Delta C^{(inc)R}_{i,j,m,t} = \frac{C^R_{i,j,m,t}}{\sum_j C^R_{i,j,m,t}} \times \Delta C^R_{i,m,t} \tag{4.3}$$

where $\Delta C^{(inc)R}_{i,m,t}$ denotes the real household consumption change for product m in country i induced by income changes, $\delta\Upsilon^R_{i,t}$ is the real income change in country i, $\beta^1_{i,m}$ is the income elasticity of demand for product m in the region country i is part of ($i \in I$), and $C^R_{i,m,t-1}$ is the previous year's real consumption of product m in country i.

Household real (disposable) income, $\Upsilon_{i,t}$, is defined as the sum of labour income from wages, $W_{i,t}$, and social transfers given by the policy scenarios, deflated by the consumer price index, $P^C_{i,t} = \sum_m \left(p^C_{i,m,t} \times \frac{C^R_{i,j,m,t}}{\sum_{j,m} C^R_{i,j,m,t}} \right)$. Currently, the model does not capture changes to other sources of income, such as remittances or capital income. This means that consumption change (through the income channel) is primarily driven by changes in wage income, with the implicit assumption that other income sources are constant.

Equation 4.2 and 4.3 capture household demand reactions by product to changes in household income.[13] The reaction is captured on the level of the consumed product m, assuming that the trade structure of final demand for a given product does not change (Equation 4.3). This means that, while the overall consumption of a product m changes as a response to income changes, the country (incl. domestic) from which this product is procured (*trade structure*) remains unchanged *in this step*. [14]

[12] One may notice that, unlike single-country input–output tables and national accounts, final demand in MINDSET does not include explicit import (M) or export (X) terms. This is because MINDSET relies on a multi-regional input–output (MRIO) framework, where trade flows are represented directly as interregional transactions between countries' final and intermediate demands, rather than as aggregated import or export components.

[13] Consumption parameters are estimated at the level of the Classification of Individual Consumption According to Purpose (COICOP). See estimation documentation for more details.

[14] Note that in the MRIO framework, the three-dimensional final demand structure *includes* domestically produced goods alongside imported goods.

Household consumption price effect

Second, the *household price effect* is given by:

$$\delta p^C_{i,m,t} = \sum_j \frac{C^R_{i,j,m,t}}{\sum_j C^R_{i,j,m,t}} \times \delta p^P_{j,m,t} \tag{4.4}$$

$$\Delta C^{(price)R}_{i,m,t} = \delta p^C_{i,m,t} \times \beta^2_{I,n} \times C^R_{i,m,t-1} \tag{4.5}$$

where $\delta p^C_{i,m,t}$ denotes the relative change of the consumer price of product m in consuming country i. $C^R_{i,j,m,t}$ is country i's real household consumption of product m produced in country j, while $\delta p^P_{j,m,t}$ is the percentage change in producer prices of product m in country j. $\beta^2_{I,m}$ is the product- and region-specific price elasticity of demand with regard to the price change in consumer prices of product m in country i, in region I.

Note that Equation 4.4 calculates a weighted average price change of producer prices for product m, given the current *trade structure*. Thus, as above, the existing *trade structure* in final demand in this step is maintained (see Equation 4.3). Importantly, due to methodological and data constraints, cross-price elasticities of demand are not explicitly included. Hence, structural changes in consumption baskets arise from product-specific income- and own-price elasticities, but price increases/decreases of one product will not directly cause the consumption of another product to change.

The consumption parameters are structurally unconstrained in the short as well as the long run. This has two implications regarding the consumption baskets as well as savings rate. First, while consumption baskets evolve path-dependently from the initial consumption structure, they lack any minimum subsistence share or adding-up aggregation condition, meaning that the model does not assume household budget exhaustion. Second, this structure implies that the savings rate is endogenous, resulting from household income and consumption changes over time.

These modelling choices for household consumption provide a reduced-form, but behaviourally plausible structure for analysing the distributional and sectoral effects of fiscal policy or climate-related shocks. Consumption change is specified as a function of changes in current income and prices - omitting any explicit wealth-holding or borrowing channel. Household balance-sheet data are often scarce or unreliable in lower-income-country settings. This approach does not assume forward-looking behaviour or intertemporal utility maximization; instead, it reflects bounded rationality, inefficient markets, and liquidity constraints (Deaton, 1992; Abhijit V Banerjee and Duflo, 2007; Mercure et al., 2016). Estimated responses often imply less-than-proportional adjustments in consumption relative to income shocks, consistent with partial consumption smoothing (Carroll, 1997). Similarly, the estimated responses to price changes effectively capture consumption inertia and limited substitutability among goods.[15]

4.1.2 Fixed capital formation (investment)

The change in real investment, $\Delta I^R_{i,m,t}$, by sector m in country i is calculated as:

$$\Delta I^R_{i,m,t} = I^R_{i,m,t-1} \times (\beta^1_{I,m} \times \delta \mathsf{GDP}^R_{i,t} + \beta^2_{I,m} \times \delta u_{i,m,t} + \beta^3_{I,m} \times \Delta \iota^R_{i,t}) \tag{4.6}$$

[15] Note that parameters are estimated by bounded regressions, described in the Estimation Documentation.

where $I^{R}_{i,m,t-1}$ is the sector's real investment in the previous period, $\delta\text{GDP}^{R}_{i,t}$ is the real year-on-year GDP change in country i, $\delta u_{i,m,t}$ is the relative change in the capacity utilization rate, which is calculated by dividing observed output with normal output (see here for how normal output is calculated), and $\Delta\iota^{R,t}_{i}$ is the percentage point change in the real lending interest rate in country i (the calculation of real lending interest rate is described below).

In MINDSET, investments in sector m are converted into final demand for capital goods produced by sector n (and m) following the investment converter (explained below, in this section):

$$\Delta\text{FCF}^{R}_{i,j,n,t} = \Delta I^{R}_{i,m,t} \times \alpha^{INV}_{m,n} \times \frac{\text{FCF}^{R}_{i,j,n,t}}{\sum_{n}\text{FCF}^{R}_{i,j,n,t}} \tag{4.7}$$

where $\Delta\text{FCF}^{R}_{i,j,n,t}$ is the absolute change in the final demand for capital goods (and services) n in country i that originate from country j, $\Delta I^{R}_{i,m,t}$ is real investment by sector m in country i, multiplied by the relevant coefficient from the investment converter $\alpha^{INV}_{n,m}$ and the proportion of the overall demand that is produced in country j. As is the case with the other final demand components, investment final demand is also subject to trade substitution.

Lastly, its important to notice how investment is considered in the national accounts, in input-output systems and in MINDSET. In the national accounts, investment (I) corresponds to the acquisition of assets used in the production process, including machinery, buildings, software, and the accumulation of inventories. When inventories are excluded, the measure reduces to gross fixed capital formation (GFCF) (Lequiller and Blades, 2014; United Nations et al., 2009).

Within the structure of input-output tables, GFCF is represented from the perspective of the producer of capital goods rather than the investor. Specifically, it appears as a final demand column, with the entry in each row indicating the value of capital goods produced by a given sector. The IO framework, therefore, does not provide information on the sector undertaking the investment but only on the origin of the capital goods (e.g., Figure 3.1).

In contrast, MINDSET conceptualizes investment from the standpoint of the investing sector. To operationalize this, the model employs an investment converter that allocates sectoral investment expenditures to the capital-goods-producing sectors. For each investing industry, the converter defines the fixed proportion of capital goods sourced from any one sector, including itself.[16] While the investment converter remains constant over time by default, the model permits exogenous modifications to reflect structural transformations in investment patterns - such as a transition from coal-based to renewable energy technologies in the electricity sector. Integration of sectoral models, such as the integration of FTT:Power (Mercure, 2012b) enables a more granular and dynamic treatment of investments for transition-relevant sectors. It also needs to be noted that investment increases the investing sector's capital stock (discounted with depreciation rates over time, see Section 4.1.2.2), which has consequences for multiple model parts, including a direct effect on employment and effects on capacity utilization (see Section 4.4).

MINDSET's specification incorporates three primary drivers of investment, past economic growth, capacity utilization rates, as well as the real lending interest rate: First, it includes an accelerator mechanism, wherein past economic growth fosters optimistic expectations and incentivises investment. This approach is rooted in the work of Kaldor

[16] The investment converter determines which sectors produce investment goods for other sectors based on Cambridge Econometrics, 2022. For example, the construction sector produces a large share of investment goods for the public sector, but the motor vehicles sector produces a large share of investment goods for transport services. The original data source is a set of bridging matrices published in the UK's national accounts. These matrices have been simplified to match the 120 sectors in MINDSET. In the conversion process, directly consumed capital goods remained unchanged, including machinery and construction-related sectors. Investment to demand conversion for the remaining shares is directed towards the sector investing, assuming that much of the remaining capital is produced within the sector (e.g., livestock, intangible capital, management of capital products).

(1957), T. Barker et al. (1980), T. Barker and Peterson (1987), and Akerlof and Shiller (2009), and is commonly employed in similar macro-econometric models such as E3ME (Cambridge Econometrics, 2022) and MFMod (Burns et al., 2019). In MINDSET, real GDP growth is used as a proxy for economic sentiment.

Second, the degree of capacity utilization plays a critical role. MINDSET allows sectors to operate above or below their normal output levels (Pollitt et al., 2017; Keen, 2021; Pollitt and Mercure, 2018). When actual output exceeds normal capacity, firms are incentivized to expand their productive capacity through investment; conversely, underutilization may lead to investment retrenchment (Scasny et al., 2009; Coad et al., 2022; Raza et al., 2023). Capacity utilization is further recognized in applied macroeconomic analysis as a key determinant of investment timing and intensity (S&P Global and Dennison, 2025).

Third, investments are influenced by the real lending interest rate, i.e., the cost of capital. In MINDSET, interest rates influence investment decisions through their effect on borrowing costs, but do not impose ex ante financial constraints. The model does not assume a priori savings–investment equality; instead, the savings–investment identity is ensured ex post. The implication of this approach is that there is no direct crowding out in the capital markets, although indirect crowding out (in capital and labour markets as well as in the goods and services markets) may occur through price effects triggered by rising investments. What distinguishes alternative theoretical frameworks is the specification of the closure mechanism, namely, the causal process by which savings and investment are reconciled (Barbosa-Filho, 2001; Barbosa-Filho, 2004).

In the loanable-funds framework, investment is constrained by the supply of savings, with the interest rate (or the rental rate of capital) equilibrating saving and investment. By contrast, in the post-Keynesian tradition, investment is autonomous: firms' investment decisions generate income, which in turn determines the level of saving required to satisfy the SNA accounting identity. The setup in MINDSET can be interpreted as consistent with either framework, depending on the specification of the financial sector.

The financial system in MINDSET accommodates investment demand through credit creation (Werner, 2016; Hein, 2007; Godley and Lavoie, 2007). Hence, interest rates do not automatically equilibrate saving and investment. Nevertheless, interest rates remain economically relevant because they affect borrowing costs. Sustained increases in investment can narrow the output gap, prompting monetary tightening and moderating investment growth (Hein and Schoder, 2011; Lavoie, 1995; Guiso et al., 2002). The inclusion of the interest rate as a determinant of investment behaviour can be consistent with both frameworks.

At the same time, MINDSET incorporates a backward-looking interest rate rule and a wedge between borrowing and lending rates due to capital market imperfections. This permits a loanable-funds interpretation: savings do not directly constrain investment because sluggish rate adjustments prevent full and immediate equilibration (Stiglitz and Weiss, 1981). Although theoretically distinct, this yields results empirically similar to the post-Keynesian closure: investment drives growth without an ex ante saving constraint.

While MINDSET does not impose ex ante financial constraints—thus ruling out direct financial crowding out in capital markets—it does allow for real crowding out effects to emerge endogenously through resource constraints and price adjustments. In this sense, the absence of a financial constraint does not imply that investment is unconstrained in real terms. When investment rises, it generates demand for capital and intermediate goods, as well as for labour, potentially driving up their prices and thereby displacing other forms of production or consumption. Such mechanisms correspond to real-economy or resource-based crowding out, which can manifest through sectoral price pressures or broader inflationary dynamics rather than through rising interest rates (Blecker, 2002; Taylor, 2004).

The reconciliation of savings and investment occurs ex post through income and price adjustments, not through the interest rate mechanism. Nevertheless, the model remains subject to real constraints in production capacity, factor availability, and intersectoral linkages, which can generate indirect crowding out effects via inflationary or relative price channels. This distinction between monetary and real-economy crowding out helps clarify that MINDSET does not deny the possibility of resource-based constraints but rather shifts the focus from financial equilibria to real adjustments within the productive structure.

Accordingly, MINDSET's treatment of money, saving, and investment is compatible with multiple closures. Under a post-Keynesian interpretation, money is endogenously created by banks and interest rates are exogenous policy instruments. Under a neoclassical interpretation, the same mechanisms reflect financial market imperfections that weaken the equilibrating role of interest rates. In both interpretations, the model rules out automatic direct crowding-out effects and emphasizes that economic growth is primarily driven by investment and effective demand, rather than being constrained by prior saving[17].

4.1.2.1 Interest rate

MINDSET incorporates two distinct real interest rates: the real lending rate, which influences investment decisions (as lower rates reduce the cost of capital and stimulate investment, as shown in Equation 4.6), and the real policy rate, which serves as a benchmark for the expected return on capital (Pasinetti, 1979). As discussed in Section 4.3, when long-term sectoral profitability falls below the real policy rate, firms may reallocate resources from productive investment to financial markets. While this mechanism represents a simplified investment hurdle, it serves to prevent sustained sectoral profit rates from falling persistently below returns implied by the policy rate.

The nominal policy interest rate follows a backward-looking Taylor-type rule with inertia (Orphanides, 2001; Orphanides, 2010). It adjusts gradually in response to deviations of lagged inflation (CPI) from an exogenous inflation target, as well as to the current output gap. The specific rule-based formulation used in MINDSET avoids assumptions of forward-looking behaviour or rational expectations, instead reflecting a monetary policy framework grounded in observable macroeconomic conditions. Central banks are thus modelled as reacting to realized inflation rather than to anticipated future developments.

The nominal lending rate is derived by applying a country-specific historical average spread (η_i) to the policy rate, capturing persistent differences in credit conditions across countries.

$$\iota^*_{i,t} = 0.5 \times (\delta P^C_{i,t-1} - \pi^*_{i,t}) + 0.5 \times \tilde{Y}_{i,t} \tag{4.8}$$

$$\iota^{*R}_{i,t} = i^*_{i,t} - \delta p^{GDP}_{i,t-1} \tag{4.9}$$

$$\iota^{R}_{i,t} = \iota^{*R}_{i,t} * (1 + \eta_i) \tag{4.10}$$

where $\iota^*_{i,t}$ is the nominal policy interest rate in country i period t, $P^C_{i,t-1}$ is the GDP deflator in the previous period, $\pi^*_{i,t}$ is the inflation target, $\tilde{Y}_{i,t}$ is the output gap, $i_{i,t}$ is the nominal lending interest rate, and η_i in the country specific historical average spread.

The inflation target is an exogenous parameter. In economies with formally adopted inflation targeting regimes, this officially announced target is used. For countries lacking an explicit target, the inflation anchor is calibrated as the higher one of either the average annual inflation rate over the preceding five years or a minimum threshold of 1%, consistent with empirical practices in monetary policy rule calibration for lower-income economies (Roger,

[17]While the model accepts that in credit creation can be a source of domestic investment, the same does not hold for imported capital goods as the model does not include an explicit exchange rate mechanism.

2010; Hammond, 2012). To prevent unreasonably high inflation expectations from anchoring long-run interest rate dynamics, any inflation target exceeding 3% is assumed to converge gradually to a long-term ceiling of 3%, aligning with international best practices (Bernanke and Mishkin, 1997).

To ensure consistency between simulated and historical macroeconomic conditions at the onset of the simulation, MINDSET applies an adjustment in the initial year. Specifically, the policy interest rate is corrected using a calibration factor equal to the difference between the endogenously determined and historically observed nominal interest rate. While this adjustment affects the use of the interest rate as a benchmark for profit expectations - particularly relevant in capital allocation decisions between financial and productive assets - it does not influence investment dynamics directly, as investment behaviour depends on changes in the interest rate rather than its absolute level (Hein and Schoder, 2011; Lavoie, 1995).

MINDSET defines the output gap as the deviation between normal output - determined endogenously in the model based on sectoral capacity assumptions (see Section 4.6) - and actual observed output at the national level. This approach follows the structuralist and post-Keynesian modelling traditions, which emphasize capacity utilization and demand-led growth over supply-side potential (Godley and Lavoie, 2007; Lavoie, 2014).

4.1.2.2 Capital stock

Capital stock is defined as the value of non-financial assets cumulated over time (Lequiller and Blades, 2014). Fixed capital assets are sector-specific, i.e., deployed capital is not mobile and cannot be transferred from one sector to another (Cohen and Harcourt, 2003).

Capital stock in the dynamic cycle is defined as:

$$K_{i,m,t} = K_{i,m,t-1} \times (1 - \gamma_{i,m}) + I_{i,m,t} \tag{4.11}$$

where K is the sectoral capital stock, γ is the country and sector-specific depreciation rate, and I is the sectoral investment. The initial value of sectoral investment is based on the above calculations, maintaining an initial rate of $0 < \frac{I}{K} < 0.05$. Note that the rate of $\frac{I}{K}$ changes endogenously following the initial annual cycle.

The initial capital stock at the country-sector level is calculated using data from GLORIA (Lenzen et al., 2017; Lenzen et al., 2021). Gross fixed capital formation (GFCF) over time is allocated to sectors based on historical output-to-capital ratios, derived from detailed EU KLEMS data (Bontadini et al., 2023). Country- and sector-specific depreciation rates are also computed from GLORIA data, based on average consumption of fixed capital[18]. These are used to construct capital stock volumes through the perpetual inventory method (Lequiller and Blades, 2014), which serves as MINDSET's initial capital stock estimate (see A.6 in the Appendix for a detailed description).

MINDSET models capital as a sector-specific, accumulated stock that evolves over time through net investment, with no mobility across sectors. This means that fixed assets are not frictionlessly re-deployable, but instead remain embedded in sector-specific production technologies. This modelling approach departs from the instantaneous capital reallocation assumption in response to marginal product differentials, and instead aligns with empirical evidence on sunk costs, sector-specific learning, and investment rigidities (Hsieh and Klenow, 2009; Guenzel, 2025; Semieniuk et al., 2022; Ramey and M. D. Shapiro, 2001; Kermani and Ma, 2023; Chester et al., 2024).

This sectoral immobility of capital introduces features such as the possibility of bottlenecks, excess capacity, and stranded assets, key phenomena in the context of economic transitions, particularly under climate change or structural

[18] These depreciation rates can be seen as an asset weighted average depreciation rate for the investing country-sector.

shocks (Semieniuk et al., 2022; Chester et al., 2024). The model is able to capture transitional dynamics, such as sector-specific over-accumulation driven by changes in demand or technology. As a result, the overall economy may exhibit low asset utilization rates, even if certain sectors experience high capital utilization and face upward price pressures.

Conversely, the same structure allows for structural unemployment and capital underutilization, particularly in lagging sectors. Finally, the use of observed depreciation rates and historical capital-output ratios to initialize capital stocks reflects the fact that investment decisions are shaped not only by current profitability signals, but also by inherited sectoral structures and technological vintages (Keen, 2021; Penrose, 1995; Guenzel, 2025).

4.1.3 Government expenditures

Unfortunately, the theoretical basis for modelling government spending is not well established (Burns et al., 2019). Therefore, the endogeneity of government spending behaviour in the model is limited, similarly to treatment in other models (Cambridge Econometrics, 2022; Raza et al., 2023). MINDSET, therefore, does not aim to provide a full public finance account, it merely uses adjustments described here to government expenditures in order to preserve *budget neutrality*[19]. Thus, government spending increases and decreases proportionally with government revenues. This is subject to change in subsequent version of the model.

First, an initial composition of government revenues is established. This setup is primarily builds on the IMF GFS database (IMF, 2024), but incorporates various other national sources as well. Government revenues are divided into seven categories: (1) income tax, (2) capital tax, (3) sales tax, (4) social security contributions, (5) international aid, (6) royalties, and (7) residual revenues. For the initial year, the share of these revenues in the total government revenue is calculated. As the model simulates the first year it allocates the amount of total government final demand to 'sources' based on these shares, therefore establishing a one-to-one relationship between captured spending (final demand) and revenue sources. This generates the initial values for $g^{inc}, g^{soc}, \ldots, g^{res}$. From this point on the model equates growth in revenues with growth in spending, assuming *budget neutrality*. Changes in government revenues, therefore, are calculated as:

[19] *Budget neutrality* of policy impacts is often assumed in modelling exercises, especially focusing on e.g., carbon pricing (Kiss-Dobronyi and Fazekas, 2022; S. Lee, Pollitt, and Fujikawa, 2019; Ergon et al., 2025; Dellaccio et al., 2022). Notably, here MINDSET assumes a more *general* budget neutrality, one that assumes that if general government revenues grow so do general government spending.

$$\Delta g_{i,t}^{inc} = g_{i,t-1}^{inc} \times \delta \Upsilon_{i,t}^{N} \tag{4.12}$$

$$\Delta g_{i,t}^{soc} = g_{i,t-1}^{soc} \times \delta \Upsilon_{i,t}^{N} \tag{4.13}$$

$$\Delta g_{i,t}^{sal} = g_{i,t-1}^{sal} \times \delta \sum_{m} C_{i,m,t}^{R} \tag{4.14}$$

$$\Delta g_{i,t}^{cap} = g_{i,t-1}^{cap} \times \delta \sum_{m} \pi_{i,m,t} \tag{4.15}$$

$$\Delta g_{i,t}^{roy} = g_{i,t-1}^{roy} \times \delta \sum_{m} Q_{i,m,t}^{R} \tag{4.16}$$

$$\Delta g_{i,t}^{res} = g_{i,t-1}^{res} \times \delta \sum_{m} \mathsf{GVA}_{i,m,t}^{R} \tag{4.17}$$

$$\Delta g_{i,t} = \Delta g_{i,t}^{inc} + \Delta g_{i,t}^{soc} + \Delta g_{i,t}^{sal} + \Delta g_{i,t}^{cap} + \Delta g_{i,t}^{roy} + \Delta g_{i,t}^{res} \tag{4.18}$$

$$g_{i,t} = g_{i,t-1} + \Delta g_{i,t} \tag{4.19}$$

All left-hand side variables are nominal price absolute change in government revenues, while the right-hand side values are as follows: $\Upsilon_{i,t}^{N}$ is the nominal national income in country i, $\sum_m C_{i,m,t}^{R}$ is national real consumption, $\sum_m \pi_{i,m,t}$ is national aggregated profits, $\sum_m Q_{i,m,t}^{R}$ is national output in royalty-paying (extractive) industries[20] and finally $\sum_m \mathsf{GVA}_{i,m,t}^{R}$ is real GDP. Thus, the relative change of these variables drives government spending proportionally to how much of government revenues are collected from that source (the share between sources can change between years).

Final demand is then calculated for country $j's$ product n from *government spending* in country i as:

$$G_{i,j,n,t}^{R} = \frac{\frac{G_{i,j,n,t}}{\sum_n G_{i,j,n,t}} \times \; g_{i,t}}{p_{j,n,t}^{0}} \tag{4.20}$$

where $G_{i,j,n,t}^{R}$ is final demand from government demand in country i and time period t for goods and services of sector n from country j, $g_{i,t}$ is the total government revenue (that is captured in the model), while $p_{j,n,t}^{0}$ is the producer price index of the good/service. Notice that final demand from government spending is also subject to trade substitution.

4.2 Trade

The multi-regional input-output structure underpins MINDSET's treatment of international trade by capturing both inter-industry and cross-border intermediate and final demand flows. This framework allows for a detailed representation of global supply chains, where goods and services flow through multiple countries before reaching final consumer demand. Trade dynamics are specified as gradual and path-dependent, reflecting empirical evidence on the rigidity

[20] Commodity exporting countries' government revenues are often strongly driven by royalties (IMF, 2020; IMF, 2015), such as from oil extraction or from other natural resources (Gunnlaugsson, Kristofersson, and Agnarsson, 2018), therefore, the model links this part of government budget to commodity outputs.

and persistence of global trade relationships (Feenstra, 2015; Johnson, 2014; Amador and Mauro, 2015). These dynamics emerge from sunk costs, firm-specific capabilities, institutional frictions, transport costs, cultural proximity and geopolitical issues that prevent instantaneous adjustments to shocks or policy changes (Melitz, 2003; Antràs and Gortari, 2020).

Trade flows are shaped by three primary drivers: initial bilateral trade patterns, relative price movements, and country-specific economic growth. These elements jointly determine changes in trade structures over time, influencing consumer prices, household consumption patterns, and production networks. Crucially, given this setup the model ignores initial price differences between the trading partners.

The baseline trade structure is calibrated using data from GLORIA, disaggregated into intermediate and final demand flows. Final goods trade is captured through the matrix $C^R_{i,j,m,t}$, representing the flow of product[21] n from country j to country i at time t, with intra-national trade (i.e., $i = j$) included. Intermediate inputs are modelled using a four-dimensional matrix $\mathbf{Z}_{i,j,n,m,t}$, indicating that product n, exported from country j, is used in the production of product m in country i. Final demand trade flows are thus three-dimensional (i, j, m), while intermediate flows are four-dimensional (i, j, n, m), reflecting the structure of nested global production processes. The evolution of these trade matrices is discussed in subsequent sections.

4.2.1 Trade substitution in final demand

MINDSET allows for trade substitution in final demand, enabling households, investors, and governments to alter the geographical sourcing of the products they consume. This mechanism captures the possibility of substituting between domestic and imported goods, as well as reallocating demand across international trade partners.

$$Y^R_{i,j,n,0} = C^R_{i,j,n,t} + FCF^R_{i,j,n,t} + G^R_{i,j,n,t} \tag{4.21}$$

$$Y^N_{i,j,n,t} = Y^R_{i,j,n,0} \times p^0_{j,n,t} \times \delta p_{j,n,t} \tag{4.22}$$

$$Y^{N*}_{i,j,n,t} = Y^N_{i,j,n,t-1} \times (1 + \beta^1_{i,n} \times \delta p_{j,n,t-1} + \beta^2_{i,n} \times \delta p_{j,n,t-2} + \beta^3_{i,n} \times \delta GDP^R_{i,t-1}) \tag{4.23}$$

Where Y^R is final demand in real values, p^0 is the price-index of the sector (initial year = 1.0), $\delta p_{j,n,t}$ is the producer (j) and product (n) specific price change, while $\delta\text{GDP}^R_{i,t-1}$ is the real GDP change in the importing (i) country in the previous period.

In order to calculate the trade substitution effects, first, final demand components are summed up (Equation 4.21) and converted to nominal price values (Equation 4.22). Then, econometric parameters are applied to nominal values of flows as described in Equation 4.23 (at the dimensions i, j, n, t) in order to calculate the new nominal trade values ($Y^{N*}_{i,j,n,t}$).

To get *constant price* values disaggregated by type of final demand (household consumption, investment, and government spending), the total real value is kept constant. Thus, the new trade nominal values to constant prices are converted (Equation 4.24), and in Equation 4.25 calculate the new trade shares ($\tau_{i,j,n,t}$). The new trade shares are then applied to the original real consumption of each product ($Y^R_{i,n,t-1}$), which yields Equation 4.27.

[21] Here, "product" encompasses both goods and services.

$$Y^{R*}_{i,j,n,t} = \frac{Y^{N*}_{i,j,n,t}}{p^0_{j,n,t} \times (1 + \delta p_{j,n,t})} \tag{4.24}$$

$$\tau_{i,j,n,t} = \frac{Y^{R*}_{i,j,n,t}}{\sum_j Y^R_{i,j,n,t}} \tag{4.25}$$

$$Y^R_{i,n,t-1} = \sum_j Y^R_{i,j,n,t-1} \tag{4.26}$$

$$Y^R_{i,j,n,t} = Y^R_{j,n,t-1} \times \tau_{i,j,n,t} \tag{4.27}$$

$$\Delta Y^R_{i,j,n,t} = Y^R_{i,j,n,t} - Y^R_{i,j,n,t-1} \tag{4.28}$$

Given that for each period t the model calculates final demand changes by i, j, n, then these changes are applied to the final demand components (assuming that the change is proportional). Hence:

$$\Delta\overline{C^R_{i,j,n,t}} = \frac{C^R_{i,j,n,t-1} + \Delta C^R_{i,j,n,t}}{Y^R_{i,j,n,t-1}} \times \Delta Y^R_{i,j,n} \tag{4.29}$$

$$\Delta\overline{G^R_{i,j,n,t}} = \frac{G^R_{i,j,n,t-1} + \Delta G^R_{i,j,n,t}}{Y^R_{i,j,n,t-1}} \times \Delta Y^R_{i,j,n,t} \tag{4.30}$$

$$\Delta\overline{FCF^R_{i,j,n,t}} = \frac{FCF^R_{i,j,n,t-1} + \Delta FCF^R_{i,j,n,t}}{Y^R_{i,j,n,t-1}} \times \Delta Y^R_{i,j,n} \tag{4.31}$$

Only limited trade adjustments across years are allowed in the model: a maximum change of 50% of trade value is allowed if $i \neq j$ and a maximum change of 25%[22] trade value is allowed if $i = j$. One might notice that previously calculated components (e.g., ΔC or ΔG) are part of the right-hand side in these equations, while the after trade calculated changes (e.g., $\Delta\overline{C}$) are part of the left-hand side. This is possible, due to the iterative nature of the model, see Section 6.4 for details.

The Trade module updates trade shares across partners while implicitly accounting for bilateral iceberg costs and other trade resistance factors. Elasticities are specific to each importer–product pair, price changes are specific to each exporter–product pair, and baseline trade levels are defined for each importer–exporter pair, thereby supporting the validity of the Armington assumption (Armington, 1969). Trade substitution in final demand is therefore modelled as responsive to relative changes in prices and economic growth across countries, consistent with empirical evidence on demand elasticity and trade reallocation dynamics (Armington, 1969; James E Anderson and Wincoop, 2003; Fally, 2015). Within this framework, shifts in trade patterns emerge endogenously as agents respond to changing competitiveness and macroeconomic conditions.

4.2.2 Trade of intermediates

Cross-country differences in economic growth and relative price movements also trigger adjustments in the intermediate trade structure in the MRIO framework. While the technology of production, represented by fixed input-output

[22] These threshold can be adjusted to satisfy specific settings.

technical coefficients or "production recipes" (D. S. Meade, 2017), is assumed to remain constant over time (with the important exception of energy efficiency/substitution, discussed in Section 5.1.2), the model allows for substitution between exporters of the same product category.

$$\alpha_{i,j,m,n,t} = \frac{z_{i,j,m,n,t}}{q_{i,m,t}} \tag{4.32}$$

$$\lambda_{i,m,n,t} = \sum_{j} z_{i,j,m,n,t} \tag{4.33}$$

$$z^{N}_{i,j,m,n,t} = z_{i,j,m,n,t} \times p^{0}_{j,n,t} \times \delta p_{j,n,t} \tag{4.34}$$

$$z^{*}_{i,j,m,n,t} = z^{N}_{i,j,m,n,t} \times (1 + \beta^{1}_{i,n} \times \delta p_{j,n,t-1} + \beta^{2}_{i,n} \times \delta p_{j,n,t-2} + \beta^{3}_{i,n} \times \delta GDP^{R}_{i,t-1}) \tag{4.35}$$

$$\theta_{i,j,m,n,t} = \frac{z^{*}_{i,j,m,n,t}}{\sum_{i,m,n,t} z^{*}_{i,j,m,n,t}} \tag{4.36}$$

$$z^{*}_{i,j,m,n,t} = \lambda_{i,m,n,t} \times \theta_{i,j,m,n,t} \tag{4.37}$$

$$\alpha^{*}_{i,j,m,n,t} = \frac{z^{*}_{i,j,m,n,t}}{q_{i,m,t}} \tag{4.38}$$

where at the end of the process, $\alpha^{*}_{i,j,m,n,t}$, is calculated which is the new input-output coefficient considering trade structure changes; $z_{i,j,m,n,t}$ is an intermediate flow from country j and sector n to country i and sector m in period t; $q_{i,m,t}$ is the output of sector m in country i, while $p^{0}_{j,n,t}$ is the producer price index, $\delta p_{j,n,t}$ is the change in producer price, $\delta GDP^{R}_{i,t-1}$ is real GDP growth in the importing country i in the previous period, $\theta_{i,j,m,n,t}$ is the calculated trade share, and $\lambda_{i,m,n,t}$ is the spending on a given intermediate flow (i.e., i, m, n specific).

This substitution mechanism operates analogously to the one used for final demand (see Section on trade substitution in final demand) but is implemented through a four-dimensional matrix, indexed by the importing country, exporting country, product sector, and producing sector, that captures the evolution of supply chain relationships over time. This approach aligns with recent empirical and modelling advances that emphasize the path-dependence and gradual restructuring of global value chains in response to macroeconomic and trade shocks (Borin and Mancini, 2020; Johnson, 2014; Mercure et al., 2016).

This specification reflects the central role of income in determining trade patterns, as established in the canonical gravity model of trade (Isard, 1954; James E. Anderson, 2011). However, it omits the explicit modelling of *effective distance* - a proxy for transportation costs, trade barriers, and other frictions - that typically features in gravity-based approaches (Isard, 1954; Tinbergen, 1962). This abstraction is justified by the model's use of an empirically grounded initial MRIO structure: since the simulation begins with a baseline state of the global economy that includes observed bilateral trade flows, frictions such as geographical distance, institutional compatibility, historical ties, and other cultural or political determinants are implicitly embedded in the baseline trade matrix (De Benedictis and Taglioni, 2011; Head and Mayer, 2014). As a result, deviations from the initial configuration occur gradually, preserving the path-dependent nature of trade relationships within global value chains.

MINDSET adopts the Armington assumption, treating goods classified within the same sector but originating from different countries as imperfect substitutes (Armington, 1969). For example, consider U.S. imports of automobiles from Japan and Germany. In this framework, a car produced in Japan is not regarded as a perfect substitute for a German-produced car. If perfect substitutability was assumed, a price increase for German cars would lead to a complete reallocation of U.S. imports toward Japanese vehicles. Instead, substitution is governed by relative price changes and the initial bilateral import shares[23].

[23] This formulation abstracts from product differentiation by origin in the sourcing of intermediate inputs. As such, country-specific

To further constrain unrealistic structural shifts, the model imposes annual bounds on trade matrix adjustments. These bounds mirror the restrictions placed on trade substitution in final demand and are intended to capture frictions in the reorganization of production networks. Specifically, domestic flows ($i = j$) are allowed to vary by a maximum of ±25% per year, while international flows ($i \neq j$) may change by up to ±50%[24]. These caps limit the speed at which sectors can restructure their input sourcing across countries, acknowledging the empirical stickiness of supply chain reconfiguration and the sunk costs involved in altering trade relationships (Levchenko, 2007; Johnson, 2014).

4.3 Producer prices

Prices play an important role in the overall setup. Producer prices are determined on the supply side, with demand agents primarily acting as price takers, adjusting only their consumption patterns in response to price changes. Consistent with the MRIO framework (Chapter 3), price effects are differentiated into direct effects and indirect effects. Direct price effects arise from cost changes within a given sector, such as due to changes in labour costs, taxation, or profit margins. Indirect price effects capture the propagation of these direct price changes across sectors through the intersectoral linkages embedded in the MRIO structure.

4.3.1 Direct price effects

In each sector n of exporting country j, the change in producer prices, $\delta p^P_{j,n,t}$, combines direct cost pressures and their propagation through supply chains. Direct price changes $v_{j,n}$ are given by:

$$v_{j,n} = \beta^1_{J,n} \times \delta Q_{j,n,t-1} + \beta^2_{J,n} \times \delta u_{j,n,t-1} + \beta^3_{J,n} \times (\pi_{j,n,t-1} - \hat{\pi}_{j,n,t-1}) \tag{4.39}$$

$$v^+_{j,n} = max(v_{j,n}, 0) \tag{4.40}$$

$$v^-_{j,n} = min(v_{j,n}, 0) \tag{4.41}$$

where $\delta Q_{j,n,t-1}$ is the previous period's absolute sectoral output change, $\delta u_{j,n,t-1}$ is the previous period's change in sectoral capacity utilization, π is the sectoral profit, $\hat{\pi}$ is the sectoral long-term profit, $v^+_{j,n}$ and $v^-_{j,n}$ are, respectively, positive and negative direct producer price effects, and where J is the region to which country j belongs to ($j \in J$).

A few additional aspects warrant attention. First, cost pass-through rates are empirically estimated and specified by sector and region, thereby reflecting differences in sectoral market environments and exposure to competition.

Second, the model allows for asymmetric pass-through: the rate at which cost *increases* are transmitted differs from that of cost *reductions*.

Third, the model accounts for deviations from normal output levels, reflecting overutilization or underutilization of firm resources. When actual output exceeds or falls short of expected levels, prices (including wages and interest rates) may rise due to queuing and auction-like effects (Cambridge Econometrics, 2022; Lewney, Pollitt, and Mercure,

quality differences or brand preferences are not explicitly captured.

[24] These threshold can be adjusted to satisfy specific settings.

2019), as well as due to increased pricing power on the supply side (ECB, 2007; Corrado and Mattey, 1997; Pollitt et al., 2017). At the macroeconomic level, empirical studies have identified a relationship between inflation and capacity utilization, whereby inflationary pressures intensify under conditions of excess demand (Corrado and Mattey, 1997).

Fourth, MINDSET incorporates target-return pricing, capturing firms' efforts to achieve a desired rate of return on capital (Lavoie, 2001; F. S. Lee, 1999). Even in sectors dominated by informal production, a minimum expected return is assumed at the industry level. The model employs a conservative benchmark: the long-term profit rate is defined as the average sectoral profit rate over the previous three years, following the definition in A. Wood (1975). In cases where the sectoral profit rate falls below the national interest rate, the interest rate is used as a benchmark for pricing decisions.

Fifth, overall sectoral output growth is considered as a determinant of pricing behaviour. This mechanism captures two effects. On one hand, sustained growth may lead to saturation and increased competition, which, given the assumption of constant cost pass-through rates, induces downward pressure on prices. On the other hand, in a context of excess capacity, rising output allows firms to benefit from scale economies and price below average cost while still achieving positive profits, as argued by Sraffa (Keen, 2021). This dynamic can further suppress price levels.

Behavioural price adjustments are assumed to occur with a time lag. This assumption follows Boggio's approach (Lavoie, 2016b), and is supported by empirical evidence from Blinder (Blinder, 1998), which suggests that firms operate under incomplete information and rely on lagged cost indicators, such as previous-period input prices, for pricing decisions. This is consistent with the broader literature on price stickiness (Blinder, 1998; Cavallo and Rigobon, 2016), which finds that firms tend to avoid frequent price changes, further justifying the use of lagged input prices in the model.

Finally, MINDSET assumes that increases in own-sector labour costs are fully passed through to prices. Since sectoral wages are independently modelled (see the Wages section), full cost pass-through prevents double-counting of wage dynamics. This ensures consistency in how labour cost shocks affect price formation.

Overall, this implementation is closer to a target-return pricing model (Lavoie, 2001), where firms adjust prices to achieve a desired profit rate. Prices are determined through a markup-over-cost mechanism, consistent with a wide range of macroeconomic and industrial organization models. This approach departs from the Walrasian assumption of instantaneous market clearing and instead assumes that firms set prices based on unit costs of production plus a markup, which may vary with demand conditions, distributional pressures, or institutional factors.

4.3.2 Indirect price effects

Indirect price effects refer to the transmission of direct cost changes through production supply chains, as captured by the structure of the MRIO matrix. These effects influence sectoral unit costs based on inter-industry dependencies. The extent to which such cost changes are reflected in a sector's producer prices depends on both its upstream supply-chain linkages and the characteristics of the market environment in which it operates, factors represented in the model by sector- and region-specific cost pass-through rates.

Within the MRIO framework, the propagation of indirect price effects is formally captured by the Leontief inverse matrix ($\mathbf{L}$), which models how shocks in one sector affect downstream costs across the production network (Miller and Blair, 2009; Leontief, 1986). However, MINDSET relaxes the assumption of full cost pass-through: it allows producer sectors to respond asymmetrically to cost increases and cost decreases through empirically estimated, sector-country-specific cost pass-through (CPT) rates. To reflect this, two distinct Leontief matrices are employed, one for rising costs and another for falling costs, leading to differentiated sectoral pass-through parameters for each case.

$$\mathbf{L}^{+} = (\mathbb{I} - \mathbf{A} \times \beta^{+})^{-1} - diag[(\mathbb{I} - \mathbf{A} \times \beta^{+})^{-1}] \tag{4.42}$$
$$\mathbf{L}^{-} = (\mathbb{I} - \mathbf{A} \times \beta^{-})^{-1} - diag[(\mathbb{I} - \mathbf{A} \times \beta^{-})^{-1}] \tag{4.43}$$

where $\mathbf{L}$ is the Leontief matrix, $\mathbb{I}$ is the identity matrix, $\mathbf{A}$ is the technical coefficients matrix, and β^{+} and β^{-} are, respectively, the elasticities, or cost pass-through rates, associated with unit cost increases and decreases[25].

Therefore, the overall price effect, accounting for direct ($v_{j,n}$) and indirect, or secondary, effects, is then defined as, for each i, m pair:

$$p_{j,n} = v_{j,n} + l^{+}_{j,n} \times v^{+}_{j,n} + l^{-}_{j,n} \times v^{-}_{j,n}, \forall(i,m) \tag{4.44}$$

where $l^{+}_{j,n}$ is an element of the $\mathbf{L}^{+}_{\mathbf{i},:,\mathbf{m},:}$ the producer country-sector specific positive Leontief matrix, a partition of $\mathbf{L}^{+}$; the same hold for $l^{-}_{j,n}$ with regards to the partition of the $\mathbf{L}^{-}$.

Each sector's direct producer price is updated through a bottom-up process, guided by empirically estimated partial elasticities that govern its own-price response. The direct price-setting mechanism captures both cost-push and demand-pull factors, and is driven by four main elements: (i) absolute output expansion or contraction; (ii) the capacity utilization gap; (iii) deviations of realized profits from a long-run sectoral benchmark; and (iv) unit labour cost growth.

Rapid output expansion creates demand-pull pressure, as firms experiencing surging orders can increase markups before facing customer substitution. Conversely, shrinking output can intensify competition, placing downward pressure on prices. Similarly, high capacity utilization rates lead to supply bottlenecks that empower firms to raise prices, while underutilized capacity encourages price reductions. Cost pressures arising from wage increases are conditioned by firms' profit shares from the previous period, higher past profitability can absorb cost increases, whereas lower margins may prompt stronger price adjustments.

The cost-plus pricing framework can be traced to Kalecki's theory of income distribution, where markups reflect firms' relative market power and the balance of class forces (Kalecki, 1954; F. S. Lee, 1999). This view has been further developed in post-Keynesian models of price formation and distributional dynamics (Lavoie, 1992; Hein, 2014). At the same time, cost-plus or markup pricing is not unique to the Kaleckian tradition. A variety of macroeconomic models also generate price-setting rules that can be expressed in markup form. For example, sticky-price New Keynesian models derive a Phillips-curve type relation in which firms adjust prices sluggishly due to menu costs, imperfect competition, or nominal rigidities (Rotemberg, 1982; Calvo, 1983; Woodford, 2003). In such frameworks, prices respond to changes in output and demand pressures, but equilibrating forces do not immediately restore full-employment conditions. Similarly, models of efficiency wages and asymmetric information generate cost-plus-type wage–price outcomes, where wages are set above the market-clearing level, feeding into markup pricing rules for firms (C. Shapiro and Stiglitz, 1984; Akerlof and Yellen, 1990).

[25] It is important to note that, even though the elasticities governing direct and indirect price effects are estimated jointly, the modelling approach remains valid. This is because direct and indirect effects are computed sequentially within each simulation year and applied iteratively, preserving the internal consistency of the price adjustment mechanism.

To reflect observed market behaviours, MINDSET explicitly separates positive and negative shocks, allowing for asymmetric pass-through and downward nominal price rigidity. This reduced-form behavioural specification captures key cost pass-through channels and embeds empirically validated rigidities, while remaining transparent, data-driven, and consistent with the input–output propagation mechanism that governs indirect price effects. Asymmetries in price transmission have been widely documented across various contexts, including energy markets (Grubb, 1995; Dargay and Gately, 1995), agriculture (Panagiotou, 2021), and broader consumer goods sectors (Peltzman, 2000). Empirical evidence generally suggests that price adjustments tend to be stronger when input costs rise than when they fall. The proposed model acknowledges that several mechanisms, including technological constraints (Grubb, 1995), market power, adjustment frictions, and menu costs (J. Meyer and Cramon-Taubadel, 2004), can contribute to this asymmetry. Accordingly, the model incorporates asymmetric transmission mechanisms to reflect these observed dynamics.

Importantly, the empirically estimated CPT parameters allow MINDSET to capture industry-specific pricing behaviours, including the possibility of zero pass-through in certain sectors where competition or institutional factors suppress price responsiveness. The CPT parameters reflect sectoral competition conditions that shape firms' pricing responses to input cost changes. This modelling approach is consistent with previous macroeconomic and post-Keynesian frameworks (Cambridge Econometrics, 2022), and follows Kalecki's original logic (Kalecki, 1942; Kriesler, 1988), which permits sector-specific markup behaviour based on differing market environments. This approach is further validated by empirical studies on competition and price formation in online markets (Cavallo and Rigobon, 2016; Cavallo, 2018).

4.4 Employment

Employment, together with the wage rate, i.e., household income from wages, serve as a primary determinant of household consumption. It plays a particularly critical role as a stabilizing mechanism in low-income economies, where labour earnings (formal or informal) constitute the bulk of household income (Abhijit V. Banerjee and Duflo, 2012).

Employment is defined in a straightforward manner. Employment data is primarily drawn from sectoral employment statistics provided by the International Labour Organization (ILO) (ILO, 2025a). Employment in MINDSET, hence, includes both formal and informal workers. However, the formal and informal economies are not explicitly modelled, such as in differing wage-setting behaviour[26]. The model adopts the ILO's standard definition of employment, which is described as follows (ILO, 2025b):

> *[Employment] comprises all persons of working age who, during a specified brief period, such as one week or one day, were in the following categories: a) paid employment (whether at work or having a job but not at work); or b) self-employment (whether at work or with an enterprise but not at work)*

Employment evolves along two dimensions: between years, reflecting structural changes in the economy, and within years, capturing short-term fluctuations in labour demand. To account for these dynamics, the model employs two distinct equations, one for inter-annual (structural) employment changes and another for intra-annual (cyclical or demand-driven) adjustments. Structurally, it responds to average sectoral wages and the accumulated capital stock, while within-year fluctuations are driven by short-term changes in sectoral output.

Between-year, the employment reaction to structural factors is given by:

[26] However, estimated behavioural parameters are context-specific and therefore a changing informality rate is indirectly embedded in the employment and wage responses.

$$\Delta N_{j,n,t} = (\beta^1_{J,n} \times \delta K_{j,n,t} + \beta^2_{J,n} \times \delta w^R_{j,n,t}) \times N_{j,n,t-1} \tag{4.45}$$

where N is the number of workers in each sector, K is the sectoral real capital stock, and w is the sectoral real average annual wage.

Structural drivers of employment, hence, reflect two key economic relationships. First, there is a negative relationship between real wages and employment, in line with both neoclassical and Keynesian economic theory (Geary and Kennan, 1982), as well as empirical evidence (Nickell and Symons, 1990; Neftci and Sargent, 1978). This captures the widely applied - though contested (Hartwig, 2017) - assumption that lower real wages can stimulate employment, while higher real wages may dampen labour demand. Second, sectoral capital stock influences labour demand. Here, the empirical estimation strategy allows for both positive and negative coefficients, acknowledging that the relationship between labour and capital can vary by sector. In some industries, capital and labour may act as complements, whereas in others they may be substitutes (Brynjolfsson, 1993; Brynjolfsson and Hitt, 1996). By estimating these relationships empirically, MINDSET accommodates this heterogeneity across economic activities. Overall, employment is modelled consistently with the approaches used in other macro-models such as E3ME and GINFORS (Cambridge Econometrics, 2022; Lutz, B. Meyer, and Wolter, 2010).

*Within-year*employment reactions to labour demand changes are driven by sectoral output growth:

$$\Delta N_{j,n,t} = (\beta^3_{J,n} \times \delta Q_{j,n,t}) \times N_{j,n,t-1} \tag{4.46}$$

In the short term, changes in employment are driven primarily by variations in sectoral output volumes (Raza et al., 2023). The elasticity between output and employment is empirically estimated and generally found to be less than one, reflecting diminishing returns to labour with increasing output. This assumption is consistent with the logic of Okun's Law (Okun, 1962), which posits that increases in output are associated with reductions in unemployment. This stylized fact has been empirically validated across a wide range of country contexts and time periods (Hashmi et al., 2021; Prachowny, 1993; Dixon, Lim, and Ours, 2017).

Total employment is bounded by the available labour force, defined as the product of the working-age population (individuals aged 15 and older) and the labour force participation rate (LPR). The working-age population is treated as an exogenous variable, based on the median projection from the United Nations (UN, 2024). In the absence of country-specific projections for LPR, the model assumes LPR evolves according to each country's historical trend. For countries where historical LPR growth is negative, the rate is held constant[27].

To reflect a *natural rate* of unemployment, including voluntary unemployment and individuals in labour market transition, the model imposes an upper bound on employment (ξ) at 99% of the labour force[28]. In each simulation cycle, total employment is checked against this constraint. If employment exceeds the limit, i.e., if the value of $N^{over}_{j,t}$ in Equation 4.47 is negative, a downward adjustment is applied to ensure consistency with labour supply constraints.

[27] For countries with declining labour force participation trends, a fixed LPR is assumed to avoid implausible long-term reductions in labour supply.

[28] This threshold can be adjusted to reflect the country-specific natural rate of unemployment.

$$N_{j,t}^{over} = LF_{j,t} \times \xi - \sum_n (N_{j,n,t} + \Delta N_{j,n,t}) \quad (4.47)$$

$$N_{j,n,t}^{adj} = N_{j,t}^{over} \times \frac{\Delta N_{j,n,t}}{\sum_n \Delta N_{j,n,t}} \quad (4.48)$$

$$\text{if } N_{j,n,t}^{over} < 0 \Rightarrow \Delta N_{j,n,t}^{*} = \Delta N_{j,n,t} + N_{j,n,t}^{adj} \quad (4.49)$$

MINDSET does not assume that binding employment constraints restrict output. Instead, when the labour supply limit is reached, the model allows labour productivity to increase endogenously to meet rising demand. In this way, output can continue to grow even in the absence of additional labour input.

It is also worth noting that this mechanism imposes a lower bound on unemployment; that is, the unemployment rate cannot fall below the exogenously defined threshold, effectively preventing negative unemployment. However, under such conditions, labour market tightness increases, leading to a stronger upward pressure on wages. To capture this dynamic, the concept of *shadow unemployment* is introduced. Even when the labour market constraint binds, the model continues to compute a counterfactual unemployment rate - one that would prevail if employment continued to rise with output. This shadow unemployment rate may, in fact, be negative and reflects the latent demand for labour that cannot be satisfied due to labour force limitations. It is this counterfactual indicator, rather than observed unemployment, that feeds into the wage-setting equation, thereby allowing wages to respond appropriately to labour market pressures even when actual employment is capped[29].

4.5 Wages

Sectoral average annual labour income is updated in each period following economy-wide and sector-specific variables in the previous period. This represents the process of annual wage adjustments (Raza et al., 2023).

$$\delta w_{j,n,t} = \beta_{J,n}^{1} \times \delta\theta_{j,n,t-1} + \beta_{J,n}^{2} \times \delta CPI_{j,t-1} + \beta_{J,n}^{3} \times \Delta U_{j,t-1} \quad (4.50)$$

where w is the sectoral average annual wage, θ is the sectoral labour productivity (or the ratio between real output and employment), CPI is the country consumer price index in the previous period, and U is the country unemployment rate in the previous period.

Moreover, wage changes are limited by two constraints: first, wage changes can be limited by setting an exogenous annual change factor λ [30]. Second, sectoral wages have an absolute lower bound, which corresponds either to the mandated minimum wage in each country (as of 2019-2025) or to the sectoral wage levels in the initial year, if the latter is lower than the minimum wage.

Therefore, wages are not given by Walrasian market-clearing, but instead by institutional, political, and macroeconomic forces (captured in the model by the baseline wage levels and the endogenous changes provided by the equation above).

[29] Notice that MINDSET does not allow for endogenous migration and labour force changes are dictated by demographic trends

[30] By default this factor is set to $\pm 25\%$. This relatively high upper-bound threshold is rarely reached in standard scenarios, and can easily be adjusted to reflect country-specific data.

MINDSET treatment of wages builds on the contributions of Kalecki, Kaldor, and the structuralist tradition (Godley and Lavoie, 2007; Taylor, 2004). Kalecki (1954) argued that in imperfectly competitive markets, firms set prices by applying a markup over unit labour costs. This makes wages both a cost driver and a source of effective demand. The markup reflects market power, implying that the wage share is shaped less by marginal productivity and more by institutional configurations and class power relations. This logic underpins the distinction between wage-led and profit-led growth regimes. In wage-led regimes, higher wages raise aggregate demand via increased consumption; in profit-led regimes, wage suppression may boost investment and competitiveness by increasing profits (Bhaduri and Marglin, 1990). Kaldor (1955) further emphasized differential saving propensities between workers and capitalists, noting that redistributing income toward wages typically raises consumption.

Structuralist extensions incorporate wage-setting into broader institutional and sectoral contexts, highlighting labour market dualism, productivity gaps, and bargaining asymmetries (Taylor, 2004). The stock-flow consistent tradition also shows that wage growth is often indexed to productivity trends and inflation expectations (Godley and Lavoie, 2007; Lavoie, 2014).

Informed by these insights, MINDSET employs a backward-looking, partial-adjustment wage-setting mechanism, driven by four observable variables: (i) sectoral labour productivity growth, (ii) consumer price inflation, and (iii) the unemployment rate. This specification aligns with empirical research showing that wages typically adjust sluggishly, not fully following marginal cost signals or intertemporal optimization (Blanchflower and Oswald, 1995; Storm and Naastepad, 2009). Rather than clearing labour markets, wages evolve under institutional constraints, nominal rigidities, and inflation dynamics, influenced by economy-wide wage coordination.

Labour productivity captures the long-run capacity for wage growth, while the unemployment rate proxies for labour market slack. This approach also echoes Keynes's insight that, in the presence of nominal rigidities, inflation can affect both real wages and real interest rates, especially in the short run (Geary and Kennan, 1982; Hartwig, 2017).

Empirical evidence from economies characterized by high informality and weak institutions confirms that wage dynamics are often inertial, heterogeneous, and associated with persistent unemployment (Loayza and Rigolini, 2011; Bosch and Esteban-Pretel, 2012; International Labour Organization, 2018; Peeters and Reijer, 2012; Van Biesebroeck, 2011; Jayachandran, 2006). By estimating behavioural parameters from sector-level data, MINDSET incorporates these structural features, including incomplete pass-through from productivity to wages and macro-level coordination failures, offering a realistic account of wage formation in contemporary labour markets.

4.6 Normal output

Normal output is the level of production that firms expect to achieve when demand conditions are normal and capacity is utilized at its normal rate. The definition of 'normal' depends on how expectations are formed and therefore varies between models. The MINDSET formulation is provided below.

Normal output serves as an expected output level, around which sectors plan their future production. It is a demand-driven, and evolving benchmark, and is a standard feature of post-Keynesian macroeconomic models going back to Kalecki (1954). In other models it may be called 'normal rate of capacity utilization' or 'normal capacity output'.

The basic idea is that firms make decisions to meet a target rate of capacity utilization, rather than full capacity utilization. This is supported by economic data that show estimates of capacity utilization persisting at 80-85% (Pollitt et al., 2017; European Commission, 2024).

Capacity is defined as more than the combination of capital and labour. Institutional factors like culture, choice of products, or product quality contribute to production capacity. Generally, firms will want to maintain standard levels in all these factors, which influences their decisions.

The implementation in MINDSET asserts that expectations about future production are determined half by historical rates of production growth and half by revealed preference from capital accumulation. The historical part is then further split into indicators that depict short-run and long-run economic growth rates. The formulation of normal output Q^* thus integrates three components with weights of $\frac{1}{4}$, $\frac{1}{4}$ and $\frac{1}{2}$, respectively. Formally, it is expressed as:

$$\begin{aligned} Q^*_{j,n,t} = \frac{1}{4} \times \frac{Q_{j,n,t-1} + Q_{j,n,t-2} + Q_{j,n,t-3}}{3} + \\ \frac{1}{4} \times \hat{Q}_{j,n,t-1} \times \Big(\frac{\frac{Q_{j,n,t-1}}{Q_{j,n,t-2}} + \frac{Q_{j,n,t-2}}{Q_{j,n,t-3}}}{2}\Big) + \\ \frac{1}{2} \times K_{j,n,t} \times max\Big\{\frac{Q_{j,n,t-1}}{K_{j,n,t-1}}; \frac{Q_{j,n,t-2}}{K_{j,n,t-2}}; \frac{Q_{j,n,t-3}}{K_{j,n,t-3}}\Big\} \end{aligned} \tag{4.51}$$

Where $Q_{j,n,t}$ is observed output in t (and $t-1$, $t-2$, $\ldots$ periods), $\hat{Q}$ is a long-term growth component, where the growth rate of previous periods is applied to the previous period's observed value, finally the third component incorporates capital productivity by multiplying the current capital stock $K_{j,n,t}$ by the highest observed capital productivity over the last three periods.[31] This setup captures the dynamics that historical production trends establish a baseline, producers plan based on constant growth assumptions, and that capital expansion is typically associated with expectations of increased production capacity.

Normal output is used in the investment and price equations. In both cases, the model compares normal (expected) output, with actually observed output. This ratio acts as a *capacity utilization* measure.

The theory of expected or normal output is based on the idea of anticipated profits and, therefore, production levels based on anticipated demand (Keynes, 1930; Keynes, 1936; Dimand, 1988). MINDSET's formulation is simpler than in comparable models such as E3ME (Cambridge Econometrics, 2022; Scasny et al., 2009; Gramkow and Anger-Kraavi, 2019; T. Barker, 1999) because of its data needs, but captures similar dynamics. While the notion of *normal output* is mostly used in the context of these exercises, the underlying concept is used and established in other approaches too. In stock-flow consistent models, such as those described by Godley and Lavoie (2007) and Godin and Yilmaz (2020), *expected sales* play a role production capacity planning, while it is a more forward looking measure, the notion is similar: producers configure their workings to an expected level of output.

The approach is further motivated by an understanding of firm-level production planning decisions. Operation management textbooks (Slack, Chambers, and Johnston, 2010; Reid and Sanders, 2010) discuss capacity measurement and demand forecasting as crucial inputs for capacity planning and to expansion decisions. Capacity measurement is understood as understanding achieved production compared to goals, as well as to demand (i.e., backlog). While demand forecasting, beyond capacity measurement, is shown to mainly consider past production (or sales) often relying on simple methods (e.g., straight line projection, exponential smoothing) for forward-looking estimates (Chikán and Whybark, 1990; Chikán and Demeter, 1993; Wermus, 2005; Amoako-Gyampah and Boye, 1998; Mccarthy et al., 2006; Hofer, Eisl, and Mayr, 2015). This underscores the formulation MINDSET uses: expectations are driven by past output and capacity measures, while differences from these expectations are evaluated retrospectively and influence expectations for the next period.

[31] This functional form is heuristic, combining historical output (first component) and its growth trend (second component) each with a weight of one-quarter, alongside capital productivity (third component) with a weight of one-half.

Hence, producers might revise their investment decisions and their price setting based on the materialized production levels compared to expectations. In the investment equations, MINDSET directly uses capacity utilization: if the ratio of $\frac{Q}{Q^*}$ increases (i.e., production is higher than expected), then production capacities might be expanded through increased investment. Rationale for this is further discussed in Section 4.1.2, but in general, it can be expected that producers invest in new capacities before their production would reach full capacity (Keen, 2021). Meanwhile, the price equation uses two factors: first, in line with Keynes' description, it uses the difference from anticipated profits (the target-return approach) to adjust prices based on observed profits. Second, it uses the difference between observed and normal output to account for inflationary effects emerging from converging to full capacity rates (Corrado and Mattey, 1997; Dray and Thirlwall, 2011). Note that these two factors, while they are rooted in the same process - a difference from expected levels of production - can result in diverging effects: a high utilization generally increases prices through the capacity channel, while the same high utilization might decrease price through increasing value added and, therefore, profit rates. The overall effect is dependent on the interaction of the estimated parameters and the initial state of the sector in question.

To summarize, normal output is *not equal* to potential maximum output (i.e., output that would be possible given full factor utilization) as empirical evidence suggests that producers rarely work with or even plan for full capacity utilization (Keen, 2021; Corrado and Mattey, 1997; Nikiforos and Foley, 2012; Pollitt et al., 2017; Dray and Thirlwall, 2011; Driver, 2000), and capacity utilization rates are usually lower still in lower-income economies (Walker et al., 2024; Egger et al., 2022). The general sense is that (1) producers keep a *"buffer"* of their full capacity when planning for production, (2) they quickly expand their capacity if approaching full utilization levels (again building in the buffer in the new planned capacity). Further discussion and model examples are available in Lavoie (2016a), Nah and Lavoie (2019), Setterfield and David Avritzer (2020), and Lavoie (2022).

4.7 Value added

The calculation of sectoral value added and gross value added is discussed here, although there are no behavioural parameters in the calculation of these variables. Nevertheless, the authors decided to discuss it here, given that the calculation relies on preceding calculations, which makes value added too behave in an endogenous manner. MINDSET structures value added to emphasize rigid cost components - taxes, depreciation, and wages - while treating profits as the residual element. This design reflects empirical evidence of short- to medium-run inflexibilities, which provide limited mechanisms to mitigate shocks to costs.

Gross value added (GVA) is calculated at the sectoral level and then aggregated across all sectors to derive gross domestic product (GDP) at the national level. In MINDSET, GVA is primarily computed in volume terms, based on total output and the structure of intermediate consumption as captured by the current MRIO matrix. This approach is consistent with the System of National Accounts (SNA) definition, which defines GVA as the difference between output and intermediate consumption (United Nations et al., 2009):

$$VA^R_{i,m,t} = Q^R_{i,m,t} \times (1 - \sum_{j,n} a_{j,n,t}), \forall (i,m) \tag{4.52}$$

where i is the producing country, m is the producing sector, and $a_{j,n,t}$ is an element of the $\mathbf{A_{i,:,m,:}}$ the producer country-sector specific technical coefficient matrix, a partition of $\mathbf{A}$. The model, however calculates nominal GVA values, such as:

$$VA^N_{i,m,t} = Q^R_{i,m,t} \times p_{i,m,t} \times (1 - \sum_{j,n} a_{j,n,t} \times p_{j,n,t}),\ \forall(i,m) \tag{4.53}$$

MINDSET also tracks the main components of gross value added separately, in line with national accounting standards (Lequiller and Blades, 2014; United Nations et al., 2009). Notably, certain components, such as compensation of workers, are computed independently within the model framework. The specification is structured as follows:

$$TAX^N_{i,m,t} = Q_{i,m,t} \times p_{i,m,t} \times t_{i,m} \tag{4.54}$$

$$DEPR^N_{i,m,t} = K^R_{i,m,t} \times p^{GDP}_{i,m,t} \times \gamma_{i,m} \tag{4.55}$$

$$PROFIT^N_{i,m,t} = VA^N_{i,m,t} - TAX^N_{i,m,t} - DEPR^N_{i,m,t} - N_{i,m,t} \times w_{i,m} \tag{4.56}$$

$$\pi_{i,m,t} = \frac{PROFIT^N_{i,m,t}}{Q_{i,m,t} \times p_{i,m,t}} \tag{4.57}$$

Where $t_{i,m}$ is the constant taxes on production rate (calculated from the initial year of data), TAX^N is the nominal value of *production taxes (net of operating subsidies)*, K^R is the real value of capital assets, with p^{GDP} is the GDP deflator and γ is the depreciation rate (again calculate from the initial year of data), $DEPR^N$ then is the nominal value of depreciation (or *consumption of fixed capital*), further N is the number of workers in the sector, w is the annual average wage rate, the two together ($L \times w$) is the *compensation of workers*; π is the profit rate; $PROFIT^N$ is then the calculated as *net operating surplus* (United Nations, 2018), which covers of elements such as interest payments on debt, dividends paid, retained earnings or purchases of new assets (Lequiller and Blades, 2014).

It is important to note how the value added (VA) calculation is structured:

- On the production side, intermediate inputs (with the exception of energy) are linked to sectoral output coefficients, implying a Leontief production structure. This assumption reflects the short- to medium-term rigidity of input–output relations, as documented in empirical input–output studies (Miller and Blair, 2009). In contrast to a CES framework, where substitution among inputs can mitigate the effects of relative price changes, the Leontief assumption emphasizes cost-push dynamics: changes in the relative price of inputs directly affect the nominal value of gross value added (VA). This underscores the model's focus on capturing structural rigidities and distributional consequences of shocks, rather than long-run substitution possibilities. Nonetheless, as shown in the Section 4.2, technical coefficients in MINDSET are affected by prices, so even if recipes are constant, there is substitution between input producers.
- The **allocation of total nominal value added** depends on multiple components. First, taxes are modelled as a constant share of nominal output (not value added). This means that if VA declines due to rising input prices, the effective tax burden further reduces the amount of VA available for other uses.
- **Depreciation** is calculated using a fixed depreciation rate, but the underlying capital stock (K) evolves based on historical investment flows. Hence, if there has been significant past investment, depreciation (as a share of VA) may increase over time.
- On the distributional side, **compensation of workers** is treated as an independently determined component of VA (as discussed in Section 4.5 and Section 4.4). Rather than adjusting as a residual to equate marginal labour productivity, wages reflect institutional and social factors and are often inflexible in the short run (Blanchard, 2017; Stockhammer, 2013). Mechanisms can explain this wage stickiness, such as collective bargaining and social partnership arrangements (OECD, 2019), efficiency-wage behaviour in which firms sustain higher wages to stabilize effort and reduce turnover (C. Shapiro and Stiglitz, 1984), or insider–outsider dynamics that reduce wage responsiveness to unemployment (Lindbeck and Snower, 1988).

- Finally, **profits** (i.e., net operating surplus) are treated as the **residual** component of value added. In the short run, profit margins absorb shocks once other VA components are accounted for. In the medium run, however, firms seek to restore target rates of return through price-setting behaviour. This mechanism links short-run distributional rigidities with medium-run adjustments in markups and profitability. (see Section 4.3).

Overall, this means that MINDSET gives priority to "cost" components of value added - compensation, taxes, and depreciation - within each year. Profit is allowed to adjust as a residual. However, between years, producers seek to restore profitability through price adjustments, thereby re-coupling profit margins with output prices over time. It also means that MINDSET follows Richard Stone's logic, assuming that volume terms of intermediates and value added sum to 1.0 (D. Meade, 2007).

Energy and emissions

MINDSET captures energy use in physical and monetary terms. Physical energy use is based on a matrix of initial-year energy flows constructed from (IEA, 2024); monetary flows of energy carriers follow (Lenzen et al., 2017) [32]. In the initial year, the model calculates both fuel use (in terajoule (*TJ*)) per flow as well as an implicit price for each of the flows (in $\frac{USD_{t0}}{TJ}$). Changes in energy consumption - allowing for efficiency and substitution - are then calculated for intermediate flows as well as household consumption (final demand). Government and investment final demand are not assumed to be price-reactive and are, hence, kept constant. Crucially, MINDSET does not differentiate between final and primary energy demand in the treatment of the flows.

A two-step setup allows MINDSET to capture both energy efficiency and substitution behaviour. Energy intensity (in $\frac{USD_{t0}}{TJ}$) considers a) the total intensity (irrespective of the fuel that is used to supply the physical demand), and b) a fuel-specific demand, influenced by relative price changes and economic activity. The first part (a) captures sectors' or households' overall consumption response to changing energy prices, i.e., energy efficiency measures. This formulation, driven primarily by economic activity and energy prices follows the tradition of Hunt and Manning (1989), Bentzen and Engsted (1993), T. Barker, Ekins, and Johnstone (1995), and Mercure et al. (2018b), with the considerable adjustment that it uses energy intensity rather than energy demand as the dependent variable. This adjustment implicates that, rather than assuming a functional form of $E \sim \beta Y + \gamma P$, the model implements $\frac{E}{Y} = \gamma P$. The practical consequence is that not just energy use changes, but *energy intensity* changes due to price effects, hence allowing for energy efficiency increases. The second part (b) captures fuel substitution effects from relative price changes between energy carriers, such as induced by policy reform (e.g., energy or carbon taxation). Substitution can also follow the varied consumption of fuels with increasing/decreasing activity[33]. This overall two-step estimation approach is adopted from the similar implementations of energy demand in E3ME (Mercure et al., 2018b; Cambridge Econometrics, 2022) and GINFORS (Lutz, B. Meyer, and Wolter, 2010).

It is important to note that energy use and resulting emission levels in MINDSET scenarios can also be informed by exogenous inputs from an energy-sector model, such as the TIMES model (an acronym for The Integrated MARKAL-EFOM1 System), or an electricity-sector model, such as the World Bank's Electricity Planning Model (EPM). MINDSET can also be fully integrated with models of dynamic technological change, such as FTT:Power (Mercure, 2012b).

The remainder of this chapter discusses how the energy efficiency and substitution mechanics are implemented, first, discussing energy use in production, or intermediate demand, and then in final demand.

5.1 Energy use in production

MINDSET captures two mechanisms for the consumption of energy in industry. First, *total energy demand* is calculated as a reaction to a weighted average energy price, such that industries' overall energy demand decreases when average energy prices rise, accounting for changes in energy intensity. Second, *energy substitution* between energy carriers (fuels and electricity) is calculated, with substitution reacting to differences in price changes between energy carriers as well as to economic activity.

[32] For more detail on the construction of this matrix see the *Energy balances processing* note.

[33] This allows the model to introduce some heterogeneity across fuel types: a higher production volume might increase the consumption of a main fuel, say petrol, but not increase the consumption of auxiliary energy, say electricity. Hence having fuels react to activity changes independently enables us to empirically capture the difference across fuels in the industrial sectors.

This setup captures structural change with regard to energy use in production[34], as it allows for changing IO coefficients, or production recipes, from the core MRIO table for energy technologies. From a policy perspective, it allows considering the transformative effects on supply chains of any policy which directly or indirectly raises the cost of energy carriers, including carbon pricing, fossil subsidy removals, or excise taxes. The adjusted MRIO table, after energy substitution effects, is carried over across years, which means that these policies have long-term effects on sectoral production structures.

5.1.1 Total energy demand of production

Total energy demand is derived from economic activity and energy intensity of economic activity, which is influenced by the weighted average price of energy:

$$\theta_{i,j,m,n,t} = \frac{e_{i,j,m,n,t}}{\sum_{j,n} e_{i,j,m,n,t}} \tag{5.1}$$

$$\delta p^*_{i,j,m,n,t-1} = \frac{1 + \delta p_{j,n,t-1}}{1 + \delta P^{GDP}_{i,t-1}} - 1 \tag{5.2}$$

$$\delta P^E_{i,m,t-1} = \sum_{j,n} \delta p^*_{i,j,m,n,t-1} \times \theta_{i,j,m,n,t-1} \tag{5.3}$$

$$\epsilon_{i,m,t-1} = \frac{E_{i,m,t-1}}{Q_{i,m,t-1}} \tag{5.4}$$

$$\epsilon_{i,m,t} = \epsilon_{i,m,t-1} \times (1 + \delta P^E_{i,m,t-1} \times \beta^1_{I,m}) \tag{5.5}$$

$$\Delta E_{i,m,t} = (\epsilon_{i,m,n,t} - \epsilon_{i,m,n,t-1}) \times Q_{i,m,t-1} \tag{5.6}$$

where $e_{i,j,m,n,t}$ is the use of fuel n originating from country j in the production of sector m in country i in period t, $\theta_{i,j,m,n,t}$ therefore is the share of fuel input n, j in total energy use of sector m, i. $\delta p^*_{i,j,m,n,t-1}$ is the GDP deflator ($\delta P^{GDP}_{i,t-1}$) adjusted price change of the energy flow in country i (adjusted with the GDP deflator of the importing country i), meanwhile $\delta P^E_{i,m,t-1}$ is the total price change of energy (aggregated) in country i for sector m. $E_{i,m,t}$ is the total energy use of sector m in country i and time period t. $Q_{i,m,t}$ is output of sector m in country i in constant price monetary units. Finally, $\beta^1_{I,m}$ is the estimated parameter of total energy demand response with respect to price changes, in region I (where $i \in I$).

5.1.2 Energy substitution in production

Once, the level of total energy use is determined, substitution between different energy carriers in production processes takes place. Fuel substitution is driven mainly by relative price changes across the different fuel types. These price differences can be driven by policy (*such as carbon pricing*) as well as other economic mechanisms (e.g., increasing cost of intermediates or increasing wage costs). The model considers four different fuel types: electricity; natural gas (including LNG); coal, lignite and related products; and refined petroleum products. Substitution is calculated as, per each of the fuel types (n):

[34] Assuming that energy is a mostly homogeneous good, with some constraints: given that MINDSET simulates changes from the initial-year energy balances, if consumption of a certain energy carrier in a given sector is zero, this fuel use remains zero in subsequent years.

$$\theta^E_{i,j,m,n,t} = \frac{e_{i,j,m,n,t}}{\sum_{j,n} e_{i,j,m,n,t}} \tag{5.7}$$

$$\theta^F_{i,j,m,n,t} = \frac{e_{i,j,m,n,t}}{\sum_{j} e_{i,j,m,n,t}} \tag{5.8}$$

$$\delta P^E_{i,m,t-1} = \sum_{j,n} \delta p_{j,n,t-1} \times \theta^E_{i,j,m,n,t-1} \tag{5.9}$$

$$\delta P^F_{i,m,n,t-1} = \sum_{j} \delta p_{j,n,t-1} \times \theta^F_{i,j,m,n,t-1} \tag{5.10}$$

$$\delta \hat{P}^F_{i,m,n,t-1} = \frac{\delta P^F_{i,m,n,t-1}}{\delta P^E_{i,m,t-1}} \tag{5.11}$$

$$e_{i,m,n,t} = e_{i,m,n,t-1} \times (1 + \delta \hat{P}^F_{i,m,n,t-1} \times \beta^2_{i,m,n} + \delta Q_{i,m,t-1} \times \beta^3_{I,m,n}) \tag{5.12}$$

where $\theta^E_{i,j,m,n,t}$ is the share of flow i, j, m, n, t in the total energy use of i, m; $\theta^F_{i,j,m,n,t}$ is the share of of flow i, j, m, n, t in the fuel-specific (n) energy use in i, m, P^E and P^F are weighted prices (energy and fuel respectively), while $e_{i,m,n,t}$ is fuel demand for i, m, n, t.

5.2 Energy use in the electricity sector with FTT:Power

The electricity sector is a central focus of the energy transition, as it is expected to undergo rapid technological change. To represent this sector within MINDSET, a model from the Future Technology Transformations (FTT) family, FTT:Power can be integrated. More detail is available in the Appendix section on the FTT:Power-MINDSET integration (Section A.5).

FTT:Power is a global, bottom-up model of electricity generation that explicitly represents competition between technologies. Rather than assuming that investment decisions are driven solely by least-cost optimization, it uses concepts from evolutionary economics and innovation diffusion theory to simulate how technologies gain or lose market share over time (Mercure, 2012b). The model accounts for factors such as learning-by-doing, investor heterogeneity, policy interventions, and system constraints, offering a more realistic depiction of how electricity generation shares might evolve under different scenarios (Mercure et al., 2018b). It draws on innovation theory and employs a predator–prey framework to describe competitive dynamics, with the pace and direction of innovation shaped by both economic development and policy interventions (Mercure, 2012b). Overall, FTT:Power offers a non-equilibrium perspective on the energy transition and the role of policies in accelerating decarbonization.

To capture the rapid technological transformation of the electricity sector and its macroeconomic and distributional outcomes within MINDSET, the technology-oriented FTT:Power model can be integrated. While the econometrically driven equations in MINDSET are suitable for capturing macroeconomic relationships, they are less capable of representing rapid technological shifts in the electricity sector. In contrast, FTT:Power is specifically designed to simulate the dynamics of technological diffusion. This integration also expands the range of policy instruments that MINDSET can address, for example, subsidies for solar photovoltaic technology deployment. Moreover, by incorporating electricity generation at the technology level, MINDSET can more accurately reflect the supply chain impacts of the energy transition. Since the intermediate input structures differ significantly between technologies (e.g., coal versus gas power plants), incorporating the detailed generation mix from FTT:Power enhances the accuracy of the economic impact assessment of the energy transition.

The main inputs from MINDSET to FTT:Power include:

1. Electricity demand - MINDSET estimates the electricity demand at sectoral level, while in FTT:Power electricity demand is given exogenously.
2. Carbon prices - in both MINDSET and FTT:Power carbon prices are defined exogenously, so the alignment of the two is essential.
3. Fuel prices - these are endogenously estimated in MINDSET, and feed into the operation costs of generation technologies in FTT:Power.

The main outputs of FTT:Power integrated into MINDSET include:

1. Electricity generation and capacity, or investments, by technology
2. Electricity sector investment by technology
3. Electricity sector employment by technology

The aim of integrating FTT:Power into MINDSET is to provide a more accurate representation of the electricity sector and thereby improve the precision of MINDSET's estimates. The separate documentation describes all steps in detail.

5.3 Energy use in final demand

5.3.1 Households

In household consumption, MINDSET assumes a substitution effect based on relative price changes and overall activity, similar to that in intermediate demand. Notably, demand changes in the model are handled through the monetary consumption effects (see household demand section in Chapter 4), therefore in energy terms MINDSET only does substitution across fuel-types, but overall demand is managed by the household demand calculation.

Households similarly react to price impacts as industry does, although the coefficients of reaction can be substantially different. Generally, in households we expect a more inelastic behaviour both for overall demand and when it comes to fuel-switching as well. In terms of substitution mechanics, however, households react in MINDSET exactly as production does, i.e., following the same process that has been described in the section on energy substitution in production.

5.3.2 Fixed capital formation

Substitution effects for energy use of fixed capital formation are not included in MINDSET. Valued fixed capital formation in the national accounts should largely contain assets that are used in the production repeatedly for more than one year (United Nations, 2018), hence energy should mostly be excluded from this category of final demand. Nevertheless, IO tables might still include values for GFCF demand for energy sectors products[35], but these are most often related not to the actual products, but intangible transfers between sectors.

[35] E.g., in 2015 in the EU27 electricity in GFCF is about 0.5% of total GFCF. See here.

Model Fundamentals

This chapter lays out the core elements that underpin the design and operation of MINDSET. While the previous chapters focused on specific accounting frameworks, behavioural equations, and the specific treatment of energy and emission flows, the discussion here provides the technical foundations necessary to understand how the model is built, parametrised, and solved. It describes the software architecture, the main data source on which the model relies, the econometric strategy used to generate region- and sector-specific behavioural parameters, and the iterative solution cycles that integrate short- and medium-term dynamics. Together, these components form the backbone of MINDSET, ensuring both analytical rigour and transparency, and enabling the model to serve as a flexible platform for applied policy analysis.

6.1 Software

Technically, MINDSET is a combination of (1) pre-processed datasets in Python's pickle format, CSV, and Microsoft Excel files, (2) a simulation program written in the Python programming language that is the core of the model, and (3) scenario files set up in Microsoft Excel. To ensure complete reproducibility, the full model, including codes, parameter and data inputs as well as concordance tables, is documented in detail in its Github repository[36]. Given the collapsibility of the model's dimensions (Section A.1), MINDSET can be run even on desktop computers with limited computational capacity; however, running the full version of the model (163 regions with bilateral linkages across 120 sectors) requires significantly more computing capacity. While theoretically it can be run in Unix environments, so far it has been only tested on Windows.

As being open-source and built on open data is a crucial goal of MINDSET, the model development has followed a process where much of the data and underlying code has been version controlled. This also means that upon release it will be open to further developments and updates will be available form its main repository, managed on Github.

6.2 Data

In its global default version, MINDSET utilizes a wide range of data to cover a large number of economic activities (sectors) and countries. However, adjusting any data or parameter inputs to country-specific data is facilitated through accessible spreadsheets and codes.

In its default application, MINDSET uses MRIO data from the open-source Global Resource Input-Output Assessment (GLORIA) database (Lenzen et al., 2017; Lenzen et al., 2021). GLORIA has 163 regions, each with 120 sectors. It was built by the University of Sydney for studying material footprint accounts, commissioned by the UN International Resource Panel. Multiple factors were considered in the selection of the default MRIO system for the model: these included geographical coverage (especially focusing on lower-income countries), detail in terms of high-relevance sectors (e.g., mineral sectors), and whether the dataset is freely accessible - to fit with the open-source philosophy of

[36] https://github.com/cpmodel/MINDSET_dynamic; access upon request.

the model. GLORIA, which follows in the footsteps of the EORA database, fulfils many of these criteria. An important caveat of using MRIO datasets is that the data undergoes matrix balancing and interpolation, especially in cases where data quality is low or the data is non-existent.

Hence, applying MINDSET to a specific country context requires checking and updating the country's data in the GLORIA system, based on countries' national accounts, supply and use and input-output tables, and other sources. This procedure ensures that sectoral aggregates, as well as production recipes, are aligned with the most recent country-specific data, while maintaining trade relations and the ability to account for regional spillovers. When preparing a country application, the first step is to collect the most recent supply and use, or input-output, tables published by the national statistics agency. Core aggregates for output, intermediate consumption, gross value added components, external trade and final demand are then updated to the model base year using countries' official national accounts data. Where the national classification is less detailed than the 120-sector GLORIA structure, it is disaggregated with auxiliary national or international datasets (e.g., FAOSTAT for agriculture, UNIDO industrial statistics, UN Comtrade and services trade data, energy balances, household and labour statistics). Once domestic production and demand is mapped to the model's sectoral classification, the updated table is then rebalanced, typically with a RAS-type procedure, to preserve national totals while keeping the original multi-regional trade structure and partner shares unless country evidence justifies an adjustment. This sequence ensures that the country block reflects the latest available national evidence while remaining compatible with the global MRIO used in MINDSET.[37]

The monetary MRIO system is extended with a network of physical flows, based on IEA and UNDESA data, see Section 3.2 for more details. To represent detailed country-level sectoral employment, the model uses ILO employment data, discussed in Section 4.4. Meanwhile, initial capital stock and investment data are based on KLEMS and GLORIA data, see Section 4.1.2.2 and Section A.6 for more detail.

Parametrization uses a data set that is more extensive in terms of temporal coverage, although *mostly* less detailed in terms of sectoral resolution. This is discussed in more detail in the next section (Section 6.3) and also in a separate note on estimation data.

6.3 Parametrization approach

Behavioural equations in MINDSET are informed by a wide range of region- and sector-specific econometrically estimated parameters. Parameter estimations are carried out on a novel compilation of sectoral panel data from various sources[38] (see the separate Data Documentation for details). The estimated parameters cover seven individual equations, representing the key dynamic relationships (see Chapter 4) differentiated between 11 regions and around 20 sectors. The theoretically bounded panel regression estimations are carried out at a sectoral level across different country- and sector-aggregations. The separate Estimation Documentation provides details on the approach, including summary statistics and results for all parameters. With the exception of large datasets, all codes and concordance tables for data processing and estimations are documented in the Github repository to ensure complete reproducibility and facilitate updating[39]

While Error Correction Models (ECMs) are well-suited to capturing both short- and long-run dynamics in economic relationships (Engle and Granger, 1987), their reliance on long time series renders the ECM approach unsuitable for

[37]Note that a re-balancing of the MRIO is possible, but not necessarily needed when country-specific data is replaced given the collapsing method and the iterative solution of the model (discussed in Section A.1 and Section 6.4).

[38]UN National Accounts Data, WDI, various KLEMS accounts, IMF, ILO, FAO, UNIDO, WIOD, REITI, ORBIS, COMTRADE, IEA, WIIW, and data from National agencies

[39]https://github.com/cpmodel/MINDSET_dynamic_estimation; access upon request.

parameterising MINDSET, as such data are not sufficiently available for low- and middle-income countries. Instead, the panel regression approach exploits both cross-sectional and temporal variation while mitigating concerns of short time series bias (Baltagi, 2005; Hsiao, 2014). Parameters are estimated by a Restricted Ordinary Least Squares with year and country fixed effects. Lagged or smoothed variables are used where appropriate to account for adjustment dynamics.

Each behavioural equation is estimated at four levels of aggregation. First, at the global level, using the full sample of countries and sectors[40], including sector fixed effects in addition to country and year fixed effects. Second, also at the global sector level, the model is estimated separately by sector using the full sample. Third, at the income group level, we stratify the sample by World Bank income classification[41] and estimate sectoral elasticities within each group. Finally, at the regional level, we estimate the model for aggregate regions defined by geographic and income dimensions.

Final parameter values for each sector-region pair are selected using a hierarchical procedure that synthesizes information across these aggregation levels. This algorithm balances statistical significance, sample adequacy, and theoretical plausibility, while avoiding overfitting. This approach is consistent with established practices in applied econometrics for heterogeneous panel data, promoting parsimony, empirical validity, and policy relevance (Im, Pesaran, and Shin, 2003; Wooldridge, 2013).

6.4 Solution cycles

MINDSET, being a two-timescale hybrid model, combines various cycles (or loops) to arrive at its reported results. These combine short-term (within-year) and medium-term (across-year) effects.

First, there is an **annual cycle**, which means that for each year of the simulation, MINDSET calculates independently, using the previous year's results as quasi-exogenous information. In this case, the solution for each of the model equations depends on the respective values of the relevant variables in $t-1$, but newly calculated values (t) does not yet factor in to the results. Importantly, at this point, the model also does not have information about the following year's value (i.e., about $t+1$); hence, agents' foresight in MINDSET is limited.

Second, the **within-year loop** is solved, which might play out multiple times within a year. The within-year loop is a Gauss-Seidel-type iterative solution (Sauer, 2012, pp. 108-112) for all variables that are allowed to change within a year. The iterative solution is expected to converge on labour compensation change, price change, and on the MRIO structure. As Table 6.1 shows many of the variables (e.g., induced investment or annual average wage) are only calculated in a between-years fashion; meanwhile, key variables (income, consumption, prices, employment, trade) are calculated in a within-year fashion. Where MINDSET uses within-year impacts, it means that it captures feedbacks within the annual cycle as well, e.g., as employment increases it will increase incomes, which will increase consumption, which boosts output, which again might increase output, but at the same time prices, which in turn can depress consumption. These balancing effects are allowed to go against each other within the year cycle until a sufficient solution is found. The model considers the solution sufficient if the convergence criteria is met[42]. The convergence criteria can be adjusted based on the modelling task at hand, but by default the convergence criteria shown in Table 6.2 are used.

[40]In the *Consumption* and *Trade* modules, sectoral indices correspond to COICOP and GLORIA classifications, respectively, whereas in other modules they correspond to ISIC Rev.4 sectors.

[41]We adopt a dynamic income classification scheme that assigns an income group to each country-year observation, while applying smoothing adjustments to reduce excessive transitions between groups over time.

[42]The convergence criteria is fulfilled if results between two within-year iterations does not change more than the threshold.

Across-years	**Within-year**
	Output
Income & consumption	Income & consumption
Investment	
Govt. consumption	
carried over	Prices
Employment (structural)	Employment (output)
Annual average wage	
Capital stock	
Normal output	
Energy intensity & fuel-mix (MRIO structure)	
carried over	Trade (MRIO structure)

Table 6.1: How certain variables are calculated in the cycles

Variable	**Convergence criteria**
MRIO matrix after trade	5%
Labour compensation change	5%
Price change	5%
Tax revenues collected (carbon pricing)	5%

Table 6.2: Default convergence criteria in MINDSET

Notably, this solution is a sufficient but not equilibrium solution as not all markets clear in a Walrasian sense. Indeed, even if global production equals demand in the goods and services, markets, this is not imposed in the factors (employment and capital) markets. Moreover, market clearing and its impacts on prices depend on behavioural responses and by how these are channelled back into the economy.

Distributional effects for workers and consumers

MINDSET's estimates of sectoral changes in consumer prices, labour demand, and labour income serve as valuable inputs for macro-to-micro, or top-down, microsimulations to understand policies' employment and consumption equality effects. The model is particularly well-placed to inform distributional outcomes as it captures the short-term frictions and related transitional costs associated with policy-induced sectoral reallocations. As a standalone tool, MINDSET disaggregates results across a wide range of economic sectors. However, this conceals the underlying equity effects across socioeconomic strata as well as the resulting changes in poverty levels. Hence, the sectoral and labour market detail in model results facilitates linkages with microsimulation tools. Results can serve as input for top-down macro-to-micro simulations such as the World Bank's Microsim model for poverty outcomes (Bourguignon, Bussolo, and Pereira da Silva, 2008). In addition, MINDSET results can be combined with the model's own simplified microsimulation modules, the Employment and the Consumption Modules[43]. The shorter-term distributional effects of employment and consumption effects are measured through augmenting MINDSET's results with countries' most recent microdata. Countries' labour force and household budget survey data are harmonized following a common MINDSET codebook, capturing all relevant socioeconomic indicators such as disposable (or market) income, consumption patterns, occupation and sector of work, gender, subnational region, and rural/urban location.[44] The sectors of work – or for consumption impacts, the product categories - serve as the linking variable between MINDSET results and the microdata. In their current version, there is no feedback from the microsimulations back to MINDSET. This chapter gives an overview of both modules.

7.1 Employment Distribution Module

The Employment Distribution Module provides a simplified version of macro-to-micro simulation models, focusing on the short- to medium-term distributional effects of policy-induced labour demand changes and how they impact different socioeconomic groups. As in more sophisticated macro-to-micro models, the main linking variable between the macroeconomic model and the microsimulation is the vector of sectoral employment changes. The microsimulation is based on countries' labour force surveys (LFS), such as from the World Bank's Global Labor Database (GLD) (Gronert and Honorati, 2026), or similar individual-level data representative of countries' populations. These are used to identify the average characteristics of workers (employed or self-employed) in each sector. Collected characteristics include the level of income[45] and respective income decile, occupation, age, gender, and region or province of work.[46]

For each group d of average workers in sector n in country j, the estimated employment impact Δl, then, follows

[43] Codes, concordance tables and documentation available on Github (https://github.com/cpmodel/mrio-microdata; access upon request)

[44] To maximize synergies with other tools, MINDSET's Consumption Module follows the same harmonization of microdata as the IMF-WB Climate Policy Assessment Tool (CPAT).

[45] Income definition (disposable, market, or labour) depends on availability in a country's microdata. Preference is given to disposable income.

[46] The MINDSET Codebook guides the harmonization of each country's survey in accordance with the Employment Module's coding program (both available on GitHub[47]). Harmonizing an LFS requires no more than one day of work. The resulting multi-country harmonized dataset supports ad-hoc requests from a wide range of countries.

from the MINDSET results:

$$\Delta l_{n,j}^{d} = \delta N_{j,n} \times lf_{n,j}^{d} \tag{7.1}$$

where δN is the relative change in sectoral employment in sector n compared to the baseline, according to MINDSET results, $lf_{j,n}^{d}$ is the number of workers of group d working in sector n, and group d can be characterized by any combination of the socioeconomic characteristics of interest (listed above).

Importantly, this method is static and is based on certain assumptions which make it particularly suitable for short- to medium-term distributional analyses. It is static in the sense that it takes as inputs only one year of MINDSET results, not a time series of sectoral effects; such as results for 2030, informing, for example, how new jobs created would likely benefit workers in different income groups differently. As a static approach, it rests on two important assumptions: first, the labour force composition remains unchanged between the survey year and the year of interest. This means the composition of workers in each subsector remains the same as in the survey year, in terms of their relative positioning in the income distribution, gender, age, and region of work. Population or other demographic changes are not accounted for. As a consequence, the Employment Module is appropriate for short- to medium-term distributional analyses. Second, the profile of new jobs and of the workers taking them is assumed to follow the same characteristics as the existing jobs and workers in each sector. As a consequence, the analysis is less suitable for policy analyses which involve large within-sector transitions, e.g., a large-scale shift from coal-fired to renewable electricity, which would likely require workers of different skills and education. Arguably, such a shift would anyway occur over the longer term, and their distributional analysis would require a level of occupational detail in the survey data which rarely exists. These two assumptions leave the Employment Distribution Module highly suitable for analysing the whole-of-economy distributional effects of any standard development policy package.

The data requirements are moderate, enabling a low-cost, yet highly granular analysis suitable for application in almost all low-income countries. The Module requires only one year's labour force survey. Yet, the analysis provides a high level of socioeconomic granularity. Survey data are being harmonized to the international standards (specifically, to the International Standard Industrial Classification of All Economic Activities (ISIC) Revision 4 and the International Standard Classification of Occupations (ISCO) 2008 revision). If surveys permit, the linking variable, sector of work, is differentiated by 88 sectors (ISIC Rev. 4 2-digit).

The granular analyses provide important insights for policy advice on labour market support measures as well as social policies by identifying losers (or losing groups) and winners, as well as potential skill mismatches. Because MINDSET takes into account the frictions which occur during sectoral and job reallocation before a new steady-state is reached, model results provide information on both the winners and losers *during* the process of reallocation:

On the one hand, MINDSET's Employment Distribution Module helps identify socioeconomic groups that may require specific support during a transition or due to climate change impacts. Job losses from policy or changing weather and hydrological patterns are often concentrated among specific socioeconomic groups. While some of these impacts might be foreseeable without additional modelling (such as the impact of climate change on low-income subsistence farmers), other similarly concentrated job losses might not be as apparent; for example, the loss of relatively well-paying jobs of low-skilled women in hospitality industries due to a reduction in tourism driven by increasing temperatures.

On the other hand, understanding the occupational and sectoral composition of new jobs can help identify which labour market support policies might be needed to seize the economic opportunities presented by a given development policy package, and to avoid economic slowdown due to skill mismatches (such as is the case in Europe McGrath (2021) and Di Pietro (2024). Based on detailed information on a country's labour force composition, the

Employment Distribution Module identifies which sectors and occupations are likely to experience the largest labour supply bottlenecks and skill shortages in achieving a given policy objective. Hence, it can provide the basis for informing targeted labour market support measures. For instance, resilient development strategies often require sizeable investment in new and better infrastructure which can lead to spikes in demand for medium-skilled craft and trade-related workers in construction (World Bank, 2022a).

7.2 Consumption Distribution Module

The Consumption Distribution Module estimates the distributional effects of both tax incidence and revenue recycling into direct household income support or infrastructure investments in a narrow monetary sense, proxied by relative changes in consumption. The method follows established approaches to consumption-focused microsimulation (Dorband et al., 2019; Vogt-Schilb et al., 2019; Hubacek et al., 2017). Country-specific household budget surveys (HBS) are combined in a consistent manner with MINDSET's macroeconomic results on price changes and, where applicable, tax revenues. The main linking variable between the macroeconomic model and the microsimulation is the vector of sector-specific price changes, and the aggregate amount of government revenues allocated to direct household income support. Countries' HBS are harmonized according to the MINDSET Codebook to identify average characteristics of households across socio-economic groups. To maximize synergies with other tools, MINDSET's Consumption Distribution Module follows the same harmonization of data as the IMF-WB Climate Policy Assessment Tool (CPAT), focusing on disposable income deciles[48] and consumption patterns of goods and services, and of different fuels and electricity.

For the **incidence from consumer price changes**, the HBS consumption aggregates need to be scaled to the MINDSET's household final demand vector, to ensure aggregation consistency between the macro and the micro results (cf. Dorband et al., 2019; Hubacek et al., 2017. A concordance table is used to match model-level products n to HBS-level consumption categories g. For each consumption category g from the household survey, the procedure allocates model-level products n ($n \in g$) into consumption category totals for each income decile d. Jointly with data on countries' total population in the year of interest (World Bank, 2024), this returns per-capita baseline consumption by consumption category g and decile d such that:

$$C^d_{g,i} = \sum_{n \in g} C^d_{n,i} \tag{7.2}$$

Similarly, relative price changes $\delta p_{g,i}$ are calculated as the weighted average price change of products n in g, using the household final demand vector entries as weights.

$$\delta p_{g,i} = \sum_{n \in g} w_{n(g),i} \times \delta p_{n,i} \tag{7.3}$$

[48] Due to data constraints, inconsistencies can exist between the income-group analyses of the Employment and the Consumption Modules: employment information is typically available at the individual level, while consumption information is reported at the household level. MINDSET operates with harmonized survey data across a large number of countries and employment and consumption effects are not assessed jointly at the household level, unlike in dedicated microsimulation exercises. Hence, the decile ranking used in the employment module need not coincide with the ranking implicit in the consumption module.

Where $w_{n(g),i}$ is household final demand vector entries as weights of n in g. The immediate overall consumption incidence, *not accounting for households' behavioural reactions* to price changes, can then be calculated for each decile d:

$$\delta C_i^d = \frac{\sum_g \delta p_{g,i} \times C_{g,i}^d}{\sum_g C_{g,i}^d} \tag{7.4}$$

The consumption incidence *with behavioural reactions* is, in the current version, calculated as:

$$\delta C_i^d = \frac{1}{C_i^d} \sum_g \delta p_{g,i} \times C_{g,i}^d \times (1 + \delta p_{g,i})^{\varepsilon_{g,i}^d} \tag{7.5}$$

Where $\varepsilon_{g,i}^d$ is the decile- and category-specific own-price elasticity of demand. The price elasticities of demand are taken from the literature (B. Meade et al., 2014; Muhammad et al., 2011), but can be adjusted to reflect MINDSET's own elasticities.

The incidence analysis of the **direct household income support** is based on country-level population data (World Bank, 2024). Let total population be POP_i, total revenue earmarked for direct household support be R_i, and let $d^* \in \{1, \ldots, 10\}$ be the highest targeted decile (so deciles $d^* \leq d$ are eligible). Thus, the targeted population is:

$$POP_{d^*,i} = \left(\frac{d^*}{10}\right) POP_i \tag{7.6}$$

The uniform per-capita transfer is then given by:

$$\tau_i = \frac{R_i}{POP_{d^*,i}} \tag{7.7}$$

Lastly, the Consumption Distribution Module allows for mapping **gains from investments in infrastructure provision** to the household level, following the approach developed in Dorband et al., 2022, also deployed in the IMF-WB CPAT. Government investments which are earmarked for infrastructure provision - water, sanitation, electricity, information and communication services, and public transport – can be simulated across consumption deciles as well. For each infrastructure category, the funds can be allocated across households without access in the baseline. Decile-specific "no-access" populations are obtained by applying access rates to population counts, where the latter are taken from countries' HBS. The allocation of potential, albeit non-monetary, 'benefits' is approximated through assumed 'transfers': For each infrastructure category, per-capita transfers are defined as the total allocation divided by the total baseline "no-access" population; the decile-level amount is then based on each decile's share of households without access. Summation over categories yields the total public-investment transfer for each decile. This approach to estimating the benefits of infrastructure access transfers abstracts in a number of ways, which are discussed in detail in Dorband et al., 2022. For example, results are intended as stylized upper-bound estimates, implicitly assuming perfectly targeted infrastructure investments. In country-specific applications, sensitivity analyses can relax this assumption by introducing leakage or administrative costs, allowing for more realistic distributional outcomes.

Discussion

This Chapter discusses MINDSET's assumptions and limitations by examining three main areas. First, Section 8.1 explores how the model represents supply-side and capacity constraints, including their implications for prices, investment, employment, and income. Second, Section 8.2 considers the model's long-term dynamics, emphasizing path dependency, adjustment processes in labour and trade, and the treatment of technological change. Finally, Section 8.3 outlines key simplifications and constraints in MINDSET's design, covering issues of sectoral disaggregation, data inputs, behavioural parameters, the financial sector, and the treatment of technology.

8.1 Supply-side constraints

While the level of production is determined by the level of effective demand, production is subject to endogenous capacity constraints through adjustments in prices, leading to adjustments in investment, employment, and income, hence, feeding into lower demand. Although MINDSET does not impose ex ante financial constraints—and thus rules out financial crowding out—higher investment can still generate real economy crowding out by increasing demand for capital, labour, and intermediate goods, raising their prices and displacing other activities. These endogenous resource constraints shift the focus from financial equilibria to real adjustments within the productive structure.

Endogenous supply-side constraints occur when factor inputs, capital stock and labour, become overutilised:

- If a sector's capacity is overutilised, i.e., output runs above normal output in the previous period, two things follow. Prices in the stretched (investing) sector rise in this period, dampening final and intermediate (through trade) demand and, if sustained, potentially leading to interest rate increases. At the same time, the sector increases investments, i.e., increasing demand for capital goods from other sectors, to add capacity for the subsequent period. Hence, prices temper demand (and hence production) and can increase capital costs, while additional investment eases the constraint.
- If the labour market tightens as the unemployment rate approaches its (exogenous) minimum, wages increase. Assuming that the overall income effect is positive,[49] household demand will increase. At the same time, wage costs raise production costs so that, with stronger demand, prices, and potentially inflation, rise. Higher prices and capital costs then lower demand and investment demand and production falls.

Hence, MINDSET does not necessarily assume full capacity utilization in the baseline, allowing for over- and under-utilization of factor inputs. Capacity utilization is measured and tracked in the model, but the constraints are 'soft' and feed into behavioural equations (see Section 4.6). For each sector in each country, the parametrisation of these equations determines both the rate of capacity utilization as well as the speed at which it reverts back to normal rates following shocks. This setup is in line with empirical evidence emerging in lower-income countries: capacity underutilization, or economic slack, can be structural - or long-term; it differs by sector and is more prevalent in lower-income environments; lastly, it can support large production increases while exerting minimal price pressures (Walker et al., 2024; Egger et al., 2022).

This approach contrasts with that in CGE models, where the level of production is determined by the supply of factor inputs, and in DSGE models, where long-term production levels are also determined by the supply of factor inputs.

[49] Whether total income falls due to lost jobs or rises due to higher wages depends on the specific parameters in a country and sector.

Although the production function approach used on both types of models has superficial attraction, it was shown in the 1970s to have no empirical basis (F. M. Fisher (1971),Shaikh (1974)).

MINDSET has a hard constraint on employment levels, meaning that the unemployment rate cannot drop below a pre-defined bound, as discussed in the Section 4.4. Additionally, when this is the case, the model still keeps track of what would have been the unconstrained unemployment rate as it creates pressure on wages.

In the modern economy, constraints are typically felt at a detailed, sub-sectoral level. Here, the limitations in sectoral disaggregation make it difficult to track possible production bottlenecks (e.g., the microchip sector is subsumed within the wider electronics sector). Within the labour market, specialized skills are becoming increasingly important, in both low- and high-income countries. The current version of MINDSET does not differentiate labour by skill type, meaning potential bottlenecks are again missed. This part of the model is expected to be revised in future versions. However, the problem with the required level of disaggregation will remain outstanding.

8.2 The long term

MINDSET is determined from an input-output framework that has been traditionally applied for short-term analysis in which supply-side constraints may be neglected. The input-output approach complemented equilibrium-based approaches that were and still are applied for long-term analysis.

The development of MINDSET into a macro-econometric model, however, provides the foundation for long-term assessment. The previous section summarizes how supply-side constraints have been added to the demand-driven system. Long-run outcomes are determined by a combination of demand- and supply-side factors.

A critical distinction in MINDSET is that the eventual long-run outcomes are the result of path-dependent processes, in which temporary shocks can have permanent outcomes. Economists have long discussed hysteresis effects, notably related to labour markets (e.g., Bluedorn and Leigh (2019)), and recovery from a negative demand shock in MINDSET does not lead to employment returning to pre-shock levels, mimicking long-term scarring effects.

Path dependency has also become a critical issue in analysis of climate change and climate change policy. The central role of technological progress in meeting emission reduction targets puts emphasis on self-reinforcing, irreversible processes that may be directly influenced by policy (Nemet, 2019; Grubb, Poncia, and Pasqualino, 2024). Climate change itself can have irreversible economic impacts (Dell, Jones, and Olken, 2012).

To the degree possible, given available data and estimation techniques, MINDSET aims to capture these effects. The existing equations go some way to doing so. There is the potential to introduce further path dependency through using asymmetrical elasticities, which has been found to be an important feature of responses to changes in energy prices (Grubb, 1995). Once people replace inefficient equipment they typically do not go back to it.

Finally, links to technology dynamics are modelled through linkages to Future Technology Transformation (FTT) models, where increasing returns to scale and learning-by-doing drive diffusion (Arthur, 1989; Mercure et al., 2016). Importantly, these dynamics also induce endogenous changes in production recipes, or structural changes, with regard to energy and labour inputs as well as across trade partners.

The relationship between short- and long-run outcomes in MINDSET thus differs from that of most other models. MINDSET is less focused on how to reach long-term steady states and more focused on how short-term outcomes determine what the long-term outcomes may be. In the model, every outcome in a given year affects every outcome in the following year.

8.3 Limitations

As a macroeconomic model, MINDSET presents a simplified version of the real-world economy. These simplifications necessarily bring some limitations to MINDSET's use and its results. This section describes some of the most important limitations of MINDSET.

Level of disaggregation
While 120 sectors present a high level of disaggregation for a macroeconomic model, it still places limitations on detailed analysis, including for climate policy. For example, production of solar panels is not defined as a sector but is part of the broader electronic equipment sector. For most purposes, MINDSET thus uses the characteristics of the electronic equipment sector as a proxy for solar panels. This approach can lead to bias in results, for example, if a country produces solar panels but not other electronic equipment. Modelling trade in sectors that combine commoditized and non-commoditized products is difficult because part of the sector effectively has an infinite price elasticity (Balistreri and Markusen, 2009). Similarly, the model cannot produce a full set of results at sub-national level. Where sub-national estimates are provided for employment results, these are derived by sharing out national totals based on labour force survey data.

Data inputs
The available level of data is linked to the data inputs. As an empirical model, MINDSET is highly dependent on using high-quality data. The data determine the starting point for the analysis, the baseline projections, and the behavioural relationships. Data used in the parameter estimation is described in the separate Data Documentation. Prior to use in a specific country, these data should be reviewed.

Behavioural parameters
A critical assumption in the model is that the behavioural parameters are assumed to remain constant, both across time and between scenarios. The method for deriving behavioural parameters is described in Section 6.3 and the separate Estimation Documentation. This approach is common to most macroeconomic models, but is particularly relevant to MINDSET because the estimated parameters replace assumptions about optimizing behaviour. In reality, behavioural patterns are context-specific, and it is not possible to predict how they will change in the future (Hodgson, 2001). The issue was first identified in Veblen (1900). In the context of policy analysis, it is often referred to as the Lucas Critique (Lucas, 1976), which explicitly recognizes that the policies used to define scenarios may change the behaviour being modelled. These criticisms are valid and highlight the limitations of predicting complex systems. By holding behaviour fixed, the econometric approach likely leads to bias towards maintaining the status quo.

Financial sector
Within MINDSET, the financial sector is represented as it would operate in 'normal' times. Banks and other financial institutions advance loans when they see profitable opportunities. This approach is empirically accurate but neglects some of the important dynamics of the financial system. For example, in times of economic booms, lending for speculative asset purchases could inflate asset values, with feedback to consumption. In economic crashes, banks may be unwilling to lend at all because of liquidity constraints.

Transition to a zero-carbon economy is likely to include both cases, because uncertainty is high and expectations could change quickly. Such occurrences would amplify wider economic effects (both negative and positive), but are not captured by MINDSET.

Treatment of technology
Addressing climate change is largely a question of getting the right technologies into the right places. For climate change mitigation efforts, the FTT modelling that is linked to MINDSET places technology at the heart of the policy question. There is an underlying assumption that technological advances will be predictable over the forecast horizon to 2050. For established technologies, predicting future prices appears possible (Farmer and Lafond, 2016), but the possibility of new breakthrough technologies (e.g., nuclear fusion, carbon direct air capture) becoming commercially viable is much more difficult to capture. Without being able to predict the direction of future technology, the uncertainty about model results thus increases toward the end of the projection period.

Exchange-rate mechanism
MINDSET 2.0 currently lacks an explicit exchange rate module, which limits its ability to capture both short- and long-run exchange rate dynamics. In the short run, trade flows have only limited effects on exchange rates, which are instead primarily driven by financial markets, interest rate differentials, speculative capital flows, and expectations (Gyntelberg, Tientip, and Loretan, 2012; Menkhoff et al., 2012; Ferrari, Kearns, and Schrimpf, 2021). Over longer horizons, exchange rates are shaped by persistent trade imbalances and terms-of-trade shocks, with important implications for competitiveness, structural change, and phenomena such as Dutch Disease (Marconi et al., 2021; Marconi, 2012; Palma, 2005). Empirical studies confirm that long-run real exchange rates are systematically influenced by inflation, productivity, openness, and financial markets (Ricci, Milesi-Ferretti, and J. Lee, 2008; Chinn, 2006). An exchange rate module could either anchor real exchange rates to balance-of-payments fundamentals, treating short-run fluctuations as finance-driven, or model how exchange rate cycles affect trade elasticities and structural transformation.

Appendix

A.1 Collapsing the MRIO structure

To facilitate fast and computationally lean model runs, MINDSET can be executed on a collapsed structure focusing only on the country of interest and its main trading partners. Given that the original GLORIA MRIO matrix has dimensions of 19680 by 19680, which translates to close to 400 million cells, calculations can be quite computationally intensive. The model code mitigates performance issues arising from this in a number of ways, including parallelization of certain calculations as well as storing and re-using intermediate results where possible. Nevertheless, solutions on the full matrix could take hours or even days. Now, given that in most cases the full MRIO is not utilized (e.g., a scenario focusing on South Africa, might have to pay less attention to induced changes in Kazakhstan), the model is able to *'collapse'* countries into more aggregated regions.

This step is done before running the model, therefore reducing the model's dimensionality, but keeping the global representation.

The process first aggregates (summing) the MRIO and final demand components, followed by a re-calculation of the country-level parameters. This re-calculation is, in most cases, a weighted average calculation, either using GVA, GDP, or employment as a weight for the country parameters.

Variable	Aggregation method
Intermediate uses	Sum
Output	
Final demand	
Energy use	
Population projections	
Value added	
Employment	
Capital stock	
Behavioural parameters	Weighted average (GDP)
Depreciation	
Government spending shares	
Unemployment rate	
Minimum wage	
Labour force participation	

Table A.1: Aggregation methods by variable

A.2 Baseline and calibration

MINDSET is primarily an **impact assessment** model, in which policies or shocks are embedded within scenarios that are then simulated. The results from these simulations are compared to a *baseline*, typically representing a *business-*

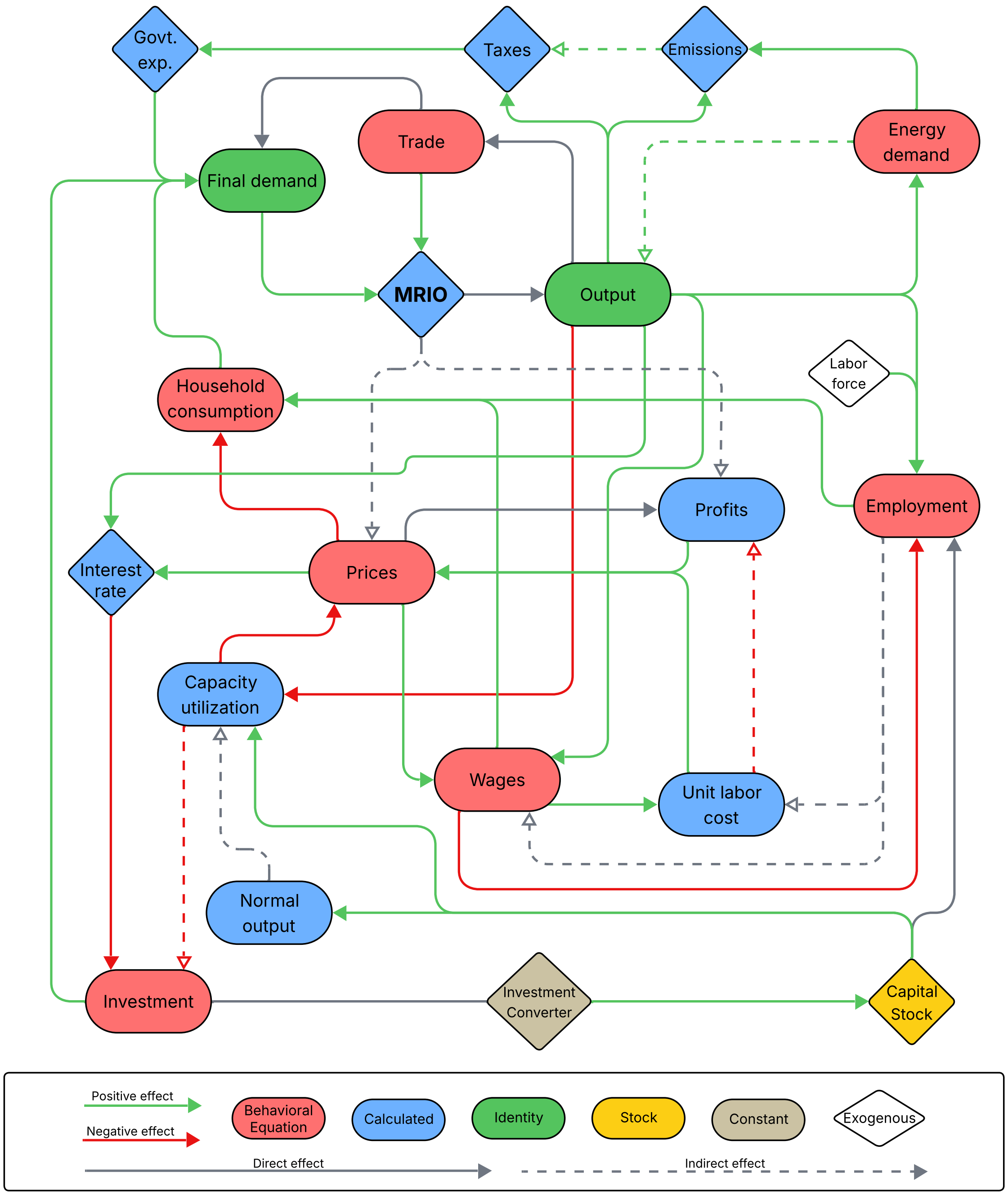

Figure A.1: Model flowchart

as-usual trajectory for the economies under consideration. This comparative approach enables the modeller to isolate the effects of specific policies or shocks from underlying baseline dynamics. Results are generally expressed in *relative terms*—that is, as deviations from the baseline—since the baseline itself evolves over time. The evolution of the baseline is determined during the process of *calibration*.

As MINDSET **is not a forecasting model**, it relies on external projections to define its baseline. In MINDSET, **calibration** refers to the process of reconciling differences between the model's *endogenous baseline* and an externally specified *long-run growth pathway*, typically by adjusting final demand in the endogenous baseline to align with the external projection, such as given by MFMOD (Burns et al., 2019). The demand adjustments required for this alignment are recorded as *residual demand*, which represents factors external to the model's endogenous mechanisms.

Calibration is conducted once at the outset of the modelling exercise through an iterative procedure. In *calibration mode*, the model simulates one year (an *annual solution*) and compares the resulting sectoral and aggregate outputs (gross value added, GVA) against the external projections. The difference—computed for each country and sector—is then propagated through the MRIO (multi-regional input–output) matrix to derive the *demand adjustment* necessary to close the output gap. In the next iteration, this adjustment is incorporated into final demand, and the model is recalculated. The process repeats until the difference between simulated output and the external projection falls below a specified tolerance (currently 2%). At that point, the demand adjustments are stored for use in subsequent simulations.

The influence of the baseline and calibration on model results is context-dependent. If calibration yields large *adjustment factors*—meaning a substantial share of a sector's final demand derives from residual adjustments—then the relative weight of endogenous dynamics in that sector's results is reduced. This is an intentional design feature: it reflects the understanding that certain demand developments in that sector cannot be captured by the model's endogenous structure and must instead be imposed externally.

A.3 List of outcome variables

This section outlines the set of outcome variables generated by the model. While the list presented here is comprehensive in terms of the model's current capabilities, additional variables and indicators may be incorporated as required for specific applications.

A.3.3 Outcomes by impact channels

Impact channels in the model represent distinct economic mechanisms through which policies or shocks influence socioeconomic outcomes. Many of the model's variables can be decomposed to show contributions from there channels, enabling a more granular attribution of effects.

For producer prices, the attribution structure differs from that of standard variables. Prices changes can be decomposed into *direct* and *indirect* effects. Direct effects arise from sector-specific dynamics, whereas indirect effects are driven by changes in the prices of intermediate inputs. Direct price changes can be further disaggregated into components attributable to changes in labour cost, output growth, deviation from expected output, deviation from profit rate and tax change.

Name	Base unit	Dimensions
Output & value added		
Economic output Gross value added (GVA)	'000 USD (2019 price)	Producing region x Producing sector x Year
Gross value added (GVA) Value added components	'000 USD (current price)	Producing region x Producing sector x Year
Profit rate Long-term profit rate	Percentage	Producing region x Producing sector x Year
Employment & wages		
Employment Average annual wage	Persons	Producing region x Producing sector x Year
National income Labour force	'000 USD (current)	Region x Year
Demand		
Final demand of households Final demand of government expenditures Final demand of fixed capital formation (FCF)	'000 USD (2019 price)	Purchasing region x Producing sector x Year
Price		
Price change Consumer price change (household)	Year-on-year percentage	Producing region x Producing sector x Year
Price index Consumer price index	2019 = 1.0	Region x Year
GDP deflator	Year-on-year percentage	Region x Year
Investment & interest		
Investment Capital stock	'000 USD (2019 price)	Investing region x Investing sector x Year
Nominal interest rate Real interest rate	Percentage	Region x Year
Trade		
Exports Imports	'000 USD (2019 price)	Region x Sector x Year x Consumption purpose
Energy & emissions		
Fuel consumption	TJ	Producing region x Demanding region x Purchasing sector x Fuel x Year
CO_2 emissions	$ktCO_2$	Emitting region x Emitting sector x Year

Table A.2: Standard outcome variables produced by the model

	Output	GVA	Employment	Demand	Investment	Trade
Fuel substitution (see Section 5.3.1)	X	X	X	X		X
Trade substitution (see Section 4.2)	X	X	X	X		X
Household price impact (see Equation 4.1.1)	X	X	X	X		X
Household income impact (see Section 4.1.1)	X	X	X	X		X
Government consumption change (see Section 4.1.3)	X	X	X	X		X
Exogenous final demand	X	X	X	X	X	X
Exogenous supply constraint	X	X	X			X
Induced investment effect (see Section 4.1.2)	X	X	X	X	X	X
Productivity change (see Section 4.4)			X			
Calibration residual (see Section A.2)	X	X	X	X		X
Labour supply constraint (see Section 4.4)			X			

Table A.3: Impact channels that can be explored by outcome variable

A.4 Constructing physical energy and GHG emission flows

This section documents the construction of a comprehensive MRIO-like database of physical energy and GHG emissions flows. The calculations are based on energy balances statistics published by the International Energy Agency

(IEA), United Nations Department of Economic and Social Affairs (UNDESA), and Emissions Database for Global Atmospheric Research (EDGAR) data (Crippa et al., 2023; UNDESA, 2025; IEA, 2024), as well as fuel-, sector- and country-specific emission factors from the Greenhouse Gas and Air Pollution Interactions and Synergies (GAINS) model (IIASA, 2020), complemented with GLORIA and COMTRADE data (Lenzen et al., 2017). It further outlines the treatment of emission intensities (CO_2/USD output) for process-related emissions in GLORIA industries. The full process, including codes and correspondence tables, is documented in detail as part of the Github repository[50]

A.4.4 Pre-processing Energy Balances

The IEA EB requires substantial pre-processing prior to its use in modelling applications. First, the raw IEA balance files are extremely large (approximately 6 GB uncompressed), primarily due to their time-series structure. Since this study focuses on a single reference year, we reduce the dataset to a cross-sectional format. Second, differences in geographical coverage between IEA and GLORIA necessitate harmonization. While IEA EB provides broad geographical coverage, certain countries are reported only within regional aggregates, whereas GLORIA requires explicit country-level representation. Accordingly, pre-processing involves both disaggregation from IEA aggregates and aggregation into GLORIA regions. Finally, to complement the IEA EB, we incorporate UNSD Energy Balance data. Although less detailed in terms of fuel and sector classifications, UNSD provides coverage for countries not individually reported by IEA.

The re-calculation of GLORIA aggregates requires combining the relative advantages of the two sources: (i) IEA provides greater detail on final consumption and fuel types, while (ii) UNSD offers more disaggregated geographical coverage. We therefore derive implied fuel flows at the country level using:

$$\hat{Y}_{f^{IEA},c,s^{IEA}} = Y_{f^{IEA},C^{IEA},s^{IEA}} \times \frac{y_{f^{UN},c,s^{UN}}}{\sum_c y_{f^{UN},c,s^{UN}}}, \tag{A.1}$$

where $\hat{Y}$ is the implied fuel use for fuel f^{IEA}, country c, and sector s^{IEA}; $Y_{f^{IEA},C^{IEA},s^{IEA}}$ is the IEA-reported fuel use for the aggregate region C^{IEA}; and the fraction represents the share of country c in the corresponding UNSD aggregate. Note that UN sector and fuel classifications are considerably less detailed than those of the IEA. Three correspondence tables are required: (1) IEA to UN country mapping, (2) IEA to UN sectors, and (3) IEA to UN fuels.

Subsequently, these implied flows are aggregated into GLORIA groups, subject to two rules: (i) calculated fuel flows are only used where IEA does not provide direct data, and (ii) in cases of multiple possible values, the mean is taken. Formally:

$$\hat{Y}_{f^{IEA},C^{GLORIA},s^{IEA}} = \sum_c f, \quad f = \begin{cases} Y_{f^{IEA},c,s^{IEA}}, & \text{if } c \text{ reported in IEA (i.e. } c = C^{IEA}) \\ \hat{Y}_{f^{IEA},c,s^{IEA}}, & \text{otherwise.} \end{cases} \tag{A.2}$$

A.4.4 Physical Fuel Flows

The construction of the physical energy flow database involves multiple steps. We begin by compiling cross-country and cross-sectoral flows of energy carriers in GLORIA sectoral classifications.

The first step is the construction of trade matrices. The IEA EB reports aggregate exports and imports but not bilateral flows. To estimate bilateral trade, we combine IEA totals with monetary trade data from GLORIA and COMTRADE.

[50] https://github.com/cpmodel/energy-IEA; access upon request

The resulting matrices are of size 163×163, representing bilateral country flows by fuel. Row and column totals are imposed from IEA EB, while the bilateral structure is inferred through the RAS algorithm:

$$r_j = \frac{\sum_i x^*_{i,j}}{X_j} \tag{A.3}$$

$$\hat{x}_{i,j} = \frac{x^*_{i,j}}{r_j} \tag{A.4}$$

$$c_i = \frac{\sum_j \hat{x}_{i,j}}{Y_i} \tag{A.5}$$

$$x'_{i,j} = \frac{\hat{x}_{i,j}}{c_i} \tag{A.6}$$

Iteration continues until the adjusted matrix $x'_{i,j}$ satisfies both $\sum_j x'_{i,j} \approx Y_i$ and $\sum_i x'_{i,j} \approx X_j$ within a tolerance of 15%.[51]

Next, we align physical flows with GLORIA sectors and monetary flows by linking IEA activities to GLORIA sectors. For flows associated with multiple sectors, we distribute values according to monetary input shares from GLORIA. Road transport fuel use is further disaggregated into residential and commercial uses (64% and 36%, respectively) following IEA Energy Efficiency Indicators[52]. Non-specified industry fuel use (INONSPEC) is redistributed across industries where reporting shares exceed global averages.

Finally, import flows are disaggregated bilaterally using the RAS-derived trade shares:

$$Y^*_{i,j,k,l,f} = Y^*_{i,z,k,l,f} \times \frac{x'_{i,j,f}}{\sum_j x'_{i,j,f}} \tag{A.7}$$

This yields the complete database of bilateral physical energy flows, expressed in terajoules (TJ).

A.4.4 Energy-borne GHG emission flows

Emission flows are calculated directly from the physical fuel use database using GAINS emission factors (IIASA, 2020), or IPCC default factors where GAINS is unavailable:

$$\text{emission}_{i,j,k,l,f} = \frac{\text{IEF}_{i,k,f}}{1000} \times 1000 \times Y^*_{i,j,k,l,f} \tag{A.8}$$

where IEF denotes the emission factor (kton/PJ). For fuels missing in GAINS, scaling factors based on IPCC defaults are applied, using anchor fuels such as coal to derive multipliers for related fuel types.

[51] This procedure is applied separately to each of the more than 50 fuels.

[52] https://www.iea.org/data-and-statistics/data-product/energy-efficiency-indicators-highlights

A.4.4 Process Emissions

Process emissions are treated separately to ensure full coverage of CO_2 emissions from industrial activity. Unlike combustion emissions, these are represented as fixed emission intensities (CO_2 per USD of output) at the sector level rather than at the flow level.

Intensities are computed as:

$$\beta_{i,k} = \frac{\text{emission}_{i,k}}{Q_{i,k}} \tag{A.9}$$

where $\text{emission}_{i,k}$ is EDGAR-reported process emissions for country i and sector k, and $Q_{i,k}$ is the corresponding GLORIA gross output (both for 2019). The resulting units are $ktCO_2$ per thousand USD of output.

A.4.4 Data Sources

- IEA Energy Balances 2023 edition (data for 2019) (IEA, 2024)
- UNSD Energy Balances (country-level data, 2019) (UNDESA, 2025)[53]
- GLORIA (data for 2019) and COMTRADE (2019)[54]
- GAINS emission factors (IIASA, 2020)[55] and IPCC default factors[56]
- EDGAR dataset for process emissions (Crippa et al., 2023)[57]

A.4.4 Correspondence Tables

- IEA products $\rightarrow$ COMTRADE (HS2017) $\rightarrow$ GLORIA sectors
- IEA flows $\rightarrow$ GLORIA sectors
- GAINS sectors $\rightarrow$ GLORIA sectors
- GAINS emission factors $\rightarrow$ IEA products
- UNSD products $\rightarrow$ IEA products

A.5 FTT:Power-to-MINDSET Integration Documentation

The transition of the global electricity sector toward low-carbon technologies is a central challenge in addressing climate change. Understanding how energy technologies spread through markets, how policies shape investment decisions, and how innovation dynamics influence long-term outcomes requires models that go beyond static cost comparisons. The Future Technology Transformations: Power (FTT:Power) model is one such tool designed to capture the complex dynamics of technological diffusion in the electricity sector (Mercure, 2012a).

[53] https://unstats.un.org/unsd/energystats/dataPortal/
[54] https://wits.worldbank.org/
[55] https://pure.iiasa.ac.at/id/eprint/17552/
[56] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf
[57] https://data.jrc.ec.europa.eu/dataset/b54d8149-2864-4fb9-96b9-5fd3a020c224

FTT:Power is a global, bottom-up model of electricity generation that explicitly represents competition between technologies. Rather than assuming that investment decisions are driven solely by least-cost optimization, it uses concepts from evolutionary economics and innovation diffusion theory to simulate how technologies gain or lose market share over time (Mercure, 2015). The model accounts for factors such as learning-by-doing, investor heterogeneity, policy interventions, and system constraints, offering a more realistic depiction of how electricity generation shares might evolve under different scenarios (Mercure et al., 2014; Mercure et al., 2018b). As explained by Mercure and Salas, 2013, it draws on innovation theory and employs a predator–prey framework to describe competitive dynamics, with the pace and direction of innovation shaped by both economic development and policy interventions. Overall, FTT:Power offers a non-equilibrium perspective on the energy transition and the role of policies in accelerating decarbonization.

The model has been applied to assess global decarbonisation pathways (Mercure et al., 2014) and resource depletion effects on energy prices (Mercure and Salas, 2013). These applications demonstrate that technology transitions are non-linear, sensitive to early policy action, and highly path-dependent. A simplification of the FTT:Power was introduced by the World Bank, FTT-FLEX, which is suitable for application as a single-country tool (Chewpreecha, Dennig, and Hansen, 2024).

To capture the rapid technological transformation of the electricity sector and its macroeconomic and distributional outcomes within MINDSET, the technology-oriented FTT:Power model can be integrated. While the econometrically driven equations in MINDSET are suitable for capturing macroeconomic relationships, they are less capable of representing rapid technological shifts in the electricity sector. In contrast, FTT:Power is specifically designed to simulate the dynamics of technological diffusion. This integration also expands the range of policy instruments that MINDSET can address, for example, subsidies for solar photovoltaic technology deployment. Moreover, by incorporating electricity generation at the technology level, MINDSET can more accurately reflect the supply chain impacts of the energy transition. Since the intermediate input structures differ significantly between technologies (e.g., coal versus gas power plants), incorporating the detailed generation mix from FTT:Power enhances the accuracy of the economic impact assessment of the energy transition.

The implementation of FTT:Power requires a broad set of data inputs to capture both the technological and economic dimensions of the electricity sector. The model relies on parameters such as capital costs, operation and maintenance costs, fuel prices, efficiencies, capacity factors, and lifetimes of generation technologies that is differentiated across regions and technologies (Mercure, 2012a). Therefore, to tailor the model to MINDSET, new regions were introduced and the technological representation was simplified. It also required aligning the two models along key economic and energy variables to ensure that their core dynamics are fully consistent. The main inputs from MINDSET to FTT:Power include:

1. Electricity demand - MINDSET estimates the electricity demand at sectoral level, while in FTT:Power electricity demand is given exogenously.
2. Carbon prices - in both MINDSET and FTT:Power carbon prices are defined exogenously, so the alignment of the two is essential.
3. Fuel prices - without the connection to a macroeconomic model, fuel prices, which are endogenously estimated in MINDSET, that feed into the operation costs of generation technologies are constant in FTT:Power..

The aim of integrating FTT:Power into MINDSET is to provide a more accurate representation of the electricity sector and thereby improve the precision of MINDSET's estimates. The main outputs of FTT:Power integrated into MINDSET include:

1. Electricity generation and capacity, or investments, by technology
2. Electricity sector investment by technology
3. Electricity sector employment by technology

The remainder of this documentation is structured as follows. The next section provides a general overview of FTT:Power, then Section A.5.5 discusseds how FTT:Power is reclassified by region and technology, while Section A.5.5 describes the linkages between the two models.

A.5.5 General Overview of FTT:Power

Within the FTT framework, investors choose among competing technologies to build new capacity, either to replace ageing infrastructure or to meet changing demand (Mercure, 2012a). Investment decisions are based on pairwise comparisons of levelised costs, conceptually similar to a binary logit model. The model accounts for investment, energy, and policy costs, which vary across locations and investors. FTT assumes that investors' perceptions of costs follow a normal distribution with a given standard deviation. Technology diffusion follows Lotka–Volterra (replicator) dynamics (Bomze, 1983), capturing market competition, investor preferences, and replacement rates. The model reflects path dependency, suboptimal decision-making, and non-marginal responses to external influences, producing the characteristic S-shaped diffusion curve seen historically. Variations in consumer preferences and technology costs determine how market shares shift between technologies as investors gradually replace their existing capacity.

The model accounts for both investor and dispatcher perspectives. Investor behaviour follows the replicator dynamics and is primarily driven by cost minimisation, while dispatchers operate technologies according to the merit order and maintain grid stability. The interactions between these two groups are interdependent—investment choices affect dispatch operations, and dispatch outcomes, in turn, influence future investments.

Systemic constraints also play a key role in shaping diffusion, particularly for variable renewable energy (VRE) sources such as solar PV and wind, which require backup or storage capacity. To capture these effects, FTT:Power integrates residual load-duration curves (RLDCs) (Ueckerdt et al., 2017), parametrised by the shares of wind and solar generation. The RLDC determines the residual load met by non-VRE technologies, along with resulting load factors, curtailment levels, and short-term storage needs. Separately, long-term storage is linked to the installed capacity of generation technologies and their utilisation rates.

FTT:Power models a portfolio of generation technologies—such as coal, gas, nuclear, hydro, wind, and solar—each characterized by its levelised cost of electricity (LCOE), learning rate, and resource availability. Investors are assumed to make choices probabilistically among technologies according to their perceived costs. The share S_i of technology i evolves according to a system of replicator dynamics equations derived from discrete choice theory:

$$\frac{dS_k}{dt} = \sum_l S_k S_l \left(F_{kl} - F_{lk} \right), \tag{A.10}$$

where F_{kl} represents the probability that investors switch from technology l to k, defined by a cost-based binary logit function (Mercure, 2012a):

$$F_{kl} = \frac{1}{1 + \exp\left(\frac{C_k - C_l}{\sigma}\right)}, \tag{A.11}$$

with C_k and C_l denoting the LCOE of technologies k and l, and σ representing investor heterogeneity in cost perception. These coupled non-linear equations yield logistic substitution patterns similar to empirically observed technology diffusion (J. C. Fisher and Pry, 1971).

Technological learning is incorporated through experience curves, in which the investment cost $C_k(t)$ of each technology declines with cumulative installed capacity $K_k(t)$:

$$C_k(t) = C_{k0} \left(\frac{K_k(t)}{K_{k0}} \right)^{-\lambda_k}, \tag{A.12}$$

where λ_k is the learning exponent derived from the learning rate $LR_k = 1 - 2^{-\lambda_k}$. This mechanism produces induced technological change: as deployment accumulates, costs fall, enhancing competitiveness and accelerating adoption (Mercure and Salas, 2013).

Natural resource constraints are captured through cost–supply curves, which relate the cumulative use of a resource to its marginal cost. As low-cost resources are exhausted, the marginal cost of extraction or exploitation rises, linking physical limits to economic evolution. This allows renewable potentials (e.g., wind, hydro) and fossil depletion to shape technology competitiveness dynamically (Mercure and Salas, 2013).

Policy instruments enter the model as modifiers of technology costs or market shares. Carbon pricing, subsidies, feed-in tariffs, and technology bans can be simulated to examine their impact on diffusion trajectories.

A.5.5 Reclassification of FTT:Power

A.5.5.1 Regional classification

Both MINDSET and FTT:Power have a highly disaggregated representation of regions. However, FTT:Power has 71 regions and MINDSET has 163. To facilitate the model integration, the FTT:Power regions, therefore, need further disaggregation.

The regional mapping from FTT:Power to MINDSET is straightforward for overlapping regions where both models include the same countries, such as EU countries. In total, 57 regions are represented in both models. The remaining MINDSET regions, which cannot be directly matched, are mapped to their corresponding FTT aggregated region or to a region with similar characteristics. This simplification is necessary to expand the regional coverage of FTT:Power and results in identical technological parameters for regions mapped to the same FTT region. However, the regional differences are generally minor among countries within the same continent and with same level of development. Furthermore, electricity generation data by technology are sourced from OWID for all 163 MINDSET regions, using historical data that correspond precisely to the MINDSET regional definitions.

Based on this mapping, the default approach generates all variables for MINDSET regions by replicating the corresponding values from their matched FTT region. However, variables that must remain region specific, such as electricity generation by technology and total electricity demand, are overwritten using historical data or inputs directly provided by MINDSET. This approach is consistent with the general operation of MINDSET, which, although it includes 163 regions, is typically run in a collapsed structure. In this setup, countries are aggregated into custom-defined regions. However, the technological parameters should be reviewed following the collapse and adjusted for the custom regions where necessary.

A.5.5.2 Technological classification

The main variable of FTT:Power is the historical generation that defines the starting point of the diffusion. Historical electricity generation data (in MWh) by country and year are sourced from Ember and Energy Institute, 2026 which has almost identical regional coverage as MINDSET. In case of the few regions that are not covered by Ember and Energy Institute, 2026, e.g., Singapore, data is sourced from their corresponding national websites.

However, Ember and Energy Institute, 2026 only provides data at a higher aggregation level than the technologies in FTT:Power Standalone. Therefore, the generation technologies used in the standalone model are aggregated following the mapping shown in Table A.4. Most variables in the integrated MINDSET–FTT:Power model are calculated as weighted averages of the original FTT:Power technology-specific values, using electricity generation as the weighting factor. This approach applies to variables expressed in relative or intensity terms, such as investment costs for new capacity (USD per kW). In contrast, variables that represent absolute quantities, such as primary energy demand, are aggregated as the sum of the corresponding technology-level values.

Although the Ember and Energy Institute, 2026 dataset does not include information on generation technologies equipped with carbon capture and storage (CCS), these technologies are nevertheless represented in the model. They are introduced with zero historical generation, ensuring that their potential future deployment can be captured once relevant policy incentives or cost reductions are introduced. Including CCS technologies in the model allows for the exploration of decarbonization pathways in which CCS becomes a viable mitigation option.

	FTT for MINDSET Technologies	FTT:Power Technologies
1	Nuclear	Nuclear
2	Oil	Oil
3	Coal	Coal, Integrated gasification combined cycle (IGCC)
4	Coal + CCS	Coal + CCS, Integrated gasification combined cycle (IGCC) + CCS
5	Gas	Combined-cycle gas turbine (CCGT)
6	Gas + CCS	Combined-cycle gas turbine (CCGT) + CCS
7	Bioenergy	BIGCC[58], Solid Biomass, Biogas
8	Bioenergy + CCS	BIGCC + CCS, Solid Biomass + CCS, Biogas + CCS
9	Hydro	Tidal, Large Hydro, Wave
10	Wind	Onshore, Offshore
11	Solar	Solar PV, Concentrated solar power (CSP)
12	Other Renewable	Geothermal, Fuel Cells, Combined heat and power (CHP)

Table A.4: Mapping of FTT:Power technologies

A.5.5 Linkages between FTT:Power and MINDSET

This section describes the integration of FTT:Power with MINDSET, detailing the bidirectional exchange of information between the two models. A schematic overview of the FTT:Power integration within MINDSET is presented in Figure A.2. The linkages between FTT:Power and MINDSET follow a similar structure to the *FTT:Power–E3ME* integration, in which key macroeconomic feedbacks are endogenised: investment, employment, energy demand, and prices co-evolve with the electricity system (Mercure et al., 2014; Mercure et al., 2018a).

The following sections describe in detail how each of the linkages is implemented, beginning with the inputs provided

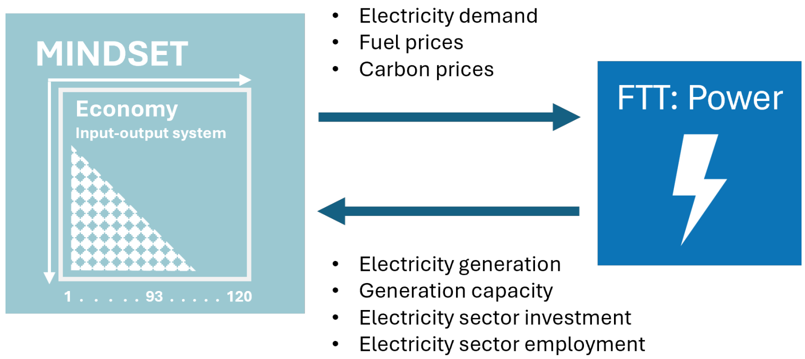

Figure A.2: High-level overview of linkages between MINDSET and FTT:Power

by MINDSET to FTT:Power, and subsequently those transferred from FTT:Power back to MINDSET.

A.5.5.1 Inputs from MINDSET to FTT:Power

This section describes the information transferred from MINDSET to FTT:Power, focusing on how electricity demand, and carbon and fuel prices are harmonized between the two models to ensure consistent scenario representation.

Electricity Demand
Electricity demand represents one of the key drivers of technological change in FTT:Power. Increasing demand necessitates the expansion of generation capacity, which in turn facilitates the diffusion of new technologies. Within MINDSET, electricity demand is endogenously estimated at the sectoral level for each region, while in the standalone version of FTT:Power it is defined exogenously. In the integrated framework, the sectoral electricity demand estimated by MINDSET is aggregated to the regional level and subsequently used as an input to FTT:Power. This input serves as the basis for determining the regional electricity generation mix.

Carbon Prices
Carbon prices are a central parameter for exploring mitigation scenarios in MINDSET. To ensure consistency between the two models, alignment on carbon pricing assumptions is essential. In both MINDSET and FTT:Power, carbon price scenarios are defined exogenously, but at different levels of aggregation. In MINDSET, carbon prices are specified by country, emitting sector (such as iron and steel), and fuel type (such as coal), whereas in the standalone FTT:Power, policies are not differentiated by fuel type.

To facilitate integration, carbon prices in FTT:Power is extended with one additional dimension, generation technology, allowing the assignment of distinct carbon prices to different fuel types, consistent with MINDSET.

In the integrated setup, fuel-specific carbon price scenarios targeting the electricity sector from MINDSET are directly used as inputs to FTT:Power. These price changes are incorporated into the carbon cost component of the Levelised Cost of Electricity (LCOE) calculation, which forms the basis for technology competition within FTT:Power:

$$LCOE_k(t) = \frac{\sum_{t=0}^{\tau_k} \frac{TI_k(t)+OM_k(t)+FC_k(t)+CC_k(t)}{(1+r)^t}}{\sum_{t=0}^{\tau_k} \frac{EP_k(t)}{(1+r)^t}}, \quad \text{(A.13)}$$

where TI_k denotes the specific technology investment cost, OM_k the operation and maintenance costs, FC_k the fuel costs, CC_k the carbon cost component associated with emissions allowances or taxes where applicable, r the technology or region specific discount rate and EP_k is the energy that the power station is expected to produce.

Fuel Prices

Analogous to carbon prices, fuel prices significantly influence the evolution of the generation mix in FTT:Power. Fuel price dynamics in MINDSET are derived from relevant upstream sectors (e.g., the gas extraction sector for natural gas). For each region, the fuel price is calculated as a weighted average of (i) the fuel price changes in exporting regions and (ii) the domestic fuel price changes, with sectoral trade flows used as weights. The resulting weighted average regional fuel prices are then provided as inputs to FTT:Power.

$$\theta_{i,j,m,n,t} = \frac{e_{i,j,m,n,t}}{\sum_j e_{i,j,m,n,t}} \quad \text{(A.14)}$$

$$\delta p^*_{i,j,m,n,t} = \frac{1+\delta p_{j,n,t}}{1+\delta P^{GDP}_{i,t-}} - 1 \quad \text{(A.15)}$$

$$\delta P^F_{i,m,n,t} = \sum_j \delta p^*_{i,j,m,n,t} \times \theta_{i,j,m,n,t} \quad \text{(A.16)}$$

where $e_{i,j,m,n,t}$ is the use of fuel n originating from country j in the production of sector m in country i in period t, $\theta_{i,j,m,n,t}$ therefore is the share of fuel input from country j in the total fuel n use of sector m, i. $\delta p^*_{i,j,m,n,t-1}$ is the GDP deflator ($\delta P^{GDP}_{i,t-1}$) adjusted price change of the fuel flow in country i (adjusted with the GDP deflator of the importing country i), such that $\delta P^F_{i,m,t-1}$ is the total price change of fuel n in country i for sector m. In the case of FTT:Power, sector m is the electricity sector.

These fuel prices serve as inputs to the fuel cost component of the LCOE calculation within FTT:Power, thereby affecting technology competitiveness and the resulting generation mix.

A.5.5 Inputs from FTT:Power to MINDSET

This section outlines the outputs of FTT:Power that are transferred to MINDSET, capturing how the technological and investment dynamics of the power sector influence the broader energy-economy system.

Electricity Generation and Capacity

The primary outputs of FTT:Power are electricity generation (MWh) and installed capacity (MW) by technology. The generation and capacity outputs from FTT:Power are used in MINDSET to estimate the supply chain impacts of the evolving electricity generation mix. A technology-specific input matrix is used to determine the inputs required for operating the various electricity generation technologies. The total output value of the electricity sector is first distributed across technologies according to the generation mix, after which the intermediate demand is estimated using the technology-specific input matrix.

$$\phi_{i,a,t} = \frac{g_{i,a,t}}{\sum_a g_{i,a,t}} \tag{A.17}$$

$$Q_{i,n,a,t} = \sum_a Q_{i,n,t} \times \phi_{i,a,t}, \tag{A.18}$$

where $g_{i,a,t}$ is the electricity generation of technology a in country i, and $\phi_{i,a,t}$ is the generation share of technology a. $Q_{i,n,t}$, where n is the electricity sector and $Q_{i,n,a,t}$ is the output of technology a in the electricity sector.

Electricity Sector Investment

The transition towards new technologies in the electricity sector not only alters intermediate demand but also entails changing investment requirements. The capital expenditures (in million USD) associated with building new generation capacities are estimated within FTT:Power for each generation technology and passed to MINDSET.

Then, the investment volumes by technology are allocated across the industries supplying capital goods to each technology-specific investment (e.g., construction, metals). Capital inputs are allocated across all sectors of the economy using a technology-specific investment converter. Through this converter, investments are translated into final demand for capital goods. The resulting investment values are subsequently used to assess the broader macroeconomic and sectoral impacts of the transformation of the electricity sector.

Electricity Sector Employment

Finally, the *direct* employment impacts associated with technological changes in the electricity sector are assessed within FTT:Power and subsequently transferred to MINDSET. FTT:Power estimates employment requirements by combining the direct labour intensity of each generation technology with the corresponding generation volumes. Direct labour intensities are applied, reflecting only the labour associated with the operation of generation capacities, in order to avoid double counting of employment impacts within the model.

The resulting year-on-year growth rates of direct employment in the electricity sector are then passed to MINDSET, where they are applied as employment growth rates for the sector. Only the growth rates, rather than the absolute employment values, are transferred. This approach ensures consistency with regional labour characteristics - such as differing levels of labour intensity - while still capturing the effects of technology-driven structural changes in the electricity sector.

A.6 Construction of the sectoral capital stock

MINDSET requires a country- and sector-specific measure of reproducible capital stock that is consistent with the GLORIA database and with the investment flows used in the model. Official capital stock statistics at this sector and country resolution are not provided in standard System of National Accounts (SNA) publications or comparable international data platforms and are not part of the GLORIA database. The capital stock is therefore inferred from observed gross fixed capital formation from GLORIA, institutional investment shares, and auxiliary evidence from KLEMS-type datasets (Bontadini et al., 2023). The outcome is an internally consistent accounting construct aligned with GLORIA rather than an official national capital stock series. It should not be used for estimations.

Data basis. Because the FCF vector identifies the producer of the investment good but *not* the investor, the central task is to transform investment by commodity into investment *by using sector*, consistent with GLORIA. The starting point is the GLORIA fixed capital formation (FCF) vector for the base year. The GLORIA output vector for the base year is later used to scale sectoral asset needs to maintain consistency with the GLORIA data. SNA tables on investment by institutional sector provide household and government shares where available. KLEMS provides, for a subset of economies, capital stock or investment by asset type and by using industry.

Step 1: FCF base year vector and asset groups. The original GLORIA FCF vector is defined over the importing and exporting countries and the traded commodity. Since the objective is to identify the investing sectors within the importing country, the exporting country dimension is dropped. This two-dimensional vector represents, for each country, the demand for investment goods by the sector that produces the capital good. Investment goods are then partitioned into broad asset categories that can be associated with specific producing sectors: (1) buildings and land, produced mainly by construction, (2) machinery, (3) vehicles, (4) information and communication technology (ICT) equipment, and (5) specialized or sector-specific equipment. Assigning sectors identifies the set of producing sectors that can supply each asset group.

Step 2: Allocation to households. Households are assumed to invest only in dwellings and thus only in construction-type assets. For the countries with SNA data, household investment shares are taken directly. For the remaining countries, the median of observed household shares is applied to construction FCF, producing a fixed split of construction investment between households, government, and industry. This step removes household investment from the construction FCF and records it as final household capital formation. Because the household share is applied only to construction, household investment in non-construction assets is underestimated, which is accepted to avoid inconsistencies in later steps.

Step 3: Allocation to government. Government is assumed to purchase only cross-sectoral assets, that is, assets produced in one sector and usable in many others (buildings, vehicles, ICT, and machinery). For these assets, the government share identified in the SNA data is subtracted proportionally from the corresponding investment goods in the GLORIA FCF vector. Government is not assigned specialised sector-specific investment goods. This step generates, for each country and investment asset, a residual FCF that is attributable to industries.

Step 4: Allocation to industries using KLEMS-based ratios. The residual industry FCF is distributed across investing industries with the help of KLEMS (Bontadini et al., 2023). KLEMS data are used to compute capital-to-output ratios by asset group and by industry (for example, buildings to output, machinery to output, ICT to output, and vehicles to output). These ratios are mapped from the original KLEMS industry classification to the GLORIA sectors. Applying these ratios to the GLORIA sectoral output vector yields implied capital needs by sector and asset group. These implied capital needs provide allocation weights that split the residual industry FCF across all using sectors that demand a given asset group. Because the allocation is done separately for each asset group, sectors that are ICT-intensive or machinery-intensive in KLEMS receive a larger share of the corresponding investment goods in GLORIA.

Step 5: Construction of the final investment converter. The allocation yields a three-dimensional matrix indexed by investing country, investment good (producing sector of the asset) and *investing* industry (or using sector). This matrix can be interpreted as an investment converter that transforms sectoral investment demand into demand for specific investment goods and, conversely, allows investment by asset group to be re expressed as investment by sector. The diagonal entries capture own-sector specialised investment that is not purchased by other sectors. This converter is the dataset that enters the MINDSET baseline and that is used to update sectoral capital stocks when investment is projected forward.

Derivation of the capital stock. Given the investment by sector and asset group, and assuming a base year in which observed or constructed investment is consistent with a steady capital-output relation, a baseline capital stock by sector is obtained by applying the asset-specific capital-to-output ratios discussed above. Since the ratios were already used as allocation weights, the resulting capital stock is necessarily consistent with the investment distribution. The model thus obtains, for each country and GLORIA sector, a capital stock level that is aligned with the base year's output and investment patterns, and that can be subjected to exogenous capital stock shocks in climate impact and disaster scenarios.

Assumptions and limitations. The procedure assumes (i) that the institutional composition of investment (households, government, corporations) observed in SNA investment by institutional sector tables for some countries can be applied to the GLORIA investment vector of countries without such tables, (ii) that household investment consists only of construction assets, (iii) that government investment is concentrated in cross-sectoral assets such as buildings, vehicles, machinery and ICT equipment, and (iv) that capital-to-output ratios by asset type and industry observed in KLEMS can be mapped to GLORIA sectors and treated as broadly stable across countries for the purpose of disaggregation. Country-specific public investment programs, and informal capital formation are not captured.

A.7 Model default dimensions

In its default, MINDSET follows the structure of the GLORIA MRIO database (Lenzen et al., 2017; Lenzen et al., 2021), although as discussed above, collapsing the model structure enables reducing dimensions (see Section A.1). The tables below present the default, GLORIA-based structure for regions (Section A.7) and for economic sectors (Section A.7).

GLORIA code	ISO alpha-3	Region name	GLORIA code	ISO alpha-3	Region name	GLORIA code	ISO alpha-3	Region name	GLORIA code	ISO alpha-3	Region name
1	XAM	Rest of Americas[1]	51	EGY	Egypt, Arab Rep.	101	MKD	North Macedonia	151	TUN	Tunisia
2	XEU	Rest of Europe[2]	52	ERI	Eritrea	102	MLI	Mali	152	TUR	Türkiye
3	XAF	Rest of Africa[3]	53	ESP	Spain	103	MLT	Malta	153	TZA	Tanzania
4	XAS	Rest of Asia-Pacific[4]	54	EST	Estonia	104	MMR	Myanmar[9]	154	UGA	Uganda
5	AFG	Afghanistan	55	ETH	Ethiopia	105	MNG	Mongolia	155	UKR	Ukraine
6	AGO	Angola	56	FIN	Finland	106	MOZ	Mozambique	156	URY	Uruguay
7	ALB	Albania	57	FRA	France	107	MRT	Mauritania	157	USA	United States
8	ARE	United Arab Emirates	58	GAB	Gabon	108	MWI	Malawi	158	UZB	Uzbekistan
9	ARG	Argentina	59	GBR	United Kingdom	109	MYS	Malaysia	159	VEN	Venezuela, RB
10	ARM	Armenia	60	GEO	Georgia	110	NAM	Namibia	160	VNM	Viet Nam
11	AUS	Australia	61	GHA	Ghana	111	NER	Niger	161	YEM	Yemen, Rep.
12	AUT	Austria	62	GIN	Guinea	112	NGA	Nigeria	162	ZAF	South Africa
13	AZE	Azerbaijan	63	GMB	Gambia	113	NIC	Nicaragua	163	ZMB	Zambia
14	BDI	Burundi	64	GNQ	Equatorial Guinea	114	NLD	Netherlands	164	ZWE	Zimbabwe
15	BEL	Belgium	65	GRC	Greece	115	NOR	Norway			
16	BEN	Benin	66	GTM	Guatemala	116	NPL	Nepal			
17	BFA	Burkina Faso	67	HND	Honduras	117	NZL	New Zealand			
18	BGD	Bangladesh	68	HKG	Hong Kong SAR, China	118	OMN	Oman			
19	BGR	Bulgaria	69	HRV	Croatia	119	PAK	Pakistan			
20	BHR	Bahrain	70	HTI	Haiti	120	PSE	West Bank and Gaza			
21	BHS	Bahamas, The	71	HUN	Hungary	121	PAN	Panama			
22	BIH	Bosnia and Herzegovina	72	IDN	Indonesia	122	PER	Peru			
23	BLR	Belarus	73	IND	India	123	PHL	Philippines			
24	BLZ	Belize	74	IRL	Ireland	124	PNG	Papua New Guinea			
25	BOL	Bolivia	75	IRN	Iran, Islamic Rep.	125	POL	Poland			
26	BRA	Brazil	76	IRQ	Iraq	126	PRK	Korea, Dem. People's Rep.[10]			
27	BRN	Brunei Darussalam	77	ISL	Iceland	127	PRT	Portugal			
28	BTN	Bhutan	78	ISR	Israel	128	PRY	Paraguay			
29	BWA	Botswana	79	ITA	Italy	129	QAT	Qatar			
30	CAF	Central African Republic	80	JAM	Jamaica	130	ROU	Romania			
31	CAN	Canada	81	JOR	Jordan	131	RUS	Russian Federation			
32	CHE	Switzerland	82	JPN	Japan	132	RWA	Rwanda			
33	CHL	Chile	83	KAZ	Kazakhstan	133	SAU	Saudi Arabia			
34	CHN	China	84	KEN	Kenya	134	SDS	South Sudan			
35	CIV	Côte d'Ivoire	85	KGZ	Kyrgyzstan	135	SEN	Senegal			
36	CMR	Cameroon	86	KHM	Cambodia	136	SGP	Singapore			
37	COD	Democratic Republic of the Congo[5]	87	KOR	Korea, Rep.[7]	137	SLE	Sierra Leone			
38	COG	Republic of the Congo	88	KWT	Kuwait	138	SLV	El Salvador			
39	COL	Colombia	89	LAO	Lao PDR[8]	139	SOM	Somalia			
40	CRI	Costa Rica	90	LBN	Lebanon	140	SRB	Serbia			
41	CUB	Cuba	91	LBR	Liberia	141	SDN	Sudan			
42	CYP	Cyprus	92	LBY	Libya	142	SVK	Slovak Republic[11]			
43	CZE	Czechia[6]	93	LKA	Sri Lanka	143	SVN	Slovenia			
44	DEU	Germany	94	LTU	Lithuania	144	SWE	Sweden			
45	DJI	Djibouti	95	LUX	Luxembourg	145	SYR	Syrian Arab Republic			
47	DNK	Denmark	96	LVA	Latvia	146	TCD	Chad			
48	DOM	Dominican Republic	97	MAR	Morocco	147	TGO	Togo			
49	DZA	Algeria	98	MDA	Moldova	148	THA	Thailand			
50	ECU	Ecuador	99	MDG	Madagascar	149	TJK	Tajikistan			
			100	MEX	Mexico	150	TKM	Turkmenistan			

Notes:

1 - includes: Anguilla, Antigua and Barbuda, Aruba, Barbados, Bermuda, British Virgin Islands, Cayman Islands, Curacao, Dominica, Grenada, Montserrat, Puerto Rico, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Sint Maarten (Dutch part), Suriname, Trinidad and Tobago, Turks and Caicos Islands

2 - includes: Andorra, Greenland, Guyana, Liechtenstein, Monaco, San Marino

3 - includes: Cabo Verde, Comoros, Guinea-Bissau, Lesotho, Mauritius, Sao Tome and Principe, Seychelles, Eswatini

4 - includes: Cook Islands, Fiji, French Polynesia, Kiribati, Maldives, Marshall Islands, Micronesia, Federated States of, Nauru, New Caledonia, Palau, Samoa, Solomon Islands, Taiwan (China), Timor-Leste, Tonga, Tuvalu, Vanuatu

5 - formerly Zaire

6 - also known as Czech Republic

7 - also known as South Korea

8 - also known as Laos

9 - formerly Burma

10 - also known as North Korea

11 - also known as Slovakia

Figure A.7: Country list, page 74

GLORIA code	ISIC Rev.4 code[1]	Sector name
1	A	Growing wheat
2	A	Growing maize
3	A	Growing cereals n.e.c
4	A	Growing leguminous crops and oil seeds
5	A	Growing rice
6	A	Growing vegetables, roots, tubers
7	A	Growing sugar beet and cane
8	A	Growing tobacco
9	A	Growing fibre crops
10	A	Growing crops n.e.c.
11	A	Growing grapes
12	A	Growing fruits and nuts
13	A	Growing beverage crops (coffee, tea etc)
14	A	Growing spices, aromatic, drug and pharmaceutical cr
15	A	Seeds and plant propagation
16	A	Raising of cattle
17	A	Raising of sheep
18	A	Raising of swine/pigs
19	A	Raising of poultry
20	A	Raising of animals n.e.c.; services to agriculture
21	A	Forestry and logging
22	A	Fishing
23	A	Crustaceans and molluscs
24	B	Hard coal
25	B	Lignite and peat
26	B	Petroleum extraction
27	B	Gas extraction
28	B	Iron ores
29	B	Uranium ores
30	B	Aluminium ore
31	B	Copper ores
32	B	Gold ores
33	B	Lead/zinc/silver ores
34	B	Nickel ores
35	B	Tin ores
36	B	Other non-ferrous ores
37	B	Quarrying of stone, sand and clay
38	B	Chemical and fertilizer minerals
39	B	Extraction of salt
40	B	Mining and quarrying n.e.c.; services to mining
41	C	Beef meat
42	C	Sheep meat
43	C	Pork
44	C	Poultry meat
45	C	Other meat products
46	C	Fish products
47	C	Cereal products
48	C	Vegetable products
49	C	Fruit products
50	C	Food products and feeds n.e.c.
51	C	Sugar refining; cocoa, chocolate and confectionery
52	C	Animal oils and fats
53	C	Vegetable oils and fats
54	C	Dairy products
55	C	Alcoholic and other beverages
56	C	Tobacco products
57	C	Textiles and clothing
58	C	Leather and footwear
59	C	Sawmill products
60	C	Pulp and paper
61	C	Printing
62	C	Coke oven products
63	C	Refined petroleum products
64	C	Nitrogenous fertilizers
65	C	Non-nitrogenous and mixed fertilizers
66	C	Basic petrochemical products
67	C	Basic inorganic chemicals
68	C	Basic organic chemicals
69	C	Pharmaceuticals and medicinal products
70	C	Dyes, paints, glues, detergents & other chem. prod
71	C	Rubber products
72	C	Plastic products
73	C	Clay building materials
74	C	Other ceramics n.e.c.
75	C	Cement, lime and plaster products
76	C	Other non-metallic mineral products n.e.c.
77	C	Basic iron and steel
78	C	Basic aluminium
79	C	Basic Copper
80	C	Basic Gold
81	C	Basic lead/zinc/silver
82	C	Basic nickel
83	C	Basic tin
84	C	Basic non-ferrous metals n.e.c.
85	C	Fabricated metal products
86	C	Machinery and equipment
87	C	Motor vehicles, trailers and semi-trailers
88	C	Other transport equipment
89	C	Repair and installation of machinery and equipment
90	C	Computers; electronic prod.; optical and precision
91	C	Electrical equipment
92	C	Furniture and other manufacturing n.e.c
93	D	Electric power gen., transmission and distribution
94	D	Distribution of gaseous fuels through mains
95	E	Water collection, treatment and supply; sewerage
96	E	Waste collection, treatment, and disposal
97	E	Materials recovery
98	F	Building construction
99	F	Civil engineering construction
100	G	Wholesale and retail trade; repair of motor vehicles and motorcycles
101	H	Road transport
102	H	Rail transport
103	H	Transport via pipeline
104	H	Water transport
105	H	Air transport
106	H	Services to transport
107	H	Postal and courier services
108	I	Hospitality
109	J	Publishing
110	J	Telecommunications
111	J	Information services
112	K	Finance and insurance
113	L	Property and real estate
114	M	Professional, scientific and technical services
115	N	Administrative services
116	O	Government; social security; defence; public order
117	P	Education
118	Q	Human health and social work activities
119	R	Arts, entertainment and recreation
120	S	Other services

Notes:

1 - only first level ISIC Rev.4 codes are shown here; more detailed correspondance available

Figure A.7: Sector list, page 75

A.8 Additional data

The data underpinning the model is primarily derived from the GLORIA database (refer to Section 3.1 in Chapter 6), which serves as the core source of multi-regional input-output data. This is further augmented with energy flow information obtained from the International Energy Agency (IEA) and the United Nations Energy Balances, which provide comprehensive and authoritative statistics on energy production, transformation, and consumption.

The model's key parameters are estimated using datasets detailed in the MINDSET Parametrization Data Documentation, ensuring transparency and reproducibility of the empirical approach. These estimation datasets incorporate a variety of economic and sectoral indicators essential for capturing the dynamic behaviour of economic agents within the model framework.

Beyond these principal data sources, the model integrates several additional datasets to establish initial conditions and to account for factors not fully represented in the core databases. These supplementary datasets enable the incorporation of contextual nuances and enhance the robustness of model calibration and are briefly described in Table A.5.

Variable	**Source**	**Use**
Share of government revenues by source	World Bank WDI, IMF GFS (IMF, 2024), OECD, national sources	Section 4.1.3 for linking government revenue drivers to government revenues and that to government spending
Investment by sector	GLORIA (Lenzen et al., 2013; Lenzen et al., 2017) & EU KLEMS (Bontadini et al., 2023)	Section 4.1.2 for establishing initial investment by sector
Capital stock by sector	GLORIA (Lenzen et al., 2013; Lenzen et al., 2017) & EU KLEMS (Bontadini et al., 2023)	Section 4.1.2 for establishing initial capital stock by sector
UN Population projections	UN Department of Economic and Social Affairs (UN, 2024)	Section 4.4 for calculating labour force developments
Labour force participation rate	World Bank WDI, ILOSTAT (ILO, 2025a)	Section 4.4 for calculating labour force developments
Unemployment rate	ILOSTAT (ILO, 2025a), IMF WEO	Section 4.4 for calculating initial labour force
Minimum wage	ILOSTAT (ILO, 2025a), news sources or lower end of wage spectrum	Section 4.5 for creating a wage floor
Policy interest rate	World Bank WDI	Section 4.1.2.1 used in interest rate calculation
Lending interest rate	World Bank WDI, IMF, EIU	Section 4.1.2.1 used in interest rate calculation
Inflation target values	World Bank WDI	Section 4.1.2.1 to be used in the Taylor-rule calculation

Table A.5: Additional data sources and their use in the model

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